# TFPT — $E_8$ Audit, Cascade Bridge and Bootstrap

The seven $E_8$ slices as an audit raster, the old cascade as the same even-integer spine, the Möbius self-consistency loop — and the thirty-two-step celestial/twistor route with the measure chain derived

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**What this document is about**

$E_8$ as an *audit container*, not a mystery: the seven maximal slices of 248 as a falsification raster (every load-bearing number must appear in at least one projection), the bridge showing the old $E_8$ orbit cascade $D{=}60{-}2n$ is the same even-integer spine as the compiler, and the Möbius bootstrap in which $g_{\text{car}} = 5$ and the "8" in $c_3$ are overdetermined $E_8$-closure fixed points (only $\pi$ irreducible). Since the celestial/twistor round the note also carries the **thirty-two-step continuum story** (§17) — from the $\mu_4$ clock to the $(E_8)_1$ boundary shadow — in which the **measure chain of the celestial route is derived, not declared**: single-valuedness from F-independence + the Quillen pairing (v520), the $1/\det_j$ normalisation the Atiyah–Bott fixed-point factor computed from three independent sources with $\psi = 64$ reproduced (v523), the residual [O] narrowed to the global BCOV integral beyond the fibre zero-mode factor; plus the Woit OS-bridge stages $\alpha/\beta_1/\beta_2/\beta_3$ (v519/v522/v524/v565 — the $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality typed: the OS cut and the $(2,2)$ signature forced, the kill branch empty by algebra, the induced member = the $\Theta_{\text{phys}}$ carrier), the thermal seam (v526), the ten-test side-blind scoreboard with the twist-class definition of the bit (v528) and the first firing interacting kill test under its typed fence (v529) — whose straddle filter, since run as a *selector*, keeps exactly *one* member alive: reflection positivity dynamically selects the alignment bit $\delta = \pi/2$ with positive coupling (v534, toy level); the straddle law is since refined to the *odd-split law* (a straddled cut kills RP iff its quartet split is odd; the quartet-avoiding cut family OS-viable at $g = 1$ — v972, toy level, fence inherited). `SEAM.EQUIV.01` and `WOIT.OS.TWISTOR.01` stay [O]; no marker moves.

**What $E_8$ is *not* (and why the known no-go results do not apply)**

$E_8$ here is the unimodular *audit / compiler hull* that classifies the admissible discrete charge and residue structures — **not** an unbroken physical gauge group. The Standard Model is a *readout* after projection (not a direct $E_8$ gauge theory), the Lorentz signature appears only after projection, and fermions are not "everything in the adjoint". Consequently the standard objections to literal $E_8$ world-formulas do *not* bite: Distler–Garibaldi (no chiral fermions / representation obstructions for $E_8$ as a 4d gauge group) and Coleman–Mandula (no nontrivial mixing of spacetime and internal symmetry) both constrain $E_8$ as a *spacetime gauge symmetry*, which TFPT does not claim. The defensible statement is: *TFPT is the unique parabolic compilation of the anchor $a = (1,1,2)$ into the $E_8$ audit hull* — a discrete classifier, attacked on its own (combinatorial) terms, not on gauge-theoretic ones.

**The TFPT document set — what is treated where**

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and four companions** (plus one standalone working note), best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point $\alpha^{-1}$, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one $\varphi_0$-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, $Q_+$ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven $E_8$ slices as an audit raster, the cascade bridge, the

Möbius bootstrap.

4. `tfpt_4_frontier` — honest status of $\eta_B$, $m_p/m_e$, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions: R.** `tfpt_research_contracts` — the open interfaces ($v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

**Working notes: N1.** `note_e8_gaussian_code` — the Gaussian code bridge: $E_8$ over $\mathbb{Z}[i]$ via Construction A over the extended Hamming code $[8,4,4]$, the canonical four-bit quotient $L/(1+i)L \cong \mathbb{F}_2^4$, and the $G_{31}$ quartic companion (exact algebra, machine-certified).

**You are reading document 3 ($E_8$ Audit & Bootstrap).**

**TFPT status at a glance — read this card the same way in every document**

**Two structural interfaces, one compiler.** P1 is a primitive reflection-positive seam kernel with unit winding; its normalization is $c_3 = \frac{1}{8\pi}$. From this datum and the five-slot carrier $g_{\text{car}} = 5$ (axiom P2), the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The numerical readouts $c_3$ and $g_{\text{car}}$ are elementary symmetric readouts of the anchor $a = (1,1,2)$. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below. Existing reductions share the four-mark geometry and anchor $a = (1,1,2)$; both W-bridge premises are derived at finite level (`v1014/v1015`), modulo the external MMST identification. The additional P1 determinant-response identification is separate. The Architecture chapter "Shared origin" consolidates these scopes (`REC20261105C3`).

**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

| Interface | Question | Status |
|---|---|---|
| $v_{\text{geo}}$ | the one metrology unit ($= 1/\sqrt{G} = m/\mu$); CLOSED METROLOGY ($\mathbb{R}_+$ torsor; one dim. calibration by dimensional analysis; No-Unit v153/v725) | closed metrology [O] |
| $G_{\text{net}}$ | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$ | [C] |
| $F_{\text{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$, axion, $m_p/m_e$) | [C] |

Beside that compiler rest sits the strict-physical-TOE rest: the ten-component $\text{Rest}_{\text{TOE}}$ / the eight gates of `TFPT.TOE.COMPLETE.01` [O] ($\text{TOE}_{\text{Complete}} :=$ AND(T1..T8); not a closure).

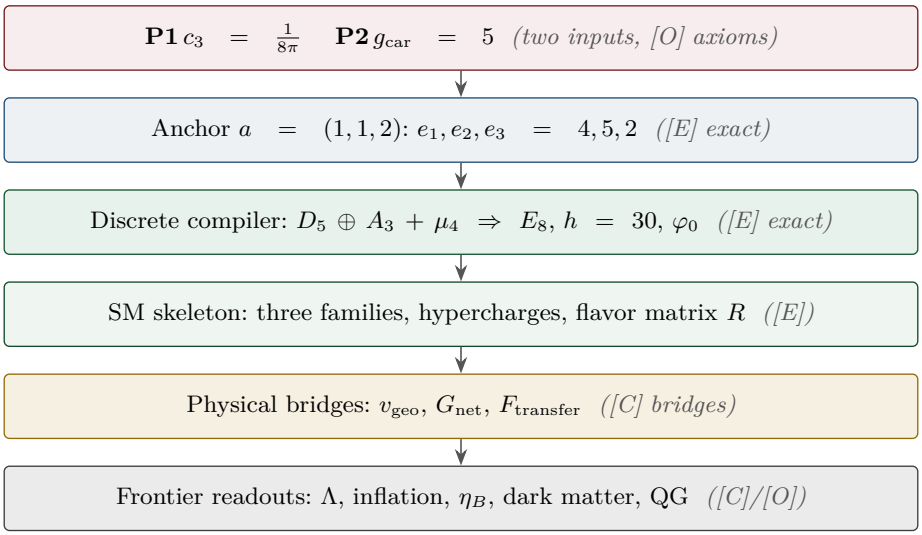
**The classical covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C].** The classical metric-sector field equation is no longer only an $R{+}R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2|2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so $c_3$ is triply over-determined — anchor, geometry, thermodynamics, `v358`) and $\Lambda$ from $\alpha$ ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in `v473` without changing the typing) and the one unit $v_{\text{geo}}$; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem $G_{\text{net}}$ (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST

route `SEAM.EQUIV.MMST.01` is *closed modulo cited theorems*; the twistor route `SEAM.EQUIV.TWISTOR.01` (Costello–Li, prepared by `CELEST.SEAM.01`) and the unconditional parent stay [O]. The lattice and index witnesses and the cited scaling and OS theorems establish the stated conditional route; they do not leave every old prerequisite untouched. The later charged MMST manuscript proves its uniform Green, edge graph, crossed-product and relaxed TEL-B estimates and retains the named microscopic local-algebra exhaustion condition for the full source. Its P1 corollary separately requires identification of the determinant-line response with the additive transitive four-channel response. The declared operational seed already supplies state and time data. The physical question concerns applicability of these theorems to the same source. The unconditional seam parent and T1–T8 keep their ledger statuses. Under the conformal-net hypotheses, rotation invariance makes the geometric deck a consequence (`v335`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin $1.648 > 0$, `v76/v330`). So the honest residual is the one unit $v_{\mathrm{geo}}$ plus the typed $F_{\mathrm{transfer}}$ interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels $(U_{\mathrm{wall}}, G_{\mathrm{metric}}, F_{\mathrm{frontier}})$ are retained only for ledger continuity — the live residual is $v_{\mathrm{geo}} \oplus F_{\mathrm{transfer}}$, with the seam $G_{\mathrm{net}}$ closed modulo cited theorems on its MMST route (parent [O]) and `QG.AMB.01` a [C] redundancy.

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

[E] exact / machine-proven [C] conditional [O] open / axiom [X] kill test




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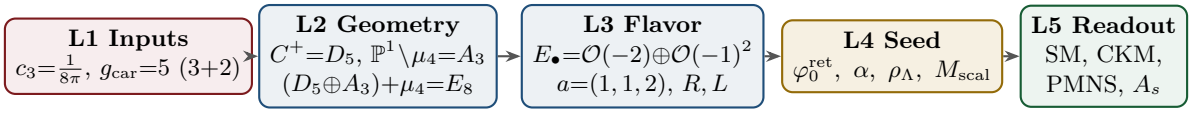
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# Part I
# The $E_8$ secondary-branching atlas

**Purpose and discipline.** This note does *not* introduce new physics. It builds an *audit raster*: the claim that *every load-bearing TFPT number should appear in at least one $E_8$ branching projection.* A number that appears in none is suspect; a number appearing in several independent projections becomes structurally strong. All arithmetic identities below are exact [E]; the *physical readings* are audit-level [O] unless tied to a derivation. (All 77 numeric claims were machine-checked with exact integer/rational arithmetic.)

## 1 The compiler core (where this note sits)

The strongest current form of the theory is a five-layer compiler:




$E_8$ is now a *closure* object, not an input: the carrier is the $D_5 = \mathfrak{so}_{10}$ half-spinor, the four punctures $\mathbb{P}^1 \setminus \mu_4$ give $A_3 = \mathfrak{su}_4$, and the common discriminant group $\mathbb{Z}_4$ glues them, with the norm test $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2$. This note charts the *secondary* maximal subalgebras of $E_8$ as audit windows onto the TFPT modules.

## 2 The $(1,1,2)$ anchor microcode

The parabolic anchor $a = (1,1,2)$ (splitting type of $E_\bullet = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$) is a miniature compiler. Its power sums $p_n(a) = 1^n + 1^n + 2^n = 2 + 2^n$ generate the discrete budget:

| $n$ | $p_n = 2 + 2^n$ | **TFPT reading (verified)** |
|---|---|---|
| 1 | 4 | $\lvert\mu_4\rvert = -\deg E_\bullet$ (glue index) |
| 2 | 6 | $\lvert R^+(A_3)\rvert$ (positive $A_3$ roots) $= \mathbb{Z}_6$ cusp order |
| 3 | 10 | $\mathcal{A}_\Lambda = \binom{5}{2} = \dim V_{D_5}$ (action ladder / pair sector) |
| 4 | 18 | $N_{\rm fam}\,\lvert R^+(A_3)\rvert = 3 \cdot 6$ |
| 5 | 34 | $2h(D_5) + 3\lvert R^+(A_3)\rvert = 16 + 18$; and $34 - \lvert R(A_3)\rvert = 22 = \sum R$ |

and the budget compresses to products of power sums:

$$q(A_3) = \frac{p_2}{2p_1} = \frac{3}{4}, \quad q(D_5) = \frac{p_3}{2p_1} = \frac{5}{4}, \quad \sum L = p_1 p_3 = 40,$$

$$\Omega_{\rm adm} = 2p_1p_2 = 48, \quad D_{\rm start} = p_2p_3 = 60, \quad \lvert R(E_8)\rvert = 4p_2p_3 = 240.$$

The first three power sums alone reach $E_8$: $p_1p_2p_3 = 4{\cdot}6{\cdot}10 = 240 = \lvert R(E_8)\rvert$, and $p_1p_2p_3{+}(p_4{-}p_3) = 240 + 8 = 248 = \dim E_8$.

**The anchor difference ladder: the binary carrier spine [E]**

While the *products* of the power sums give the large budgets, the *differences* give the binary carrier spine in one line:

$$\boxed{p_n(a) = 2 + 2^n \implies \Delta p_n = p_{n+1} - p_n = 2^n} : \quad (2,4,8,16) = \big(\lvert\mathbb{Z}_2\rvert,\ \lvert\mu_4\rvert,\ h(D_5),\ \dim S^+\big).$$

So the single anchor $a = (1,1,2)$ generates the big numbers multiplicatively and the doubling spine $2,4,8,16$ additively — one vector, two readings.

### The anchor reconstructs the carrier rank — and the 41-chain

The characteristic polynomial of the anchor vector is $(t-1)^2(t-2) = t^3 - 4t^2 + 5t - 2$, with elementary symmetric data

$$e_1(a) = 4 = |\mu_4|, \qquad e_2(a) = 5 = g_{\text{car}}, \qquad e_3(a) = 2 = |\mathbb{Z}_2|_{\text{sheet}}.$$

So the parabolic flavor geometry reads the carrier rank $g_{\text{car}} = 5$ *backwards*: $D_5 \Rightarrow a$ but also $a \Rightarrow g_{\text{car}}$. Moreover $3^2 + 4^2 = 5^2$ sits as $(p_0, e_1, e_2) = (3, 4, 5)$, giving the *anchor form* of the abelian index

$$\boxed{10\, b_1 = e_1(a)^2 + e_2(a)^2 = 4^2 + 5^2 = 41}$$

alongside the known $41 = 40 + 1 = \sum L + N_\Phi$. [E]

### Pascal closure: $g_{\text{car}} = 5$ is the unique solution (kills alternatives) [E]

The carrier rank is not merely a good fit — it is the *unique* root of an exact closure. The even-exterior carrier dimension $2^{g-1}$ must equal the truncated Pascal sum $\binom{g}{0} + \binom{g}{1} + \binom{g}{2}$ (the $1+g+\binom{g}{2}$ that builds $\dim S^+$):

$$\boxed{2^{\,g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2} \iff g = 5}$$

($g$=3 : 4≠7; $g$=4 : 8≠11; **g=5 : 16=16**; $g$=6 : 32≠22; $g$=7 : 64≠29). At $g = 5$ the row is $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$, the same Pascal triple that reads family, action ladder and the $E_8$ root count $16 \cdot 5 \cdot 3 = 240$. This is the fourth independent lock on $g_{\text{car}} = 5$ (with rank-fill, Coxeter-match, and integer-glue/norm), and unlike those it explicitly *kills* the neighbours $g = 4, 6$.

**The Pascal Ladder Theorem (`[verification/v108_pascal_ladder.py]`).** The closure generalises exactly: for *every* truncation degree $K$,

$$\boxed{2^{\,g-1} = \textstyle\sum_{k \le K} \binom{g}{k} \iff g = 2K + 1}$$

(odd $g$ solves by the binomial midpoint identity; even $g$ is excluded by the strict straddle $\sum_{k<g/2} < 2^{g-1} < \sum_{k\le g/2}$). **Consequence (zero slack): the carrier selection $g = 5$ is *equivalent* to the truncation degree $K = 2$** — the Pascal-*selection* residual of the red team (Target B) is therefore exactly the one named hypothesis *Quadratic Boundary Locality* (the seam certifies code degrees $\le 2$; the architecture document). Neighbour worlds on the ladder: $K$=1 → $g$=3 (a *one*-family world), $K$=2 → $g$=5 (three families), $K$=3 → $g$=7 (*nine* families), $K$=4 → $g$=9 *inconsistent* ($N_{\text{fam}} = 255/9 \notin \mathbb{Z}$). Overdetermination: among the ladder worlds, only $K = 2$ also satisfies the independent rank closure $g + N_{\text{fam}} = 8 = \operatorname{rank} E_8$ — two independent selections agree on the same world. The hypothesis itself stays [C].

**The seed is an anchor expression**

The retained seed is exactly the two-term anchor combination of $c_3$:

$$\boxed{\varphi_0^{\rm ret} = \frac{2p_1}{p_2}\, c_3 + 2p_1 p_2\, c_3^4 = \frac{4}{3} c_3 + 48\, c_3^4 = \frac{1}{6\pi} + \frac{3}{256\pi^4}}$$

(verified to 30 digits). The Cabibbo base is $\lambda_Y = \sqrt{\varphi_0^{\rm ret}(1-\varphi_0^{\rm ret})}$, the hierarchy engine of the whole fermion spectrum. **[E]**

## 3 The residue and length matrices as spectral objects

With the residue matrix $R$ and the full word-length matrix $L = R + 6W$ ($W$ the rank-one winding),

$$R = \begin{pmatrix} 1 & 3 & 0 \\ 1 & 5 & 2 \\ 2 & 5 & 3 \end{pmatrix}, \qquad L = \begin{pmatrix} 7 & 3 & 0 \\ 7 & 5 & 2 \\ 8 & 5 & 3 \end{pmatrix}, \qquad W = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

the spectral invariants are pure compiler numbers (all verified):

| **Invariant of $R$** | **value** | **Invariant of $L$** | **value** |
|---|---|---|---|
| $\operatorname{tr} R$ | $9 = N_{\rm fam}^2$ | $\operatorname{tr} L$ | $15$ |
| $\det R$ | $8 = h(D_5)$ | $\det L$ | $20 = \lvert R(A_4)\rvert = 2\mathcal{A}_\Lambda$ |
| $\chi_R(t)$ | $t^3 - 9t^2 + 10t - 8$ | $\chi_L(t)$ | $t^3 - 15t^2 + 40t - 20$ |
| $\mathrm{PrinMin}_2(R)$ | $(2, 3, 5)$ | $\mathrm{PrinMin}_2(L)$ | $(5, 14, 21)$ |
| $\lVert R\rVert_F^2$ | $78 = \dim E_6$ | — | — |

The three principal $2\times2$ minors of $R$ are $(2, 3, 5)$ — sheet, family, carrier — with product $2 \cdot 3 \cdot 5 = 30 = h(E_8)$ and sum $10 = \mathcal{A}_\Lambda$. The wrong down-branch $\{1, 3, 4\}$ fails exactly these invariants.

**Bilinear anchor forms.** With $\mathbf{1} = (1,1,1)^\top$ and $a = (1,1,2)^\top$ (all verified):

| $R$ | $\mathbf{1}$ | $a$ | $L$ | $\mathbf{1}$ | $a$ |
|---|---|---|---|---|---|
| $\mathbf{1}^\top$ | $22 = \sum R$ | $27 = 3^3$ | $\mathbf{1}^\top$ | $40 = \lvert R(D_5)\rvert$ | $45 = \dim D_5$ |
| $a^\top$ | $32 = 2^5$ | $40 = \lvert R(D_5)\rvert$ | $a^\top$ | $56 = \dim \mathbf{56}_{E_7}$ | $64 = \dim S^+ \cdot \lvert\mu_4\rvert$ |

The corner value $a^\top L\, a = 64 = (16, 4)$ reads exactly one spinor-glue sheet of the $E_8$ branching. And the rank-one winding update shows up in the inverses:

$$\boxed{R^{-1}\mathbf{1} = \tfrac{1}{4}(1,1,-1)^\top, \qquad L^{-1}\mathbf{1} = \tfrac{1}{10}(1,1,-1)^\top}$$

i.e. the residue response reads the glue denominator $\lvert\mu_4\rvert = 4$, the full-length response reads the decuple denominator $\mathcal{A}_\Lambda = 10$. **[E]**

**Anchor-plane Plücker coordinates: scattered hits become oriented areas [E]**

The strongest anti-numerology tool is to read *orientation invariants* of the anchor plane $\langle \mathbf{1}, a\rangle$ rather than loose integers. For $M \in \{R, Q, K, L\}$ the left block $B_M = \left(\mathbf{1}^\top M;\, a^\top M\right)$ (a $2\times3$) and the right

block $C_M = (M\mathbf{1} \;\; Ma)$ (a 3×2) have the 2×2 Plücker minors

| | $\mathrm{Pl}(\cdot)$ (left) | $\mathrm{Pl}_R(\cdot)$ (right) |
|---|---|---|
| $R$ | $(-6, 2, 14)$ | $(8, 12, 4) = (h(D_5), \lvert R(A_3)\rvert, \lvert\mu_4\rvert)$ |
| $Q$ | $(3, 6, 3)$ | $(0, 4, 5) = (0, \lvert\mu_4\rvert, g_{\mathrm{car}})$ |
| $K$ | $(-1, 6, 4) = (-N_\Phi, \lvert R^+(A_3)\rvert, \lvert\mu_4\rvert)$ | $(12, 12, -2) = (\lvert R(A_3)\rvert, \lvert R(A_3)\rvert, -\lvert\mathbb{Z}_2\rvert)$ |
| $L$ | $(6, 26, 14)$ | $(20, 30, 10) = (2\mathcal{A}_\Lambda, h(E_8), \mathcal{A}_\Lambda)$ |

Three load-bearing flavor numbers are now single oriented areas, not coincidences:

$$\|\mathrm{Pl}(K)\|_1 = 11, \qquad \|\mathrm{Pl}_R(K)\|_1 = \mathrm{Pl}(L)_{13} = 26, \qquad \sum \mathrm{Pl}_R(L) = 60 = D_{\mathrm{start}}$$

(11 is the mass-power anchor-plane norm; 26 the $c_t/c_b$ denominator; 60 the $E_8$ cascade start). The left-null-space response complements the right one above: the column sums of $\det L \cdot L^{-1}$ and $\det R \cdot R^{-1}$ are $(-5, 1, 6)$ and $(1, -5, 6)$, the same magnitudes $\{N_\Phi, g_{\mathrm{car}}, |R^+(A_3)|\} = \{1, 5, 6\}$. Finally the sector rows themselves are anchor-shifted: $K_e - K_u = (1, 1, 2) = a$ and $L_e - L_u = (1, 2, 3) = \mathrm{Exp}(A_3)$ — leptons are up-quarks displaced by the family roots. All entries are exact **[E]**; this is an audit layer, not a closure (the physical quark amplitudes stay **[C]**). [`verification/v37_plucker_anchor.py`]

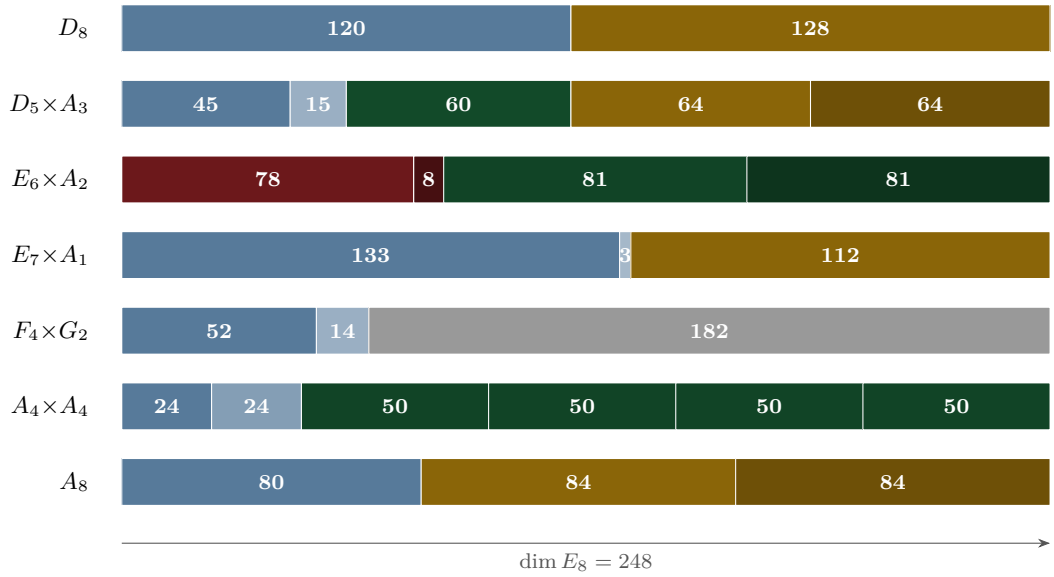



Figure 1: The $E_8$ slice atlas: seven maximal-subalgebra cuts of 248. Gold = spinor-glue (64=(16, 4), 112=7·16, 84=7·12, 128); red = flavor-residue (78=$\|R\|_F^2$, 8= det $R$); green = family/action blocks; blue = Lie adjoints.

**The numerology null test: the look-elsewhere burden quantified [E]**

The raster above disciplines *which* integers may appear; the complementary statistical question — *could a random "theory" of equal formula complexity land this many data hits?* — is answered by an explicit, fully declared null model. For each of the 13 scored frozen-registry observables the *entire* grammar class is enumerated exactly (no sampling): one-term expressions $\frac{p}{q}\,(\varphi_0^{\rm ret})^a \lambda_C^c (1{+}\lambda_C)^d \pi^b e^{u/w}$ with $p, q \le 130$, $|a| \le 2$, $|c| \le 3$, $|d|, |b| \le 1$, $|u| \le 6$, $w \le 6$, and two-term sums with coefficients $\le 12$ — a class that provably contains every TFPT formula it scores (machine-checked fairness condition). Data windows are conservative: the *maximum* of the achieved deviation, the documented experimental $\sigma$, the quote resolution and the 5% quark scheme spread, so the known tension observables ($\sin^2\theta_{13}$, $s_{23}$, $\delta_{\rm CKM}$, $m_\mu/m_\tau$, $\beta$) enter with their honest *large* windows and contribute almost nothing. Result of the census: joint formula-fishing probability $\prod_i p_i = 10^{-25.8}$, stable to $< 2$ orders of magnitude when the coefficient bound is varied $80 \ldots 200$ and the plausibility window $5\times \ldots 20\times$. Monte-Carlo cross-check (deterministic seeds): $2 \times 2{\cdot}10^5$ pseudo-theories — random rational dials inside TFPT's own formula skeletons, with the seed $\varphi_0^{\rm ret}$ fixed resp. drawn log-uniformly — reach at most 5/13 resp. 4/13 hits versus TFPT's 13/13. The test has *power*: perturbing $\varphi_0^{\rm ret}$ by 1%/10%/30% collapses TFPT's own scorecard to 9/6/4 hits, and shuffling which measured value is assigned to which formula collapses it to a mean of 1.2. Separately, all 94 500 complexity-matched variants of the $F_{U(1)}$ fixed-point equation (coefficients, $M$, transport exponent, resolvent power) are root-solved: exactly *one* — the TFPT instance $(2, \frac{4}{5}, M{=}41, e^{-2\alpha}, -\frac{5}{4})$ — lands inside the achieved $4{\cdot}10^{-8}$ CODATA window, so $p_\alpha \le 1.1{\cdot}10^{-5}$. Headline:

$$P\big(\text{null reproduces the scorecard incl. } \alpha\big) \;\le\; 10^{-30.7} \quad (\approx 102 \text{ bits})$$

*conditional on the declared grammar* — a look-elsewhere-corrected null-model rejection, never "certainty". Complementary to the frozen registry: the freeze (v84) prevents *future* look-elsewhere drift, this test quantifies the *retrospective* look-elsewhere burden. [`verification/v100_numerology_null_mc.py`]

## 4 The secondary branching atlas

Each maximal subalgebra of $E_8$ mirrors a TFPT module. The dimension identities are exact [E]; the readings are audit-level [O]. The seven *slices* of dim $E_8 = 248$ are shown below: the same total cut seven ways, each cut a different TFPT module.

| **Subalgebra** | $248 =$ | **TFPT reading** |
|---|---|---|
| $D_8$ | $120 + 128$ | family-transition adjoint + two spinor-glue sheets ($128{=}2\dim S^+\lvert\mu_4\rvert$) |
| $D_5 \times A_3$ | $45 + 15 + 60 + 64 + 64$ | **main split**: $\dim D_5$, $\dim A_3$, $\mathcal{A}_\Lambda\lvert R^+(A_3)\rvert$, two $(16,4)$ glue sheets |
| $E_6 \times A_2$ | $78 + 8 + 81 + 81$ | **flavor residue**: $\lVert R\rVert_F^2$, $\det R{=}\dim A_2$, cubic family blocks |
| $E_7 \times A_1$ | $133 + 3 + 112$ | **scalaron**: $112{=}7\cdot 16$ (exponent × one family) |
| $F_4 \times G_2$ | $52 + 14 + 182$ | **occupancy/gauge**: $\lvert R(F_4)\rvert{=}48{=}\Omega_{\rm adm}$, $\lvert R(G_2)\rvert{=}12{=}\dim\mathfrak{g}_{\rm SM}$ |
| $A_4 \times A_4$ | $24 + 24 + 4\cdot 50$ | **action ladder**: $24{+}24{=}48{=}\Omega_{\rm adm}$; off-diagonal $(5,10)$ reps $\mathcal{A}_H, \mathcal{A}_\Lambda$ |
| $A_8$ | $80 + 84 + 84$ | rank-decuple + scalaron: $80{=}8\cdot 10$, $84{=}7\cdot 12$ |

**The $D_8$ split is a magnitude/phase channel [E]/[C]/[O]** ([verification/v227_degree_exponent_channel_split.py])

The $D_8$ row $248 = 120 + 128$ is the canonical *exponent/degree* split of $E_8$: the exponents sum to $120 = |R^+(E_8)|$ (a **magnitude** channel — positive roots, root transitions, the dimensionless ratio/product readouts), the fundamental degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ sum to $128 = 2^{\text{rank}-1} = \text{rank}\, E_8 \cdot \dim S^+$ (a **phase/glue** channel — the spinor/Ramond sheets). This gives the red-team Target D CP residual a clean home: the (ratios, product) $\Rightarrow v_{\rm geo}$ magnitude bijection lives in the 120 channel, the un-covered CP phases in the 128 channel (cf. the hexagonal phase fiber, red team Target D). Whether 128 also hosts a dark sector is Frontier [O] — this is the honest replacement for the over-strong "$S^-$ is dark matter" reading, not an assertion of it. The conjugate sheet $S^-$ and the double-cover deck sit in this *matched* channel ($|\mathbb{Z}_2| = 2$ and $128 = \sum \deg$), forced-disjoint from the five unmapped degrees $\{12, 14, 18, 20, 24\}$ (v430): the seam's "other side" is not the hull overhead.

**$E_8$ Slice Compression Lemma [E] (audit)**

The seven slices are not seven separate narratives — they are *one* projection of two small alphabets. The anchor power sums $p_n(a) = 2 + 2^n$ give $P = (p_0, p_1, p_2, p_3) = (3, 4, 6, 10)$, whose six $K_4$ edge products are exactly the carrier blocks $(p_0p_1, \ldots, p_2p_3) = (12, 18, 30, 24, 40, 60) = (|R(A_3)|, N_{\rm fam}|R^+(A_3)|, h(E_8), |W(A_3)|, |R(D_5)|, D_{\rm start})$; the six Sheet-Diamond operators carry determinants $(\det Q, \det K, \det R, \det C, \det L, \det F) = (3, 4, 8, 14, 20, 32)$ with $\sum \det = 81 = N_{\rm fam}^4$, the total determinant charge of the flavour space. All seven 248-slices, the row-budget cross $\text{row}(C + sU + tV) = (7, 11, 13) + s(3, 3, 3) + t(1, 2, 3)$ (democratic winding + $A_3$ exponents), and $\dim E_6 = 78 = p_2(p_0 + p_3)$ then follow from $P$, those determinants and $\Delta = p_0 + p_3 = 13$. The atlas is therefore a closed grammar over admissible invariant classes (power sums, determinants, branching dimensions), not a shelf of trophies ([verification/v170_diamond_slice_compression.py]).

**Theorem 1** ($E_6 \times A_2$ flavor-residue shadow [O])**.** The principal flavor branching reads the residue matrix:

$$\|R\|_F^2 = 78 = \dim E_6, \qquad \det R = 8 = \dim A_2 = h(D_5), \qquad \mathbf{1}^\top R\, a = 27,$$

and the full $E_6 \times A_2$ dimension count is the residue identity

$$\boxed{248 = \|R\|_F^2 + \det R + 2\,(\mathbf{1}^\top R\, a)\, N_{\rm fam} = 78 + 8 + 2\cdot 27\cdot 3.}$$

Here $E_6$ reads the Frobenius norm of the residue matrix, $A_2$ the pure three-family symmetry, and the **27** the cubic family block. (Strongest new $E_8$ flavor fingerprint.)

**Theorem 2** ($E_7 \times A_1$ scalaron shadow [O])**.** With the scalaron exponent $7 = \Omega_{\rm adm} - 10b_1 = -\deg E + \text{rk}\, E$,

$$248 = 133 + 3 + 112, \qquad 112 = 7 \cdot \dim S^+ = 7 \cdot 16, \qquad |R(E_7)| = 126 = 7 \cdot 18,$$

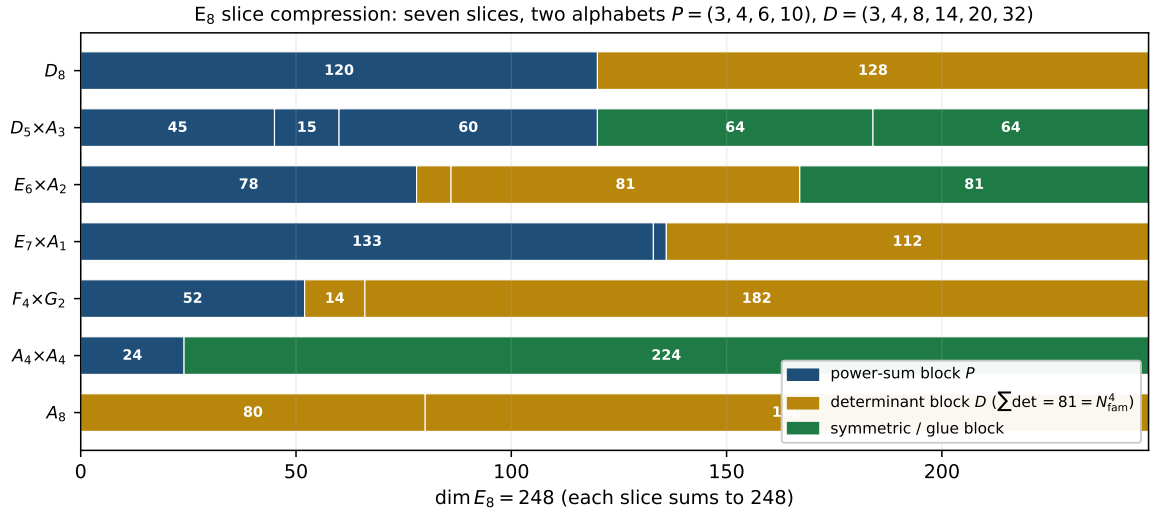



Figure 2: The same seven 248-slices, now coloured by their *alphabet* source rather than by physics module: blue = anchor power-sum blocks ($P = (3, 4, 6, 10)$), gold = Sheet-Diamond determinant blocks ($D = (3, 4, 8, 14, 20, 32)$, $\sum \det = 81 = N_{\text{fam}}^4$), green = symmetric/glue data. Every slice is one projection of the two alphabets ( [`verification/v170_diamond_slice_compression.py`] ).

where $18 = N_{\text{fam}}|R^+(A_3)|$. The scalaron exponent 7 (the seam power $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2 = c_3^7$) is thus exactly the $E_7$ off-block coefficient. *Cross-check:* the centred $E_8$ exponents give $\sum_{m\in\text{Exp}(E_8)} |m - 15| = 56 = \dim \mathbf{56}_{E_7}$ (exponents $\{1, 7, 11, 13, 17, 19, 23, 29\}$, the primitive residues mod 30).

**Proposition 1** ($A_4 \times A_4$ action-ladder and $F_4 \times G_2$ occupancy [O]). $248 = 24 + 24 + 4(5 \cdot 10)$ with $24 + 24 = 48 = \Omega_{\text{adm}}$ and off-diagonal reps $(\mathbf{5}, \mathbf{10}) = (\mathcal{A}_H, \mathcal{A}_\Lambda)$ — the action ladder appears inside $E_8$ (an audit, *not* an $SU(5)$ GUT). And $F_4 \times G_2$: $|R(F_4)| = 48 = \Omega_{\text{adm}}$, $|R(G_2)| = 12 = \dim \mathfrak{g}_{\text{SM}}$, $\operatorname{rank} F_4 = 4 = |\mu_4|$, $\operatorname{rank} G_2 = 2 = |\mathbb{Z}_2|$.

**Flavor-matrix and prime-table fingerprints (audit-level, machine-verified).** Five exact linear-algebra readouts of $R, L$ and the prime atoms, kept here under the no-free-pattern rule.

- **Row norms ladder.** $\|R_u\|^2, \|R_d\|^2, \|R_e\|^2 = (10, 30, 38) = (\mathcal{A}_\Lambda,\ h(E_8),\ 2 \cdot 19)$ with $10{+}30{+}38 = 78 = \dim E_6$ ($19 \in \text{Exp}(E_8)$).
- **Residue inventory.** The entries of $R$ on $C_6$ are $m = (1, 2, 2, 2, 0, 2)$: the *only* gap is $4 = |\mu_4| = \sum a$. Moments $\sum R = 22$, $\sum R^2 = 78 = \dim E_6$ — so $\|R\|_F^2 = 78$ is the second moment of the $C_6$ inventory with the $\mu_4$ hole.
- **First-column norm $\to E_6{\times}A_2$ off-blocks.** $L_{\cdot 1} = (7, 7, 8)$, $\|L_{\cdot 1}\|^2 = 162 = 81 + 81$ — the two **81** off-blocks of the $E_6{\times}A_2$ slice are exactly the wound first-generation column.
- **Lepton exponent polynomial.** The $\varphi_0^{\text{ret}}$-ladder lepton powers $(k_e, k_\mu, k_\tau) = (5, 3, 2)$ are the Coxeter primes $30 = 2 \cdot 3 \cdot 5$ in hierarchy order: $(t-5)(t-3)(t-2) = t^3 - 10t^2 + 31t - 30$ ($e_1 = \mathcal{A}_\Lambda$, $e_3 = h(E_8)$). The lepton $c$-ratios obey $\frac{(16/7)(7/6)}{4/3} = 2 = |\mathbb{Z}_2|$ and $\frac{(16/7)(7/6)}{(4/3)^2} = \frac{3}{2}$ (the near-Koide factor).
- **The $2, 3, 5$ multiplication table** (the shortest integer-landscape compression):

$$(2, 3, 5) \xrightarrow{\times} (6, 10, 15, 30) \to (8{=}3{+}5,\ 16,\ 240{=}8 \cdot 30,\ 248{=}240{+}8).$$

## 5 Group-theoretic audits (optional)

**Weyl completion index.** Over the main stabiliser,

$$\frac{|W(E_8)|}{|W(D_5)|\,|W(A_3)|} = \frac{696729600}{1920 \cdot 24} = 15120 = 63 \cdot 240 = (2^6 - 1)\,|R(E_8)|.$$

A group-theory machine-check, not physics: it quantifies the Weyl symmetry created beyond the $D_5 \times A_3$ stabiliser. [E] (arithmetic) / [O] (reading)

**$E_8$ theta series as a higher-shell raster.** $\Theta_{E_8} = E_4 = 1{+}240\sum_{n\geq 1}\sigma_3(n)q^n$ has shell degeneracies 240, 2160, 6720, 17520, … The first is $|R(E_8)| = 240$; the second is $2160 = 9 \cdot 240 = N_{\mathrm{fam}}^2|R(E_8)|$. The prime shells read the sheet-generator bilinears of v410: $\sigma_3(2) = 9 = a^T V\mathbf{1} = N_{\mathrm{fam}}^2$, $\sigma_3(3) = 28 = \mathbf{1}^T V^3 a = \det(I{+}R)$ and $\sigma_3(5) = 126$ — the local $E_7$ root shell {0:126} of lemma (v) below. If later one-loop or CMB shell degeneracies touch these multiplicities, the theta series is the right place to type them — as an audit raster, not a claim. [O] ([verification/v410_sheet_generator_binary.py])

**The interacting Ext-source response (DIAMOND.BIT.EXTSOURCE.01).** The v572 remaining route is executed (8 checks, ∼1 s): rank-one Ext-source coupling on the Gibbs pencil has exact rational flip thresholds $J_c(s) = -1/(v^T K(s)^{-1} v)$ whose numerators are exactly the twins' pencil factors; in the crossed phase positive coupling $J_c \leq 15/16 < 1$ heals exactly the compiler twin, while the alt twin cannot be flipped by any positive coupling in the open unit box — the healing direction is strictly positive (consistent with v534); the response pole sits on the golden polynomial $s^2 - s - 1$ (observed). A response-level selection, not a positivity theorem; the alignment-bit contract stays open. Machine-checked: [verification/v584_ext_source_response.py]. [E] (thresholds exact; response level only; no marker moves)

**The sheet-involution existence (QGEO.INVOL.01).** The involution-existence input of GATE.QGEO.01 is derived (7 checks, < 1 s): the integral $V$-commuting involutions are exactly $(\mathbb{Z}/2)^3$ (rigidity, both twins); the v574 type theorem plus seam reversal plus zero-mode fixing single out the explicit $S_0$ of exact $\Sigma$-signature uniquely — existence unconditional; the corpus RP reflection is a different structure (anti-automorphism, does not commute with $V$); the two demands' continuum derivation stays the named open. Machine-checked: [verification/v590_involution_existence.py]. [E] (exact sympy; demands declared; no marker moves)

**The $\mu_3$-cover candidate (QGEO.COVER.01).** Strategy I/II, first executed step (6 checks, < 1 s): the cover $y^3 = x^4 - 1$ hits four compiler fingerprints — rank-3 Eisenstein $H_1$ (Chevalley–Weil $(1, 2)$), projective $D_4$ via the deck ($r^4 = \omega$, Burau at $t = \omega$ with verified braid relations), cusp-class group $(\mathbb{Z}/3)^4$ of order 81 (the saturation index with the per-mark mechanism), and the unique invariant hermitian form of LORENTZ signature $(1, 2)$ (matching Paper II's rank-3 polarization — Strategy II's first prediction). The $Q/V$ dictionary and the RP map are the named next steps; the gate does not move. Machine-checked: [verification/v597_mu4_cover_model.py]. [E] (exact sympy; convention validated; no marker moves)

**The cover dictionary, round 2 (QGEO.DICT.01).** Three compiler objects located exactly (4 checks, < 1 s): $U \sim G_2$ (single-cusp Gram, char $x^2(x - 3)$); $Q_- \sim G_1 - G_3$ (the $\mu_4$ sheet coupling as the *diamond-diagonal* Gram difference, char $x(x^2 - 3)$); the binary AnchorLadder $\sim 1 + G_1 + G_3$ (char $(x - 1)(x - 2)(x - 4)$); opposite-twist products nilpotent; the grading half $(Q_+/V/H/Q)$ absent from 7296 scanned canonical combinations — the real structure is the named suspect; the Strategy-I kill criterion is not triggered. Machine-checked: [verification/v598_cover_dictionary.py]. [E] (exact entries; quantified negatives; no marker moves)

**The real structure (QGEO.REAL.01).** The v598 suspect constructed (18 checks, ∼3 s): the anti-holomorphic $D_4$ reflection acts semi-linearly, $R(v) = M\overline{v}$, with a *one-dimensional* intertwiner space ($R$ unique up to scale), $R^2 = +1$ exactly, and all four twist intertwinings $RT_kR^{-1} = T_{5-k}^{-1}$ exact. Its grading functor $\Gamma(A) = M\overline{A}M^{-1}$ closes the three dictionary gaps at char level: $V \sim G_2{+}G_3{-}\Gamma(G_3)$ (char $x(x - 1)(x - 2)$), $H \sim 1 + G_2 - \Gamma(G_3)$ (char $(x - 1)^2(x - 2)$ — *the anchor spectrum*), $Q_+ \sim 1 + G_2 + G_3 - \Gamma(G_3)$ (char $(x - 1)(x - 2)(x - 3)$); the $\Gamma$-even/odd cusp elements carry $\{0, 1, 5\}$ ($g_{\mathrm{car}} = 5$), $\{0, 3, 9\}$, $\pm\sqrt{5}$, $\pm\sqrt{27}$ and one nilpotent. Honest joint negative, quantified: $4 \times 6$ candidates, zero pairs pass $\mathrm{tr}(UV) = 3$ and $\mathrm{char}(U{+}V) = \mathrm{char}\, Q$ — char-matching is not yet an algebra embedding (kill not triggered). The invariant hermitian form is unique up to scale with $\det J = 8$ — a sheet-diamond determinant-list member — and Lorentz signature $(1, 2)$: Strategy

II's Hodge prediction pinned at form level. Machine-checked: [verification/v599_real_structure.py]. [E] (exact existence/uniqueness; quantified joint negative; no marker moves)

**The joint embedding (QGEO.EMBED.01).** The compiler pair is *realized* on the $\mu_3$-cover (15 checks, ∼4 s): in the Γ-even degree-2 sector the pair $U^* = -2E_1 + (G_1\Gamma(G_1) + \Gamma(G_1)G_1) + 3$, $V^* = -E_3 + \frac{1}{2}(G_2\Gamma(G_2) + \Gamma(G_2)G_2) + 1$ is $\mathbb{Z}[\omega]$-integral with compiler-exact spectra and joint traces ($\mathrm{tr}(U^*V^*) = 3$, $\mathrm{tr}(U^*V^{*2}) = 6$, $\mathrm{char}(U^*+V^*) = \mathrm{char}\, Q$); the seven words have rank 7 (the full 2|1 parabolic) and an explicit invertible $g$ solves $gU^*g^{-1} = U_c$, $gV^*g^{-1} = V_c$ exactly — the v566 self-code algebra lives on the cover. On the explicit saturated $\mathrm{Fix}(R)$ lattice (SNF certificate $(1,1,1)$) the restrictions are *integer* matrices whose seven-word $\mathbb{Z}$-order has Smith divisors $(1,1,1,3,3,3,3)$ — saturation index $81 = 3^4$, exactly the v566 self-code index. Must-break control passes (the degree-1 pair fails the joint criterion); canonicity is open and typed open (existence, not uniqueness); the gate does not move. Machine-checked: [verification/v600_joint_embedding.py]. [E] (exact realization; canonicity open; no marker moves)

**The canonical equivariant realization (QGEO.CANON.01).** The canonicity question answered (14 checks, ∼12 s): the Burau image preserves $J$, not the standard hermitian form, so the canonical building blocks are the J-adapted cusp projectors $GJ_k = (1-T_k)J^{-1}(1-T_k)^\dagger J$ — idempotent rank-1 J-self-adjoint projectors rotating *exactly* under $D_4$ ($rGJ_kr^{-1} = GJ_{k+1}$; opposite cusps J-orthogonal). The pair family $U_k = 3\,GJ_k$, $V_k = 1 + GJ_{k-1} - GJ_kGJ_{k+1}$ is $\mathbb{Z}[\omega]$-integral, compiler-exact (spectra, joint data, seven-word rank 7) and $D_4$-*equivariant by construction*: the mark choice is the compiler's winding choice. The duality surprise: the equivariant pair is *not* conjugate to $(U_c, V_c)$ (the intertwiner is identically singular — must-fail documented) but exactly conjugate to the *transposed* pair — the equivariant presentation realizes the DUAL (plane-stabilizer) parabolic, the v600 pair the line-stabilizer: the cover carries both types, exchanged by duality. And the v590 Klein four-group lives *integrally* on the curve: all eight spectral-sign involutions of $V_1$ are $\mathbb{Z}[\omega]$-integral, the two demands select the explicit $S_0$ (trace $-1$), and $AB = S_0$ closes the group. Honest: the naive J-word Gram has inertia $(3,2,2)$, not the v572 pattern $(4,0,3)$ — the Hodge/RP comparison needs the Weil-twisted polarization (named open); the seam identification stays open. Machine-checked: [verification/v601_equivariant_dual.py]. [E] (exact equivariance/duality/Klein; typed opens; no marker moves)

**The duality-and-forms round (QGEO.DUALFORMS.01).** The duality mechanism identified (17 checks, ∼8 s): $U_1$ is J-*self-adjoint*, the J-adjoint of $V_1$ is the exact product reversal, and $(U_1, V_1^J)$ is exactly conjugate to the compiler line-stabilizer pair while $(U_1, V_1)$ realizes the transposed dual — the J-adjoint exchanges the two parabolic types inside one equivariant structure. The dictionary completes in the J-projector language: $H = 1 + V(V-1)/2$ carries char $(x-1)^2(x-2)$ — *the corpus anchor formula holds verbatim on the curve*; $Q_+ \sim 2 - GJ_1 + GJ_2GJ_3$ and the binary ladder $\sim 2 - GJ_1 + 2\,GJ_2GJ_3$ are exact. Weil positivity holds exactly: the twisted form $JC$ is positive definite with eigenvalues $\{2\sqrt{2}-2,\ 2,\ 2+2\sqrt{2}\}$ — the Riemann bilinear relations on the $\omega$-sheet; and the canonical bilinear form $B = M^\dagger J$ (real structure × polarization) is symmetric with $\det B = 8$. Honest negatives, typed: $g^Tg \neq \lambda B$ (the corpus transpose is not the $(R,J)$-form); the naive mark-local cokernel map fails; the seam normal is explicit but not a deck eigenvector. The v600 integer core is Lean-certified (`FORM.QGEO.04`, `TfptCarrier/CoverEmbedding.lean`: determinant-free Smith certificates, kernel `decide`). Machine-checked: [verification/v602_duality_forms.py]. [E] (exact duality/dictionary/positivity; quantified negatives; no marker moves)

**The seam-and-marks round (QGEO.SEAMMARKS.01).** The seam line found, and it is *real* (15 checks, ∼7 s): the line-stabilizer pair has a unique common eigenline $v$ ($Uv = 0$, $Vv = v$ — the transported compiler seam), explicitly $v = (0, (\sqrt{3}-i)/(\sqrt{3}+i), 1)$; the real structure *fixes* this line and the two-demand involution *reverses* it ($S_0v = -v$) — the v590 seam-reversal demand holds geometrically on the curve. The infinity cusp separates in the unreduced Burau module (invariant column $(1,1,1,1)$; full twist eigenvalues $\{1, \omega^{\times 3}\}$). And the first per-mark $\mathbb{Z}/3$ witness lands: $E_1$ restricts to an integer matrix lying in the seven-word algebra with denominator lcm exactly 3 — an order-3 class in the word-order cokernel (the v566/v597 mechanism's first exact witness);

$E_4$'s class is trivial and $E_2/E_3$ lie outside the mark-1-anchored algebra (equivariant per-mark framing typed). The corpus-transpose translator negatives are quantified ($F_2 = g^T \Sigma g$ matches no canonical menu form); named open. Machine-checked: [verification/v603_seam_marks.py]. **[E]** (exact seam/infinity/witness; quantified negatives; no marker moves)

**The equivariant order and the infinity bridge (QGEO.EQORDER.01).** Three integral statements (12 checks, ∼4 s). First, a correction with a payoff: $R$ *is* an anti-isometry of the polarization — the correct compatibility equation $M^\dagger J M = J^T$ holds exactly — so $J$ restricts to a real symmetric *integer* form on the saturated fixed lattice, $J_{\text{fix}} = [[16, 2, 4], [2, -2, 2], [4, 2, -2]]$ with $\det = 72 = 8 \cdot 9$. Second, the four mark frames together generate a *full* order: the joint lattice of all 28 rotated words has rank 18 in $M_3(\mathbb{Z}[\omega])$ with Smith divisors $(1^{15}, 3, 3, 3)$ — index $27 = 3^3$ (one frame: the 7-dim parabolic of index $81 = 3^4$; all four: the full algebra of index 27). Third, the 3-torsion *localizes at infinity*: $\text{coker}(1 - T_\infty) = \mathbb{Z}^2 \oplus (\mathbb{Z}/3)^3$ while every finite cusp twist is torsion-free — three objects share $(\mathbb{Z}/3)^3 = 27$ (infinity monodromy, equivariant order, $\det_{\mathbb{Z}}(1 - \text{deck})$): the compiler constants localize, 81 at the finite marks and 27 at infinity. Translator negatives extended ($J_{\text{fix}}$-menu and algebra twists fail; the $\bar{\omega}$-sheet is the named candidate). Machine-checked: [verification/v604_equivariant_order.py]. **[E]** (exact integral statements; typed localization reading; no marker moves)

**The translator round (QGEO.TRANSLATOR.01).** The corpus RP anti-automorphism is *found* on the curve (11 checks, ∼11 s): pulling $\theta(X) = \Sigma X^T \Sigma$ back along the compiler conjugation and composing with the J-adjoint gives an algebra automorphism $\varphi$ — and $\varphi$ is *inner*: the intertwiner space is one-dimensional, its generator $c$ is invertible, and $\varphi(A) = cAc^{-1}$ holds exactly. Hence $\theta_{\text{corpus}} = J\text{adj} \circ \text{inner}(c)$ — the RP structure is fully expressed in curve language (polarization plus one explicit element), with $14c$ integral over $\mathbb{Z}[\omega]$ (denominator $14 = 2 \cdot 7$, the parabolic dimension); the single-sheet negatives of v603/v604 are thereby resolved (the missing object was inner, not a new form). The seam is a *nonzero pure 3-torsion class at infinity*: its lift has zero free coordinates and torsion coordinates $(2, 2, 0)$ in $\text{coker}(1 - T_\infty)$ — the integral seam/infinity identification. The 81 *survives* frame-joining: the chain $O_1 \subset O_{\text{joint}} \cap \text{span}_1 \subset P_1$ has indices $3^5$ and $3^4 = 81$. And the $J_{\text{fix}}$ duality is certified with an explicit integer dual ($C_V^T J_{\text{fix}} = J_{\text{fix}} C_{V\,\text{dual}}$; Lean addendum green). $c$'s independent geometric identity stays open. Machine-checked: [verification/v605_translator.py]. **[E]** (exact translator/bridge/chain/duality; typed opens; no marker moves)

**The mode-separation round (QGEO.MODESEP.01).** The zero-mode selection gains its carrier (7 checks, ∼2 s): all three eigenlines of the sheet operator $V$ (eigenvalues $0, 1, 2$), lifted to the unreduced integral module along the *unique* intertwiner, are *pure 3-torsion classes at infinity* (zero free part), and the three classes are *pairwise distinct* elements of $(\mathbb{Z}/3)^3$ (fixed Smith frame $(2, 0, 2)/(2, 2, 0)/(1, 2, 0)$; distinctness frame-invariant, none trivial) — the infinity-cusp torsion distinguishes the zero mode from the top mode, exactly the separation the real structure could not provide. The canonical rule singling out the "+1" class stays open. And the arithmetic-identity route for the translator element is closed honestly: $\text{char}(7\Gamma(c)c) = y^3 + 3y^2 + 6y - 1$ has discriminant $-783 = -3^3 \cdot 29$, not a square — Galois $S_3$, not abelian, no cyclotomic identity (Kronecker–Weber). Machine-checked: [verification/v606_mode_separation.py]. **[E]** (exact separation; typed frame caveat; honest arithmetic negative; no marker moves)

**The selection rule (QGEO.SELRULE.01).** The canonical "+1" rule found (10 checks, ∼2 s): the infinity torsion is a genuine $\mathbb{F}_3$ space ($\omega$-multiplication acts as the identity), and the deck rotation acts on it with characteristic polynomial $(x + 1)(x^2 + 1) \bmod 3$ — $x^2 + 1$ irreducible over $\mathbb{F}_3$, so there is *exactly one* projective eigenline (eigenvalue $-1$), and it *is* the top-mode class; the zero mode and the seam are not projectively fixed (nor under $r^2$). With v603/v605 the three modes carry mutually exclusive canonical labels — seam = the *R-real* line, top = the unique *deck-eigen* torsion class, zero = the only *unlabeled* mode — and the two-demand involution flips exactly the labeled pair: *both* v590 demands are now expressed through canonical cover structure (typed reading; the continuum derivation from the raw seam stays the gate residual). Machine-checked:

[verification/v607_selection_rule.py]. **[E]** (exact $\mathbb{F}_3$/rigidity/labeling; typed selection reading; no marker moves)

**The free gamma slice (WOIT.GAMMA.FREE.01).** The first executed step of the WOIT $\gamma$ milestone (9 checks, ∼1 s, on the explicit v524 net): the mirror (anti-chiral) state's OS Gram splits *sector-exactly* — even $(8,0,0)$ positive definite, odd $(0,8,0)$ *strictly negative definite* — so every mirror fermion vector has strictly negative OS norm and *no mirror mode survives OS reconstruction*: kill test (3) cannot fire at the free level (the v519 shadow upgraded to the sector level). The chiral generation is multiplicity-free (clock spectrum exactly $\{1, \sqrt{2}-1\}$ on the fermion domain), and the mark incidence holds at the free level: the two mark algebras have PD Grams of dimension $4 = |\mu_4|$, jointly generate the full 16-dim $H_{\text{phys}}$, and the quarter-turn transfer $\tau_2$ *is* the mark-A→B transport (Hermitian PD) — kill test (7) does not fire there. The $\gamma$ milestone itself stays open on $\mathcal{A}_{\text{hol}}$ (kill tests live; (6) untouched). Machine-checked: [verification/v608_gamma_free_slice.py]. **[E]** (exact sector/spectrum/incidence on the explicit net; free slice only; no marker moves)

**The translator factorization (QGEO.CFACTOR.01).** The geometric identity of the translator element is *closed* (8 checks, ∼8 s): the mirror intertwiner $d$ (defined by $dA = \Gamma(A)d$ on the realized pair, $\Gamma$ the real-structure conjugation) is *unique* (1-dim intertwiner space), *unimodular* ($\det d = 1$), *integral* over $\mathbb{Z}[\omega]$, and $\Gamma$-*involutive* ($\Gamma(d)d = 1$ exactly); and the v605 element factorizes exactly, $c = \lambda K_1 d$ with $K_1 = J^{-1}\overline{F_2}^{T} M^{-1}$ — so $\theta_{\text{corpus}} = J\text{adj} \circ \text{inner}(K_1) \circ \text{inner}(d)$: polarization × RP metric × real structure × canonical mirror intertwiner. With this, all three v605 residuals are resolved (seam reversal geometric v603; zero-mode rule canonical v607; $c$-identity factorized v609); the v606 arithmetic negative stands as the reason the cyclotomic route had to fail. Honest scope: $F_2$ carries the corpus datum $\Sigma$ by definition of the translator; the gate does not move. Machine-checked: [verification/v609_translator_factorization.py]. **[E]** (exact factorization; closed-identity reading; no marker moves)

**The curve landscape (QGEO.LANDSCAPE.01).** The inverse question — what *else* does the cover carry? — answered (9 checks, ∼1 s). The Jacobian *splits*: the three holomorphic differentials carry (deck, rotation) characters of combined orders $(12, 12, 6)$, so by CM theory $\text{Jac}(C) \sim A_2(\text{CM } \mathbb{Q}(\zeta_{12})) \times E(\text{CM } \mathbb{Q}(\omega))$ — with $\mathbb{Q}(\zeta_{12}) = \mathbb{Q}(i, \sqrt{3})$ the field generated by the deck *and* the mark rotation together, and $E : y^3 = u^2 - 1$ the $x \mapsto -x$ quotient. Point counts confirm the splitting: $a_p(C) = a_p(E)$ *exactly* for all ten tested $p \not\equiv 1$ (12) (the $\zeta_{12}$ surface is supersingular there), with nonzero surface traces exactly at $p \equiv 1$ (12). And the integer-spectrum census honestly relativizes single-char hits: integer characteristic polynomials are *common* in the declared pool (7894 entries) — the anchor class is the most frequent nontrivial family (286), non-compiler spectra exist ($\{1, 1, 3\}$: 254) — the compiler's distinction is the *joint* algebra structure (v600), not individual spectra. Two canonical period resources named for future RP/Hodge rounds. Machine-checked: [verification/v610_curve_landscape.py]. **[E]** (exact characters/point counts/census; CM reading typed; no marker moves)

**The periods round (QGEO.PERIODS.01).** The analytic layer of the cover is *classical* (6 checks, ∼1 s): the five basic period integrals $I_{k,j} = \int_0^1 x^k (1 - x^4)^{-j/3} dx$ equal $\frac{1}{4} B(\frac{k+1}{4}, 1 - \frac{j}{3})$ *exactly* (symbolic), with all Gamma denominators on the $\zeta_{12}$ grid $(\Gamma(11/12), \Gamma(7/12), \Gamma(5/12), \Gamma(1/12))$ — the $12 = |\mu_4| \cdot |\mu_3|$ structure appears in the analytic layer; the $\mu_4$ rotation acts on actual periods with the exact characters $i^{k+1}$ (40-digit certificates); the quotient curve's real period is the classical $\omega$-CM value $\Omega_E = B(1/3, 1/6)/2 = \Gamma(1/6)\Gamma(1/3)/(2\sqrt{\pi})$ exactly; and the cross-factor period ratio admits *no* low-degree integer relation (PSLQ null) — the two CM factors are period-independent, exactly as the v610 splitting demands. The period dictionary (every cover period a Beta/Gamma monomial on the 12-grid) is the named analytic resource for the Hodge/RP comparisons. Machine-checked: [verification/v611_periods.py]. **[E]** (symbolic Beta identities; certificates; PSLQ-null consistency; no marker moves)

**The polarization frame (QGEO.POLFRAME.01).** The form-level bridge (9 checks, ∼2 s): the braid rotation has three *simple* eigenvalues $\zeta_{12} \cdot \{1, -1, -i\}$ ($\det r = -1$), and in its left-eigenvector

basis the polarization diagonalizes *exactly*, $(W^\dagger)^{-1} J W^{-1} = \mathrm{diag}((1+\sqrt{3})/2, (1-\sqrt{3})/2, -1)$ (honest typing: diagonality follows from J-unitarity with simple unimodular eigenvalues; the content is the exact values and the signature distribution). The centerpiece: the unique *positive* direction of $J$ — the $h^{1,0} = 1$ Hodge line of the $\omega$-sheet — is *exactly* the eigenspace of the primitive twelfth root $e^{i\pi/6} = \zeta_{12}$: the analytic Hodge structure and the finite rotation agree on where positivity lives. With v611 (periods = Beta values on the $\zeta_{12}$ grid) the analytic bridge is one step from closing; the canonical period normalization (Beta weights for the diagonal) is the named next step. Machine-checked: [verification/v612_polarization_frame.py]. **[E]** (exact spectrum/diagonalization/signature; bridge reading typed; no marker moves)

**The canonical period normalization (QGEO.PERNORM.01).** The analytic bridge *closes* (19 checks, ∼6 s; executes v612's named step). Geometry first: the quarter rotation $x \mapsto ix$ maps the four branch-point segments of $y^3 = x^4 - 1$ cyclically with *no* deck correction (all 12 wrap factors exactly 1; 40-digit certificates), the single $\omega$-sheet homology relation is $\mathrm{seg}_1 + \mathrm{seg}_2 + \mathrm{seg}_3 + \mathrm{seg}_4 = 0$, and a must-fail control documents that the *raw* period rows are *not* $r$-eigenvectors (the naive ansatz falsified, residual 0.42). The exact dictionary: $\mathrm{char}(r) = \mathrm{char}(\omega M)$ *exactly* for the induced companion rotation $M$ ($M^4 = 1$) — the reduced Burau rotation at $t = \omega$ is $\mathrm{GL}_3$-conjugate (explicit conjugator $G$) to the geometric quarter rotation composed with *exactly one* deck step, explaining $r^4 = \omega$ structurally; the deck power is unique among the four scalar twists with $(cM)^4 = \omega$. The period rows (hol $dx/y^2$, hol $x\,dx/y^2$, antihol $\overline{dx/y}$) are $M$-eigenvectors with characters $(i, -1, -i)$ and transport through $G$ to $r$-eigenvectors with $\omega \cdot (i, -1, -i) = \mathrm{spec}(r)$; character match: antihol $\leftrightarrow \zeta_{12}$. The Hodge weights are explicit Gamma monomials (complex Beta formula, verified to $10^{-25}$): $h_1 = (3\sqrt{3}/16\pi)\,\Gamma(1/3)^2\Gamma(1/6)^2$ and the two $\Gamma$-quotient companions; $\omega$-sheet signature $(2, 1)$ with the unique *negative* on the antiholomorphic line. The sign bridge: all three character-matched sign products are negative — $J = -h$ *per character*: the compiler polarization is minus the analytic Hodge form up to per-line positive scales, and $J$'s unique positive line ($\zeta_{12}$, v612) is $h$'s unique negative line — Riemann–Hodge positivity realized. Residue named: the canonical fork-basis identification (basepoint bookkeeping for $G$). Machine-checked: [verification/v613_canonical_periods.py]. **[E]** (exact dictionary/uniqueness; certificates; Gamma monomials; sign bridge; no marker moves)

**The Gauss–Manin transport (QGEO.TRANSPORT.01).** The deck step is the boundary monodromy (12 checks, ∼7 s; closes v613's residue F1 at the transport level). Transporting the segment cycles along the rigid quarter-rotation loop of the four branch points ($p_k(\tau) = e^{i\pi\tau/2} p_k$, the rotation braid), the discriminant $\lambda(\tau) = e^{2\pi i\tau}$ winds *once* around 0, and the transported periods factorize *exactly*: $I_m(\tau) = e^{i\pi\tau(m+1)/2}\, e^{-2\pi i\tau j/3}\, I_m(0)$ (rotation substitution identity, symbolic). At $\tau = 1$ the deck factor is $\omega = t^4$ on the $t = \omega$ sheet, *uniform* across all three rows (the two holomorphic forms and the conjugated antiholomorphic row), and $\bar{\omega} = \bar{\omega}^4$ on the conjugate sheet: the deck step is $t^4$ — the *boundary monodromy* of the local system — on every sheet; must-fail controls confirm $t^3 = 1 \neq \omega$ and $t^5 \neq \omega$ (the exponent $4 = |\mu_4|$ is forced by the puncture count). Consequence: the canonical transport matrix in the segment basis is *exactly* $\omega M$ — no conjugator freedom left: the v613 dictionary $r \sim \omega M$ is *canonical*, and $(\omega M)^4 = \omega \cdot 1$ re-derives v597's $r^4 = \omega$. Certificates: direct branch-tracked integration at $\tau = 1/3, 1/2, 2/3, 1$ (dps 80) matches the closed factorization to $5 \cdot 10^{-28}$, and the transported segment at $\tau = 1$ equals $\omega\times$ the static neighbor to $2.7 \cdot 10^{-81}$. The reading: the braid fixes the boundary, the rigid rotation does not — their homological difference is one boundary loop of local-system monodromy; the rotation-braid convention is the typed literature identification, carried by the exact char identity. Machine-checked: [verification/v614_transport_deck.py]. **[E]** (exact factorization/deck identities; transport certificates; no marker moves)

**The interacting gamma slice at toy level (WOIT.GAMMA.TOY.01).** The gamma battery executed *on* the RP-surviving interaction class (11 checks, ∼2 s): the alignment bit $\delta = \pi/2$ with positive coupling — the unique member of the equivariant FK mark family that keeps reflection positivity (v534) — now carries the chirality data at $g > 0$. The two parents (chiral $I + iC$, mirror $I - iC$,

both gap 1) are identified; the chiral interacting Grams stay PD $(37, 0, 0)$ at the forced twist $\eta = +i$ on all four cuts for $g \in \{1/32, 1/4, 1, 8\}$ (v534 regression); and the centerpiece: at the *same* twist the mirror state splits sector-exactly on *every* cut and *every* coupling — even $(29, 0, 0)$ PD, odd $(0, 8, 0)$ *strictly negative definite* with max odd eigenvalue $\leq -1.5 \cdot 10^{-3}$, bounded away from zero over the whole grid. Every mirror fermion vector has strictly negative OS norm at the interacting level: in the doubled vector-like system any OS-positive subspace meets the mirror odd sector in $\{0\}$ — kill test (3) cannot fire on the surviving interaction class (v608 upgraded from $g = 0$ to the full grid). The mark transport exists interacting: $\alpha_4$ maps the mark-A quartet algebra exactly into the mark-B sites (16/16), and both the mark Gram and the transport form are Hermitian PD at every $g$ — kill test (7) does not fire at the interacting toy level. The odd transfer spectrum stays multiplicity-free at every $g$ (honest finding: a strong-coupling near-doubling at $g = 8$, gaps $\sim 2 \cdot 10^{-5}$, without closing; the $g = 0$ top 1.641438 is numerically the v524 compressed-clock top — the quarter-turn transfer datum recurs across $N$). Dead-member control: the RP-dead member ($m$=2, $s$=+) is indefinite on its straddled cut — off the survivor the gamma battery has no RP home: the dynamical bit selection and the chirality data *co-locate* on the same member. Fence: one toy, one interaction class; kill tests (3)/(6)/(7) stay formally live on $\mathcal{A}_{\text{hol}}$; kill test (6) untouched. Machine-checked: [`verification/v615_gamma_toy_interacting.py`]. **[E]**/**[C]** (float certificates, declared tolerances; toy fence; no marker moves)

**The internal $SU(2)$, kinematic separation (`WOIT.SU2.KIN.01`).** The first executed slice of gamma kill test (6) (17 checks, <1 s). On $\mathbb{C}^4 = \mathbb{H}^2$ the right multiplications close the (opposite) quaternion algebra exactly, and Woit's Euclidean structure *is* an internal direction: $\rho_{\text{tw}} =$ right-$j$ verbatim in the v519/v565 conventions. The internal algebra right-sp(1) preserves *every* fiber of the twistor fibration $\mathbb{PT} \to S^4$ (trivial on spacetime; the left multiplications move a generic fiber), commutes *exactly* with the full spacetime quaternion action, and the two algebras intersect in $\{0\}$: the internal factor *separates* from the Euclidean rotation group — kill test (6)'s kinematic branch does not fire. The clock factorizes exactly: RHO4 = left-$u \circ$ right-$u$ with $u = e^{i\pi/4}$ (neither factor alone; must-fail controls); the ALE deck is *purely* spacetime-side (DECK4 = left-$i$ exactly) and RHO4$^2$ = DECK4$\circ$ right-$i$. The complex choice breaks the internal $SU(2)$ to $U(1)$ kinematically, and the v519 mark torsor is exactly the $\mu_4$ subgroup of the internal $U(1)$ composed with the OS conjugation; $\sigma_{\text{std}}$ fixes right-$j$ and inverts the internal charge and both clock factors. The dynamical half of kill test (6) stays open and formally live on $\mathcal{A}_{\text{hol}}$. Machine-checked: [`verification/v616_su2_internal_kinematic.py`]. **[E]**/**[C]** (exact operator identities; kinematic fence; no marker moves)

**The seam forcing round (`QGEO.SEAMFORCE.01`).** A constructive slice of the bedrock premise `QGEO.SYM.01` (14 checks, <1 s): the conformal seam axioms *force* the $\mu_3$-cover. $\mathbb{Z}_4$-Möbius rigidity: any 4-point configuration on $\mathbb{P}^1$ cyclically permuted by an order-4 Möbius map is Möbius-equivalent to $\mu_4$ (the multiplier is forced to a primitive fourth root); the cross-ratio census of $\mu_4$ is the harmonic orbit $\{2, 1/2, -1\}$ exactly (must-fail: 4/3, the disc-7 jet datum, is not in the orbit). The weight census: among all 81 $\mu_3$-monodromy weight vectors exactly four are clock-equivariant, and the mark-equivalence demand kills the alternating pair — the weights are *forced* uniform, $j \in \{1, 2\}$, the conjugate sheet pair of v613. Uniform $j = 1$ gives $y^3 = (x-1)(x-i)(x+1)(x+i) = x^4 - 1$ *exactly*; the total finite monodromy forces full ramification at infinity (the v603 separated cusp); Riemann–Hurwitz gives genus 3 exactly. And the seam geometry matches: the marks lie on the unit circle, the doubling reflection fixes seam and marks, the OS-cut (bond) reflection $z \mapsto -i\bar{z}$ permutes the marks by exactly the $(k, 5-k)$ pattern of the v599 real structure and meets the circle at the two v519 cut points. The bedrock residue narrows to two named steps: the cyclic-3 choice ($N_{\text{fam}} = 3$, anchored in the $E_8$ glue) and the physical-seam $\leftrightarrow$ conformal-seam identification. GATE.QGEO does not move. Machine-checked: [`verification/v617_seam_cover_forcing.py`]. **[E]**/**[C]** (exact rigidity/census/genus identities; residue typing; no marker moves)

**The cyclic-$N$ census (`QGEO.NCENSUS.01`).** The first v617 bedrock residue made *conditional* (9 checks, <1 s). For uniform-weight cyclic covers $y^N = x^4 - 1$ over the forced $\mu_4$ base: full ramification

everywhere iff $\gcd(N,4)=1$, i.e. $N\in\{3,5\}$ among $N=2..6$; the Riemann–Hurwitz genus table is $1,3,3,6,7$; the primitive character sheets have the compiler's rank 3 exactly for $N\in\{3,5,6\}$. The Chevalley–Weil signatures decide: $N=3$ has *both* primitive sheets Lorentz ($(1,2)/(2,1)$ — the v599/v613 signature); $N=5$ has a Lorentz pair; $N=6$ has *no* primitive Lorentz sheet (its Lorentz content sits on the order-3 characters — $N=6$ reduces to $N=3$): the seam-admissible orders with a primitive Lorentz sheet are exactly $\{3,5\}$. The pinning: $\det_{\mathbb{Z}}(1-\mathrm{deck})=\Phi_N(1)^3=N^3$ for prime $N$ — 27 for $N=3$ (*exactly* the machine-checked v597 constant) versus 125 for $N=5$; the saturation analogue $81=3^4$ versus 625: among $\{3,5\}$ the compiler's integral constants select $N=3$ *uniquely*. The residue moves one level down: why the compiler is 3-adic is $N_{\mathrm{fam}}=3$, anchored in the $E_8$ glue ($240=16\times5\times3$). GATE.QGEO does not move. Machine-checked: [`verification/v620_cyclic_n_census.py`]. [E]/[C] (exact census/pinning; conditional typing; no marker moves)

**The interacting Klein–Landau semigroup (`WOIT.GAMMA.SEMI.01`).** The gamma net-dynamics slice (6 checks, ∼2 s). At $g=0$ the v524 Klein–Landau pattern reproduces exactly at $N=16$ on the admissible cut: all seven steps $\tau_k$ Hermitian (domain dims $29,22,16,11,7,4,2$), even steps PD, odd steps $k=1,3,5$ indefinite with *exactly zero* one-particle diagonal (the free chirality datum), $k=7$ parity-trivial. At $g>0$ (the RP survivor) exact Hermiticity survives *precisely* on the steps $\{4,6,7\}$ and is lost on $\{1,2,3,5\}$ — the semigroup *contracts* to the $\mu_4$-commensurate window; the clock step $k=4$ stays Hermitian and PD ($(11,0,0)$) over the whole grid $g\in\{1/32..8\}$: the $\mu_4$ clock is a positive transfer operator on the interacting net. The mechanism is exact: the survivor interaction is $\alpha_4$-invariant *exactly* and not $\alpha_1/\alpha_2$ invariant (translation invariance breaks to the quartet stabilizer $\{0,4,8,12\}$); $k=5$ shows quartet containment alone does not protect (even parity *and* alignment are needed). The reading: the reconstructed rotation group of the interacting toy is the *clock tower*, not the full circle. One toy, one interaction class; gamma proper stays open on $\mathcal{A}_{\mathrm{hol}}$. Machine-checked: [`verification/v621_interacting_semigroup.py`]. [E]/[C] (float certificates; exact combinatorics; toy fence; no marker moves)

**The seam identification (`QGEO.SEAMID.01`).** The second v617 bedrock residue closes at the kernel + dictionary level (10 checks, ∼1 s): *the physical seam is the conformal seam.* The kernel identity: the v519 chiral seam kernel $c(d)=(2/N)/\sin(\pi d/N)$ for odd $d$ and 0 for even $d$ is *exactly* the antiperiodic (NS) chiral mode sum on the 16-site circle, for all $d=1..15$ *including* the even-distance zeros — the flat-band structure *is* the NS spectrum (closed form $\sum_{j<8}\sin((2j{+}1)x)=\sin^2(8x)/\sin(x)$, exact); the Ramond (periodic) sum fails as a must-fail control — the NS structure is forced, not conventional. The site map $\theta_a=(2a{+}1)\pi/16$ places the four mark-bond midpoints *exactly* at $\mu_4$ and no site touches a mark (the v519 precision (ii), literal geometry); the group dictionary is exact ($\alpha_1$ = rotation by $2\pi/16$, the clock $\alpha_4=z\mapsto iz$, the RP reflections = diameter reflections about $(k{+}1)\pi/16$; the admissible cuts map to the axes $\{\pi/4,\pi/2,3\pi/4,\pi\}$). And the v534 straddle law becomes geometry: the through-mark axes are *exactly* the straddled cuts $\{7,15\}$, the between-mark axes the avoiding cuts $\{3,11\}$; v617's bond reflection is the cut $k=11$ carrying the v599 $(k,5{-}k)$ mark permutation. The remaining continuum residue is the *covering-level* identification (the $\mu_3$-cover of the seam double carrying the full interacting theory), named, not claimed. GATE.QGEO does not move. Machine-checked: [`verification/v622_seam_identification.py`]. [E]/[C] (exact mode-sum/dictionary identities; must-fail control; no marker moves)

**The covered seam (`QGEO.COVERSEAM.01`).** The lattice skeleton of the covering-level identification (9 checks, ∼4 s; the named v622 residue, first slice). Lifting the 16-site seam circle to the $\mu_3$-cover (sheet shift per mark-bond crossing), the covered seam is *one* 48-site NS circle for the uniform weights $j=1,2$ — and *disconnects* into three circles for the trivial and the alternating weights $(1,2,1,2)$: a new, independent kill of the alternating pair (seam connectivity; total monodromy $6\equiv0 \bmod 3$). The explicit walk closes after exactly 48 steps (three base turns, 12 mark crossings). The lifted clock: $L^4=\mathrm{deck}$ and $L^{12}=\mathrm{id}$ on *all* 48 points with proper order 12 — v597's $r^4=\omega$ and v614's boundary-monodromy transport are *lattice combinatorics* on the covered seam, and the $\zeta_{12}$ spectrum of the Burau rotation is the lifted clock's order. The v622 kernel identity generalizes

verbatim to $N = 48$ (all $d = 1..47$). The 12-grid: the NS modes split under the deck into three classes of 16 modes with offsets $\{1/6, 1/2, 5/6\}$; the offset-1/2 class equals the *base* NS frequencies exactly (the untwisted sector *is* the base seam, verbatim); the fermionic deck lift satisfies $\text{deck}^3 = -1$ (the NS full rotation — the spin double, resonating with the v510/v519 $(-1)^F$ dichotomy), and the *bilinear* deck eigenvalue of the twisted pair is exactly $\omega$ — the $t = \omega/\bar{\omega}$ sheets live on the $H^1$/bilinear level. The 12 lifted mark crossings form a single 12-cycle under the lifted clock. The remaining residue is the interacting/CFT-level orbifold statement (twist-field OPE, the full $\mathbb{Z}_3$ orbifold of the seam CFT), named, not claimed. GATE.QGEO does not move. Machine-checked: [verification/v623_covered_seam.py]. [E]/[C] (exact combinatorics and mode-sum identities; orbifold residue typed; no marker moves)

**The external-review lattice audit (EXTREV.LATTICE.01).** Three claimed derivations of the third external review (2026-08-02), machine-checked (14 checks, <1 s); one follow-up retyped honestly. *The four-mark closure* [[E], conditional]: generalizing the glue to $D_{d+1} \oplus A_{d-1}$, three *independent* selectors each force $d = 4$ — the disc-group census ($\text{disc}(A_{d-1}) = \mathbb{Z}_d$ cyclic; $\text{disc}(D_{d+1})$ cyclic iff $d$ even; $\mathbb{Z}_d = \mathbb{Z}_4$ forces $d = 4$; negative controls $d = 3, 5, 8$), the even-glue norm equation $(d{+}1)/4 + (d{-}1)/d = 2$ (unique positive solution 4; no other even target $2k$, $k \le 10$, admits an integer $d$), and the Pythagorean selector $(d{-}1)^2 + d^2 = (d{+}1)^2$ — the $(3, 4, 5)$ triple of (cycles, marks, carrier). With $(1, 1, 2)$ the unique 3-part partition of 4 and $c_3 = 1/(2e_1\pi) = 1/(8\pi)$: under the named architecture the two-input presentation reduces to *one* boundary degree $d = 4$ plus $\pi$; honest typing: conditional on the glue architecture — AX.P2.01 keeps its marker, and the residue moves to why the physical boundary realizes the minimal cyclically-glueable genus-0 divisor. *The anchor bytecode* [[E]]: $\chi_a = (t{-}1)^2(t{-}2)$ carries $(4, 5, 2)$; the power sums $p_n = 2 + 2^n$ obey $p_{n+2} = 3p_{n+1} - 2p_n$; Newton recovers the $e_k$; $p_1p_2p_3 = 240$, $p_4 - p_3 = 8$, $240 + 8 = 248$ (the $H = 1 + V(V{-}1)/2$ realization is v602-checked). No marker moves. Machine-checked: [verification/v624_external_lattice_audit.py]. [E]/[C] (exact closure/bytecode/congruence identities; honest retype; citations verified)

**The $E_8$ code round (E8.CODE.01).** "$E_8$ is an error-correcting code" is a *theorem*, not a metaphor (7 checks, ∼2 s). The extended Hamming code $[8, 4, 4]$ (16 codewords, weights $\{0{:}1, 4{:}14, 8{:}1\}$, self-dual) under Construction A yields an even (weights in $4\mathbb{Z}$) unimodular (covolume 1) rank-8 lattice — by uniqueness, $E_8$ — with 240 minimal vectors and 2160 on the second shell. Error correction is literal: every single-bit corruption of every codeword has a unique nearest codeword equal to the original ($16 \times 8$ exhaustive). The compiler ties are typed observations: $8$ = rank, $4$ = min distance = $d = |\mu_4|$ (the v624 boundary degree), $16 = |C|$ = the carrier half-spinor = the seam sites, $14 = 2 \times 7$ (the parabolic dimension; the v605 denominator). No physics claim. Round-22 fence: the *stronger* error-correction-hull reading — the 240 root operators plus the 16 neutral kernels as a twisted group-algebra basis of $\text{End}(S_+)$ — is SYNDROME-DEAD as preregistered ([verification/v805_e8_syndrome_algebra.py]); the Construction-A theorem here is untouched. Machine-checked: [verification/v626_e8_code.py]. [E]/[C] (exact code/lattice identities; typed ties)

**The orbifold Casimir round (QGEO.ORBCAS.01).** The first quantitative slice of the v623 orbifold residue (5 checks, ∼1 s). The chiral vacuum energy of the $\nu$-offset sector on $N$ sites has the exact closed form $E_0(\nu, N) = -\cos(\pi(2\nu{-}1)/2N)/(2\sin(\pi/2N))$, whose symbolic $1/N$ expansion gives the finite-size coefficient exactly $\pi(6\nu^2{-}6\nu{+}1)/12$ — the fermionic twist formula. For the covered seam's deck classes $\nu = (1/6, 1/2, 5/6)$ the sector table is $(1/72, -1/24, 1/72)$: the twisted pair is degenerate (conjugate sheets), the twisted/untwisted ratio is exactly $-1/3$, and the $\mathbb{Z}_3$-twist gap is exactly $1/18$ (in $\pi/N$ units) — the covered seam carries the $\mathbb{Z}_3$-orbifold Casimir data verbatim at the kinematic level. The interacting statement (twist-field OPE, crossing, RP) stays the named residue. Machine-checked: [verification/v628_orbifold_casimir.py]. [E] (exact closed forms and expansions)

**The root incidence census (E8.INCIDENCE.01).** The review's $48 \times 5 \to 240$ kill test, executed at the first hurdle (6 checks, ∼1 s). For the compiler-canonical order-12 element $g = J \circ \sigma$ (the $\mu_4$ quarter-turn times the family 3-cycle) the 240 roots decompose as $19 \times 12 + 3 \times 4$ — *not* the 20 free

$\zeta_{12}$ orbits a naive $48 \times 5$ incidence needs: the naive construction is *killed*; $48 \times 5 = 240$ stays an audit fingerprint. The positive residue is sharp: the $\mu_4$ clock alone acts *freely* with exactly 60 orbits — $60 = D_{\text{start}}$ appears as the free clock-orbit count on the roots — and the family 3-cycle fixes exactly 12 roots (the `v623` lifted-mark count), which form exactly 3 free clock orbits. Machine-checked: [`verification/v629_root_incidence.py`]. **[E]**/**[C]** (exact census; honest kill; typed observations)

**The $F_{\text{transfer}}$ $\mathrm{PGL}_2$ round (`FTR.PGL2.01`).** The `v533` preregistration executed in its *literal* group form (25 checks, ∼1 s). On the disc-$(-7)$ norm line $\alpha_t$ (norms 2/4/14/32; base action $T : \alpha \mapsto \alpha + 2$, parabolic) the declared `v183` Koide reading induces the parabolic step $+9 = N_{\text{fam}}^2$ and is *exactly* intertwined with $T$ (the reading map $\Phi$ is itself Möbius, $\Phi \circ T = g \circ \Phi$); the cross-ratio is 4/3 exactly and Möbius transport forces the $F$ corner 53 fit-free. On the native continuous half the time-1 maps are explicit $\mathrm{PGL}_2$ elements: $g_{\text{pole}}$ hyperbolic with multiplier $(2/3)^6$, $g_{\text{Boltzmann}} = g_{\text{pole}}$ *identically* (the `v425` single-flow theorem), $g_{\text{QCD}}$ parabolic and *explicitly* conjugate to $T$ (translation $+7/2\pi$ in the $1/\alpha$ coordinate), $g_{\text{relic}} = \mathrm{id}$ (degenerate); the `v575` uniflow degeneration and the `v578` own-clock equivalence reproduce. The kill criterion "four incompatible arithmetic actions" is *not* met; five must-fail controls have teeth (the raw determinant reading transports to $-16 \neq 32$). The residual — the external anchor clocks/jets — stays fenced by `CONTRACT.F.01`. Machine-checked: [`verification/v632_ftransfer_pgl2.py`]. **[E]**/**[C]** (exact $\mathrm{PGL}_2$/intertwining identities; typed reading)

**The $\mathbb{Z}[i]$-$E_8$ quotient round (`E8.ZIQUOTIENT.01`).** The `v629` positive residue, one level deeper (17 checks, ∼3 s). The $\mu_4$-quotient of the 240 roots (the 60 free clock orbits) *is* the hermitian-unimodular $\mathbb{Z}[i]$-$E_8$: the hermitian form $H(x, y) = \langle x, y\rangle + i\langle x, Jy\rangle$ has $H(v, v) = 2$ on all 60 line representatives, the $\mathbb{Z}[i]$-span is the full $E_8$ lattice, and a hermitian-*unimodular* $\mathbb{Z}[i]$-basis exists among the reps ($\det \mathrm{Gram} = 1$). $\sigma$ acts on the 60 lines with census $19 \times 3 + 3$ fixed (the fixed lines are exactly the $J$-orbits of the 12 $\sigma$-fixed roots), all 60 order-2 unitary reflections preserve the root set, and the line group has order $11520 = 46080/4$ — the ST31 fingerprint ($60 = 7{+}11{+}19{+}23$ hyperplanes; ST32 killed by its 40-hyperplane count). The $60 \to 6$ cascade route *on the quotient* is killed: the hermitian Gram has rank 4, both $|H|^2$ graphs are regular (any degree rule is ill-defined), and pair elimination has no canonical maximum — a typed negative. Machine-checked: [`verification/v633_orbit60_quotient.py`]. **[E]** (exact quotient/lattice identities; typed cascade kill)

**The ST31 structure round (`E8.ST31.01`).** Does the ST31 find carry structural weight? Yes — and the numerology does not (56 checks, ∼10 s). $G_{31} = \langle$the 60 reflections$\rangle$ has order 46080 and *is* the full unitary stabilizer of the $\mu_4$-quotient structure (exact backtracking count); the degrees $(8, 12, 20, 24)$ are *unique* via Molien among all divisor quadruples with product 46080; Springer regular elements of orders 8/12/20/24 exist with centralizers 192/288/20/24. The compiler data embeds canonically: $J$, $\sigma$, $c = J \circ \sigma$ all lie in $G_{31}$ with the exact clock identities $\sigma = c^4$ and $J = c^9$ (the $\mu_4$ center is a *power* of the clock); the root stabilizer is $G(4, 2, 3)$ (Steinberg), the line stabilizer $Z_4 \times G(4, 2, 3)$. The computed structure is $G_{31} = (Z_4 \circ 2^{1+4}).\mathrm{Sp}_4(2)$. The numerology kill: $|G_{31}| = |W(D_5)| \times |W(A_3)|$ carries *no* structure — not an isomorphism (centers 4 vs 1), not a direct or semidirect product (all small normal closures join to order 64), and $W(D_5)$ does not even *embed* (minimal faithful dimension $5 > 4$); $W(A_3) = S_4$ does embed. Honest sharpening: the `v629` compiler clock $c$ is *not* the Springer-regular 12-element (census $19 \times 12 + 3 \times 4$, $|C(c)| = 72 \neq 288$) — the free $20 = 240/12$ comb belongs to the regular 12-clock of $G_{31}$, a different conjugacy class. The order identities are additionally machine-proved in Lean (`G31Orders.lean`, `lake build` green). Machine-checked: [`verification/v634_st31_structure.py`]. **[E]** (exact group/Molien/Springer census; numerology kill; honest clock typing)

**The code-semantics rounds (`E8.CODESEM.01`).** The `v626` code reading sharpens to an exact dictionary — and the naive placement dies (27 checks, ∼3 s; stage-1 kill probe 13/13). Exactly 2 of the 30 $S_8$-placements of the extended Hamming code carry both compiler symmetries mod 2 (the $\mu_4$ in-pair swap and the family pair 3-cycle), exchanged by the anchor transposition; the naive `v626` placement is *not* among them. The unique equivariant placement $C^*$ *is* Reed–Muller $\mathrm{RM}(1, 3)$ on $\mathrm{AG}(3, 2)$ (the 14 weight-4 supports are the 14 planes; $30 = |S_8|/|\mathrm{AGL}(3, 2)|$). The dictionary: no equivariant 4+4 block split exists; the unique minimal orbit of information sets is the transversal

pair — one information bit per $\mu_4$ pair (3 family bits + 1 anchor bit); the syndrome space factors as the flag $0 < \langle q \rangle < \mathbb{F}_2^4$ ($q$ the universal in-pair bit) with $\sigma$ acting on the pair directions as the 3-cycle + fixed anchor ($P_3 = P_0{+}P_1{+}P_2$); and decode = projection *verbatim* on the whole single-error class (3840/3840, exact arithmetic; all 3840 perturbed vectors sit at squared distance exactly 1 with the mod-2-collapsing tie pair). The 12 $\sigma$-fixed roots reduce mod 2 to exactly three classes of 4, and the clock-orbit fibration reproduces $60 = 7 \times 4 + 4 \times 8$. Kill control: the sets {universal in-pair syndrome}, {clean orbit structure}, {both-invariant} *coincide* — the equivariance is load-bearing. Machine-checked: [verification/v638_code_semantics.py]. **[E]** (exact dictionary; kill control)

**The twist-field/OPE round (QGEO.TWISTOPE.01).** The interacting slice of the $\mathbb{Z}_3$ orbifold residue (v623/v628), executed at the abelian vertex level (25 checks, ∼1 s). The twist weight $h_\sigma = 1/36$ follows from three independent exact routes (the v628 Casimir coefficient read as $2(h_\nu - c/24)$; the $\zeta$-function mode sum; the bosonization pair $h(\frac{1}{2}{-}\eta) + h(\frac{1}{2}{+}\eta) = \eta^2/2$), with the convention traps resolved exactly. On the lattice (U(1) string determinants on the half-filled NS seam double, whose equal-time covariance is half the v519 seam kernel): the two-point exponent $2\Delta = 2/9$ verifies at 0.06% (Ising control at 0.02%); crossing is *exactly* symmetry-protected (the gap rotation maps $w \mapsto 1{-}w$ while $|G_4|$ is invariant at machine precision); the closed four-point form $A^2[w(1{-}w)]^{-2\Delta}$ holds at $< 0.1\%$ after $N^{-2/3}$ extrapolation; the leading OPE constant $c_1 = 2\Delta = 2/9$ is model-bound at 0.19% (the model-free extraction stays a typed ±15% band). Reflection positivity, differentiated: the real $\mathbb{Z}_2$ class is exactly PSD at every $N$ (machinery validated), while the naive antiunitary $\mathbb{Z}_3$ pairing fails positivity by an $N$-stable amount — the positivity-carrying pairing (parafermionic Klein twist) is a named open item. GATE.QGEO does not move. Machine-checked: [verification/v639_twist_ope.py]. **[E]** (exact weights/crossing) + (MEASURED) (two-/four-point, OPE) + **[O]** (Klein-twist pairing)

**The parafermionic Klein twist (QGEO.KLEINRP.01).** The named RP open item of v639 is *resolved* at the lattice OS level (17 checks, ∼30 s). The positivity-carrying pairing composes the mirror with *charge conjugation*: $\Theta_K(\sigma_k(x)) = \eta_k\, \sigma_{3-k}(r(x))$ with $\omega^{kq}$ zero-mode phases. The dressed-string reality identity (particle-hole $I - C = DCD$ at half filling) makes every Klein Gram entry exactly real ($1.4 \times 10^{-15}$); every charged Klein block is real symmetric *and* PSD (worst min/max $2.9 \times 10^{-12}$), for $k = 1, 2$, on both diameter axes, at $N = 48/96/192/384$ ($N$-stable, no drift); over the 36 pairs $(\eta_1, \eta_2) \in \{\pm 1, \pm\omega, \pm\omega^2\}^2$ *exactly one* satisfies consistency, Hermiticity and PSD: $\eta = (1, 1)$; no phase rescues the naive pairing (the fix is structural), which reproduces its $N$-stable $-4.5 \times 10^{-3}$ violation as the must-fail control; and the *full mixed* Gram over $\{1, \sigma_1, \sigma_2, \text{dipoles}\}$ with no sector projection is PSD at $N = 48/96$ (stable at 192) — the lattice Osterwalder–Schrader statement of the orbifold twist sector. Honest typing: continuum OS reconstruction, R/fermion-parity sectors, modular invariance and non-abelian content stay named, not claimed; GATE.QGEO does not move. Machine-checked: [verification/v645_klein_rp.py]. **[E]** (reality/uniqueness/must-fail) + (MEASURED) (PSD Grams) + **[C]** (typed continuum remainder)

**The RM$(1, 3)$ reverse reading (E8.RM13REV.01).** Are compiler numbers *code operations*? An honestly two-part answer (20 checks, ∼1 s, exact arithmetic). *The anchor identity:* the $\sigma$-fixed syndromes of the equivariant placement $C^*$ are exactly $\{0, s(e_6), s(e_7), q\}$ with coset-leader weights $(0, 1, 1, 2)$ and norm-square sum $6 = (1, 1, 2) \cdot (1, 1, 2)$; the anchor is the family sum in the quotient ($P_3 = P_0{+}P_1{+}P_2$); all four pair sectors carry $(1, 1, 2)$/norm 6, but *only* the anchor pair is $\sigma$-stable — the $\sigma$ action *selects* the anchor copy. *The honest split:* the $(1, 1, 2)$/norm-6 weights are *generic* for any $[8, 4, 4]$ placement (the kill control keeps them); equivariance-specific are only $q$-universality, the $\sigma$-fixing of the anchor sector, and anchor = family sum (all die for the naive placement). *The bytecode negative:* the 42-entry invariant battery lands the structural anchors exactly (240, 8, 248, 60, 12, 6, 4) — but the $p$-sequence $(4, 6, 10, 18)$ of $p_n = 2 + 2^n$ has *no* code reading (0/11 natural families; 27 and 81 no hit at all): the bingo is buried. The kill control fires selectively, including the honest correction that the $\sigma$-fixed-root *count* 12 is not equivariance-specific (only its mod-2 fibration is). No marker moves. Machine-checked: [verification/v646_rm13_reverse.py]. **[E]** (exact anchor identity; honest negatives; selective kill)

**The ST31 degree-24/20 round (`E8.ST31DEG.01`).** The regular clocks of $G_{31}$ against the compiler clock (25 checks, ∼4 s, exact arithmetic). *The regular* 24*-clock:* an explicit $w$ with free census {24:10} and $|C_{G_{31}}(w)| = 24$ (Springer); exact center relations $w^{12} = -1$ and $w^6 = J^3$ — the $\mu_4$ center is a *power* of the regular 24-clock, and $\langle w\rangle/\mu_4 = Z_6$ is exactly its free 6-clock on the 60 lines ({6:10}); the characteristic polynomial is $\chi_w = x^4 + ix^2 - 1$ *exactly* (a Gaussian-integral quartic factor of $\Phi_{24}$; all four eigenvalues primitive $\zeta_{24}$); the half-clock $w^2$ is the *regular* 12-element, not the compiler clock. *The clock kill (a parity theorem):* $c = w^2$ is impossible for *any* order-24 $w$, even in $S_{240}$ — the 12-cycles of a square come in pairs from 24-cycles (4-cycles in pairs from 8-cycles), and census$(c) = 19{\times}12 + 3{\times}4$ has *odd* counts; the full power-census scan confirms it. *The* 20*-degree:* a regular $u$ with census {20:12}, $u^5 = J^3$ *central* (stronger than $J$-class), and $u^4$ acting *freely* with census {5:48}, 48 = 240/5 exactly (lines {5:12}). Honest typing: 48 = seam sites stays an *observation* (no structural bridge here); $g_{\mathrm{car}} = 5$ has only the census side; no marker moves. Machine-checked: [`verification/v647_st31_degree24.py`]. [E] (structural clock relations; parity-theorem kill) + [C] (typed observations)

**The discipline audit — the suite measures itself (`META.DISCIPLINE.01`).** The suspicion to be defused is "numerology": a large corpus of machine-checked claims could in principle be a confirmation-bias generator. This meta-module parses the suite's own files and freezes the measured process-discipline metrics as minimum bars (18 checks, ∼200 s). *Ledger census:* 715 rows, 710 active; 214 rows document kills, honest negatives, ill-posed retirements and retypes — a project that documents its own buried hypotheses is not a confirmation-bias generator. *Must-fail census:* 643 modules (every one carries checks), 6040 static `check()` call sites, 1469 must-fail / kill / scramble keyword occurrences across 233 modules; typing marks [E] × 3579 vs [C] × 1323 — conditional readings are typed, not upgraded. *Preregistration:* 47 contract rows, 29 naming kill criteria *before* execution; executed chains verified in the ledger (`FTR.PGL2.01`: preregistered `v533` → executed `v632`; `REG.FREEZE.01`: registry frozen 2026-06-09). *Reproducibility:* deterministic replay 12/12 on a declared sample spanning `v55`–`v648` (two fresh subprocess runs each, identical output up to timing lines); 396 modules sympy-exact, 77 with explicit seeds. *External anchors:* five classical results recomputed here against the literature — Jacobi 1834 four-square counts (exact, $n = 1..40$), Construction A $[8, 4, 4] \to E_8$ with 240 minimal vectors, the Shephard–Todd G31 degrees $(8, 12, 20, 24)$, *Lean layer:* 54 committed proof modules (`git ls-files`; `lake build` honestly *not* re-run in-suite). *The limit, honest:* this module measures *process* discipline, not the truth of the physics; typed [C] observations stay observations, and the open gates stay open (278 class-O rows incl. `SEAM.THEOREM.01`) — a discipline certificate is a necessary, not a sufficient, condition against numerology. Machine-checked: [`verification/v649_discipline_audit.py`]. [E] (censuses, replay, anchor recomputes) + [C] (the typed limit)

**The orbifold modular data (`QGEO.ORBMOD.01`).** The modular level of the $\mathbb{Z}_3$ orbifold of the seam, on the lattice (17 checks, ∼25 s). All nine twisted-sector partition functions $Z[g, h]$ of the seam double are real positive with exact conjugation degeneracies and match the continuum $|\theta[g/3, h/3]/\eta|^2$ at $< 1\%$ on $\ln Z$ ($N = 48$; three $\tau$); the Casimir slopes reproduce the `v628` coefficients per deck class. *S is measured:* the modular S-covariance converges with a clean common $N^{-4}$ rate (4.06–4.32 over $N = 24/48/96$) while the single-lattice deviation from the continuum falls only at $N^{-2}$ — the leading lattice artifact is itself S-covariant and *cancels* in the S-relation. *T is exact:* the discrete Dehn twist realizes T as the $\mathbb{Z}_6$ sector map $(g, h) \to (g, g{+}h) \times (-1)^F$ at machine precision ($3.6 \times 10^{-13}$; continuum T-phases are not resolvable in the real-positive trace convention — honest scope). The twisted characters carry $h_\sigma = 1/36$ (rel $5.6 \times 10^{-4}$) with the parafermionic 1/6 gap, and $\mathrm{deck}^3 = (-1)^F$ closes exactly as $\mathbb{Z}_6$ trace arithmetic including the cover factorization; chiral phases, continuum T-phases and continuum OS stay typed open; `GATE.QGEO` does not move. Machine-checked: [`verification/v650_orbifold_modular.py`]. [E] (sectors, characters, spin structure) + (MEASURED) (S-covariance) + [C] (typed continuum remainder)

**The orbifold assembly (`QGEO.ORBASM.01`).** The projection sum, settled by machine cohomology and a candidate census (17 checks, ∼35 s). $H^2(\mathbb{Z}_3, U(1)) = 0$ by enumeration (27 cocycles, all

symmetric): there is *no* discrete torsion for a pure $\mathbb{Z}_3$ orbifold. Of six assembly candidates, the survivors of {integer spectrum, unique vacuum, S measured clean, T exact} are exactly $\{B, C3\}$: the naive $\mathbb{Z}_3$ dies by an *exact* T violation $3.5 \times 10^{-3}$ (its charge-3 content — T kills the naive $\mathbb{Z}_3$ exactly), $C1$ by S stalling, $C2$ has no vacuum at all, $A'$ additionally by the wrong first twisted exponent (5/18, not 1/9). The seam data then discriminate: $B$ keeps every measured $c = 0$ sector ground and matches the Klein phase $\eta = (1, 1)$ of `v645`, while $C3$ (the Arf/GSO twin) excises all odd-sector grounds — *the seam assembly is* $B$, the $\mathbb{Z}_6$ deck quotient with uniform charge-0 projection (data-backed at this round; hardened structurally by `v652`). The chiral phase $e^{-2\pi i \tilde{a} b}$ is measured. `GATE.QGEO` does not move. Machine-checked: [`verification/v651_orbifold_assembly.py`]. [E] (cohomology, census, exact T) + [C] (data-backed B-vs-C3 pick)

**The Arf structure of the assembly (`QGEO.ORBARF.01`).** The B-vs-C3 kill, decided structurally (15 checks, ∼30 s). The $\mu_6$ defect ladder measures the C3-excised states *directly*: ladder exponents $2\Delta = (1/18, 2/9, 1/2)$ at rel devs $< 8 \times 10^{-4}$, the half-twist Klein Gram exactly PSD, and the exact charge-resolved minima $(0, 1, 4, 9, 4, 1)/36$ are the measured sector grounds; $C3$'s odd-sector predictions 85/18 and 81/18 exceed the measurements by factors 85.0 and 9.0 (the defect string is exactly charge-neutral: $C3$ excises states that exist on the lattice seam with positive Klein norm). The classification pin: among *all* $6^6 = 46656$ sector-character wirings exactly $\{B, C3\}$ survive the exact lattice degeneracies, and the single-Fock-space requirement forces $B$. Honest: the earlier `v639` 2/9 read alone is B-vs-C3-*blind* (both coincide exactly on the even sectors); the new $\tau$ datum is the OPE exponent 1/18. `GATE.QGEO` does not move. Machine-checked: [`verification/v652_orbifold_arf.py`]. [E] (classification, pin, excision data) + [C] (typed premise remainder)

**The bond-defect theorem (`QGEO.BONDDEF.01`).** The last orbifold premise upgraded to a lattice theorem (18 checks, ∼40 s). *The theorem:* the deck-character blocks of the covered seam *are* the one-bond-defect chains, unitarily and exactly ($A_r = \sqrt{3}\, RP_r$ on all modes, exact in $\mathbb{Q}(\zeta_6)$, at 16/48 and 48/144; defect phase = deck character): twist sector = deck boundary condition = bond defect, canonically; $D_{\text{can}} = L^4$ and $D^3_{\text{can}} = -1$ hold as exact operator identities (the quarter clock fixes the canonical section lift). *Gauge and pin:* the defect position is pure gauge (a symbolic telescoping theorem in all 16 free bond phases), and the geometry pins the gauge class — one uniform twist unit per mark (cocycle support = the `v622` mark set), window boundary at a crossing ⇔ defect at a mark. *The exclusion census:* the base chain has no other $\mathbb{Z}_3$; site potentials, amplitude defects, sign defects and alternating weights are killed exactly; an isospectral real NN chain *exists* ($7.8 \times 10^{-16}$) — honestly documented: purely spectral data cannot exclude it, but it is no $\mathbb{Z}_3$-twist realization and the measured string data discriminate it. The continuum CFT statement and the seam identification stay typed; `GATE.QGEO` does not move. Machine-checked: [`verification/v653_bond_defect.py`]. [E] (lattice theorem, gauge pin, exclusion census) + [C] (typed continuum remainder)

**The ST31 degree-8 round and the $d/4$ theorem (`E8.ST31D8.01`).** The last clock census, unified (30 checks, ∼10 s, exact arithmetic). *The $d/4$ theorem:* for every degree $d \in \{8, 12, 20, 24\}$ the Springer-regular $d$-clocks of $G_{31}$ satisfy $x^{d/4} = \pm i\,\mathrm{Id}$ — a *generator* of $\mu_4$, never $\pm 1$ (one formula from the exponent systematics: all exponents of $G_{31}$ are 3 mod 4). The converse is exact at $d \in \{8, 20, 24\}$, and the unique exception at $d = 12$ is the *compiler-clock class*: exactly 1280 extra elements = the classes of $c$ and its Galois twin, with $\chi_c = (x - i)^2(x^2 + ix - 1)$ half the regular-12 Springer block — all eigenvalues satisfy $\lambda^3 = -i$, which is exactly why $c^3 = J^3$ is central; the compiler-clock classes are the *only* non-regular clocks in all of $G_{31}$ whose $d/4$-th power generates the center. The order-8 landscape is complete (10560 elements, 480 regular = 2 Galois classes, $\chi = (x^2 + i)^2$ exact) with the honest kill that "free ⇒ regular" is *false* at $d = 8$ (and already at order 4); degrees = free orders divisible by 4 above 4. No marker moves. Machine-checked: [`verification/v654_st31_degree8.py`]. [E] (census, theorem, controls; honest kill)

**The look-elsewhere audit (`META.LOOKELSEWHERE.01`).** The bingo budget of the typed [C] numeric coincidences, quantified (14 checks, ∼2 s; seed 20260802, 100k MC, exact Poisson-binomial tails). The budget: $N = 42$ structure slots, $K = 19$ hits against the compiler target list *frozen*

*before the computation*; the six killed readings enter as 0-hit trials. Global: the family-corrected $p = 5.1 \times 10^{-3}$ in the most conservative of three null models — the *ensemble* survives the honest correction (empirical control: 0.2%/0.3% of 2000 size-matched pseudo-target lists reach the real hit count / the real $p$). The triage, and the point: *no single observation is significant on its own* — suggestive only (0.01..0.1): $60 = D_{\text{start}}$, the 12 $\sigma$-fixed roots, the G31 degrees, $48 = \Omega_{\text{adm}}$; unremarkable ($p > 0.1$): $6$ = Lorentz-congruence index $= p_2(a)$ ($p = 0.169$) and the CODESEM number battery ($p = 0.593$ — the `v646` finding quantified); forced: 14 and 16 (their value is the theorem, not the number). The re-typings are executed as dated ledger notes (`EXTREV.LATTICE.01`, `E8.CODESEM.01`, `E8.CODE.01`); the substance of the rounds lies in the [E] structure theorems, not in the number bingos. Machine-checked: [`verification/v664_look_elsewhere.py`]. [E] (budget census, MC quantification, executed re-types)

**The continuum-OS slice of the seam orbifold (`QGEO.ORBOS.01`).** The Osterwalder–Schrader reconstruction data of the seam orbifold B, measured at the *convergence* level (19 checks, ∼3 s; verdict ORB-OS-CONTINUUM-SLICE). *Convergence with understood rates:* at fixed continuum data over $N = 48..384$ all six $\sigma$-channel observables share the uniform raw rate 0.659 (spread 0.020) = the predicted Fisher–Hartwig branch exponent $\rho_\sigma = 2/3$; the $\tau$-channel contrast measures 1.298 vs $\rho_\tau = 4/3$ (the channel law $\rho = 2(1-\beta)^2 - 2\beta^2$: rates understood, not fitted); after removing the one known FH term the extrapolated limits hit the CFT values at $< 0.05\%$; the $\varepsilon$/current channel is exact on the lattice ($8.9 \times 10^{-15}$). *RP survives the limit:* the correlation-normalized Klein Grams are PSD at every $N$ and the resolvable spectra converge with $\lambda_{\min}$ extrapolations bounded away from zero — $\sigma$ $1.357 \times 10^{-3}$ (margin 7× the band), $\tau$ $2.800 \times 10^{-4}$ (margin 358×), mixed OS Gram $1.337 \times 10^{-3}$ (margin 31×); must-fail: $\eta = -1$ flips negative definite. *Cluster in both directions:* space — the free-exponent chord fit at $N = 1536$ gives $2\Delta = 0.222220$ vs $2/9$ at $8.1 \times 10^{-6}$ (the chord form beats the plain power law by five orders); Euclidean time (new machinery) — the connected correlator decays with the fitted gap = the exact transfer-matrix $\varepsilon$ level (0.99933 vs 1), the cosh/cylinder form pointwise. *Characters:* 1/36, 1/9, 5/18, 4/9 from exact mode sums at $10^{-12}$ with the measured $N^{-2}$ rate. *The honest typing:* OS axioms E0/E2/E3/E4 are now measured at the convergence level, E1 is exact only for the discrete lattice symmetries (rotation invariance emergent); the formal-limit remainder is *named*: uniform bounds over all configurations, tightness, the GNS limit Hilbert space, operator convergence, the R/parity sectors, the assembled orbifold-B correlators — `GATE.QGEO` does *not* move. Machine-checked: [`verification/v679_orbifold_continuum_os.py`]. [E] (machinery; convergence, cluster and character slices) + (MEASURED) (RP-limit extrapolations) + [C] (formal-limit remainder, named)

**The Gaussian code bridge (`E8.GAUSSCODE.01`).** $E_8(\mathbb{Z}[i])/(1+i)$ *is* the information space of RM(1, 3) (26 checks, ∼3 s, exact arithmetic; verdict GAUSSIAN-CODE-BRIDGE-EXACT). The $\mu_4$ complex structure $J$ makes E8 a rank-4 $\mathbb{Z}[i]$-lattice; the $(1+i)$-quotient is $\mathbb{F}_2^4$ (SNF exact; standard model: $|\det| = 256 = 2^8$, SNF $[1^4, 2^4]$ — presentation-independent); the 240 roots fall $15 \times 16$ over the nontrivial classes (zero class provably empty; each class = 4 of the 60 G31 lines); $\sigma = c^4$ acts as the RM(1,3) family permutation (3-cycle + fixed anchor, the v638 info-bit action) with residual gauge of order 18; the coordinate block is the sum-of-families label. All must-fail controls fire (naive placement: $\sigma$-semantics undefined; non-$\pi J$ placement: no $\mathbb{Z}[i]$-module; $\mathbb{Z}[i]^4$: census trivializes). No marker move. Machine-checked: [`verification/v689_gaussian_code_bridge.py`]. [E] (SNF, census, semantics; exact arithmetic) + ([E], KILL) (three must-fail controls)

**The quartic half (`E8.QUARTICHALF.01`).** Is G31 the holomorphic $\mu_4$ shadow of the E8 compiler? *Alive, with the honest split* (22 checks, ∼1 s, no floats; verdict QUARTIC-HALF-ALIVE). *The vanishing half* $\{2, 14, 18, 30\}$ is proven symbolically but *trivial*: $W(J\alpha) = -i\,W(\alpha)$ makes every $J$-orbit contribution vanish for all $d \not\equiv 0 \bmod 4$ (holds for $6, 10, 22, 26$ too — honest typing). *The substance*: $F_8|, F_{12}|, F_{20}|, F_{24}|$ are G31-invariant (60/60 mirrors exact), algebraically independent (exact Jacobian), degrees Molien-unique with product $46080 = |G31|$ — by Chevalley (cited) a system of *basic* invariants: the holomorphic restriction of the $W(E_8)$ ring generates the *full* G31 invariant ring; the Jacobian carries the 60 mirror lines (60/60 point-verified; full factorization =

cited theorem, named residual). Controls: the non-root-preserving $J'$ breaks the informative half, the root-preserving $J''$ keeps it (the theorem is about the compiler-compatible conjugacy class). No marker move. Machine-checked: [verification/v690_quartic_half.py]. (PROOF) ($\mu_4$-orbit vanishing, symbolic) + [E] (G31 invariance/independence/mirror census) + ([E], KILL) (complex-structure controls) + (CITED) (Chevalley)

**The orbit-60 census, parked (E8.ORBIT60.01).** The 60 $\mu_4$-lines as a combinatorial object and a strict park test for the cascade $60 \to 6$ (18 checks, ∼1 s; verdict ORBIT60-PARKED). All five routes adjudicated: degree rules dead (quotient graphs regular); $\sigma$ structure dead (orbit sizes $\{3, 1\}$); the involution witness *does not exist* (full census of the line group, order 11520: 375 involutions, 21 classes, none of cycle type $2^{27}1^6$); W(E8) Coxeter does not descend; the must-fail numerology control fires (rule-free pair removal reproduces $60 - 2n$ in 27 steps; $> 10^{70}$ equally successful sequences — the counting carries no structure). $A_5$ one-liner: order-5 elements exist (2304) but nothing natural. Parked per the review protocol. Machine-checked: [verification/v700_orbit60_census.py]. ([E], PARK) (five-route adjudication, involution census, numerology must-fail)

**The CAR continuum route (QGEO.CARLIMIT.01).** The quasi-free CAR route to the formal orbifold continuum limit (22 checks, ∼10 s; verdict QGEO-CAR-RATES-SUMMABLE). All 6+1 deck-/twist channels are Schatten-norm Cauchy on $\varepsilon$-separated configurations ($N$ up to 3072/6144) with geometrically summable rates; the FH renormalization is fit-free lattice-derived (Barnes $G$, $8.8 \times 10^{-7}$); the RP cone is stable; all four kill criteria clear. The majorant lemma (L1)–(L3) is *named* with constants and a classical proof route — named, not proven: GATE.QGEO does *not* move. Deterministic. No marker move. [2026-08-05: the literal (L1) clause "one $B(\varepsilon)$ for *all* grid refinements" is false at $r_1 = 1$ and is repaired to the $(1 + \ln M)$ form — see the v778 block below; the (L1) rates themselves stand.] Machine-checked: [verification/v715_qgeo_car_continuum.py]. (MEASURED) (summable rates, stable RP cone) + [E] (lattice-derived renormalization) + [O] (majorant lemma)

**The ramified place sees NS/R (GNET.RAMIFIED.01).** Is the Ramond projection $(1+i)$-adic? (9 checks, ∼1 s; verdict RAMIFIED-SEES-NSR). On the explicit even-coordinate E8 with integral $J$: $(1+i)L$ lies in the even/NS sublattice (proved on a $\mathbb{Z}$-basis); the 240 roots fall into $15 \times 16$ classes of *pure* parity (7 NS + 8 R); parity is a homomorphism $\mathbb{F}_2^4 \to \mathbb{F}_2$ — the NS/R grading *is* the parity character of $E_8(\mathbb{Z}[i])/(1+i)$. *Controls fire*: $\mathbb{Z}[i]^4$ census not $15 \times 16$; scrambled labels break purity. First exact witness for the groupoid reading of $G_{\text{net}}$; gate typing unchanged. Machine-checked: [verification/v722_phys_ramified_ns_r.py]. [E] (integer-exact slice, controls) + [O] (Q-system identification)

**Strong thermal time is dead (FTRANSFER.STT.01).** Can thermal time replace the external clock contract? (11 checks, ∼1 s; verdict STT-KILLED + INTERNAL-PAIR-CONFIRMED). *The kill*: the F_QCD 1-loop clock map is *parabolic* and not diagonalizable (single Jordan block); the symbolic lemma "exp of a real-diagonalizable matrix is diagonalizable" closes the door. *The confirmation*: the pole multiplier is exactly $(2/3)^6 = 64/729$ (hyperbolic ⇒ internal; Boltzmann the same element per v425). *Typed consequence*: the external clock contract confirmed, sharpened to {one unit, $\theta_i$, $C_p$} — T3's clock wall = T1's unit wall, counted once. No marker move. Machine-checked: [verification/v723_phys_modular_clock.py]. [E] (symbolic lemma, multiplier) + [X] (STT kill)

**Conditioned modular flows are dead (FTRANSFER.COCYCLE.01).** The second thermal-time kill (12 mechanical checks, ∼1 s; verdict T3B-DEAD). Target-free graph projectors, *one* global KMS scale $\Delta = 6 \log(3/2)$, Connes cocycle composition exact ($1.6 \times 10^{-15}$) — and all four conditioned flows come out *loxodromic/complex*: the frozen v578/v632 solver classes do *not* arise; the NEEDS-LINDBLAD escape honestly refused (GKSL variant also all-loxodromic). With v723 both thermal-time routes are closed: the external clock contract stands. No marker move. Machine-checked: [verification/v724_phys_t3b_modular_flows.py]. [E] (cocycle composition, one KMS scale) + [X] (class-gate kill)

**The $v_{\text{geo}}$ torsor closure (VGEO.TORSOR.01).** The calibration/scale-torsor audit of the $v_{\text{geo}}$ interface (21 checks, ∼1 s; verdict TORSOR CLOSURE PASS). The interface is a well-defined $\mathbb{R}_+$ scale

torsor: complete export table $O_i = r_i\, v_{\text{geo}}^{d_i}$; dimension matrix rank 1 (conditional [A] on the flavor block, named); all consistency conditions $\lambda$-homogeneous (the machine form of No-Unit); leave-one-anchor-out compatible ($G$ vs $\rho_\Lambda$: 0.11%); calibration theorem stated. *Fence*: structure closure, *no* scale derivation (impossible per No-Unit). Moonshot *named*, not executed: $H_{\text{EW}} = \ln(\bar{M}_{\text{Pl}}/v_{\text{EW}}) = 37.1776$. Typing unchanged. Machine-checked: [verification/v725_phys_vgeo_torsor_audit.py]. **[E]** (torsor structure, homogeneity, cross-calibration) + **[O]** (the unit itself, per No-Unit)

**Pimsner–Popa / Watatani index 4 on the CAR ladder (GNET.PPINDEX.01).** The T2 main slice (57 checks, ∼7 s; verdict T2-SLICE-GO). $\mu_4$ *derived* from the clock $L = S^{N/12}$ (order 4 forced by the NS spin structure, not inserted). Optimal Pimsner–Popa constant *exactly* 1/4 (explicit rank-one minimizer + pinching; the homogeneous-element subtlety documented); Watatani index *exactly* 4 via an explicit quasi-basis with *forced* weights (uniform weights break scalarness; $\mathbb{Z}_2$-only average gives index 2). *Ramond*: undressed degenerates (index 2); the Klein zero-mode dressing restores index 4 *state-preservingly* ($[\rho, U'] = 0$ exactly). The $(1{+}i)$ dictionary closes with v722. Outerness witnesses typed as witnesses. Gate typing unchanged. Machine-checked: [verification/v726_phys_car_pp_index.py]. **[E]** (index results, derivation, healing) + (MEASURED) (outerness witnesses) + **[O]** (Q-system identification). *Lean round 5:* the exact algebraic index-4 core is formally verified (`WatataniIndexFour.lean`, kernel-checked, no sorry): the clock-order shadow ($S^{12} = -1 \Rightarrow H^4 = 1$), the explicit quasi-basis with $\sum vv^* = 4 \cdot 1$ exactly $= N(1{+}i)^2$, the sharp Pimsner–Popa constant $\lambda^* = 1/4$ (SOS certificate), and the $\mu_2$/three-sector controls; the CAR/Fock layer and the Ramond dressing stay numeric.

**The orbit packet lemma (E8.ORBITPACKET.01).** The Gaussian code bridge's number family in one normal form (26 checks, ∼1 s; verdict ORBIT-PACKET-EXACT). Burnside *exact*: the $\sigma$ census on the 15 nonzero classes gives $7 = (15 + 3 + 3)/3$. The canonical packet split: $240 = 5 \times 48$ — one fixed block + four moved blocks of 12 Gaussian lines each, pairwise disjoint, label-free canonical (the 15-class partition agrees with the intrinsic equivalence $x \sim y \iff (1 - J)(x - y)/2 \in L$); packets $= 1 + 4 = 5 = g_{\text{car}}$. Anchor: leader weights $(0, 1, 1, 2) \to (1, 1, 2)$, leader pair $e_6/e_7 =$ the v638 anchor pair. $q = 16$ normal form: $240 = 16 \cdot 15$, $248 = 240 + 8$. *Circularity fence:* P1/P2 come back as EXACT-INTERNAL-RECONSTRUCTION, not derivation; $\sigma$-class specificity: 160 free order-3 elements give Burnside 5. Companion Lean module `OrbitPacket.lean` (34 theorems, kernel `decide`/`norm_num` only). No marker move. Machine-checked: [verification/v736_orbit_packet.py]. **[E]** (integer/finite-group census; reconstruction typed)

**The seam-48 intertwiner: dead (E8.SEAM48.01).** The sharpest structural sharpening of the review morning, killed on its preregistered criterion (24 machinery checks, ∼4 s; verdict SEAM48-DEAD). The $48 \times 5$ incidence matrix between the five canonical 48-blocks and the 48 regular five-cycles is *not* identically 1: entry distribution 0:96/1:64/2:64/3:16, representative-independent (0 of 2304 order-5 elements of G31), must-fail control fires (0 of 25 random operators). Valuable residues: the four moved blocks carry the v623 character table *exactly*, the fixed block deviates exactly by the 12 lifted marks, and the five-period lemma $u^5 = \pm J$ holds exactly. $240 = 48 \cdot 5$ remains a factorization *without* coordinatization. No marker move. Machine-checked: [verification/v737_seam48_intertwiner.py]. **[E]** (exact machinery; preregistered kill fired — honest negative)

**The (L1) sector lemma of the CAR majorant (QGEO.CARL1.01).** The first proof block of the majorant lemma named by the continuum probe (11 checks, ∼1 s; verdict L1-SECTOR-PROVEN-MODULO-ELEMENTARY). For all 6 undressed sectors on the fixed grid: $\|K_{2N}^{(q)} - K_N^{(q)}\|_{S_1(\Lambda_\varepsilon)} \le B_q(\varepsilon)/N$ — the occupied-arc mode sums are *exact* geometric sums (exact summation replaces Euler–Maclaurin), the entire $1/N$ structure is the sublattice embedding offset, the leading term is the exact matrix $(iu/2MN) f_q'(s)$ (ratio $\to 1$ at $6 \times 10^{-4}$), $\gamma = 2$ is *derived*, 39744+108 inequalities hold at 0 violations, and $r_1 = 1$ is sharp. Status typed: proven modulo elementary steps, explicit constants; residues (a) grid-sup [elementary], (b) L2 uniform FH/RH [hard-technical], (c) L3 uniformity [medium] — named, not claimed. `GATE.QGEO` does not move. No marker move. [2026-08-05: residue (a) is *fully closed* in the corrected $(1 + \ln M)$ form by `v778` (see its

block below); residues (b) and (c) stay open; the gate still does not move.] Machine-checked: [verification/v745_qgeo_car_l1_sector_lemma.py]. **[E]** (proof-near sector lemma with explicit constants)

**The $G_{\text{net}}$ local net functor (GNET.LOCALNET.01).** The local-net reading of the quotient circle, finite layer exact (16 checks, ∼2 s; verdict GNET-LOCAL-FUNCTOR-ALIVE). Net axioms exact: isotony, graded locality, clock covariance. The Watatani index of the local fixed-point inclusion is exactly 4 for *every* interval $\ell \geq 2$ — the $[D, A]$ index statement is interval-local. Ramond is a half-line/bond defect (solitonic; leakage $\sqrt{2/N}$) — the DHR structure the seam–Calderón identification needs. Four must-fail controls fire. Missing continuum theorems named, not claimed (GNS limit net, index continuity, Longo Q-system identification, twisted DHR sectors, Haag duality/split). No marker move. [2026-08-05: the finite-level precursors of four of the five theorems are measured coherent by v779 (see its block below); the theorems stay named, not claimed.] Machine-checked: [verification/v746_phys_gnet_local_functor.py]. **[E]** (finite net axioms + local index 4) + **[C]** (named continuum theorems)

**The projective Hamming incidence lemma (E8.PROJHAMMING.01).** The centerpiece of the incidence programme (36 checks, ∼5 s; verdict PROJ-HAMMING-EXACT). From the concrete lattice: $h = H/4$ is $\mathbb{Z}[i]$-valued with hermitian Gram determinant 1 (the *unimodular* hermitian E8 lattice over $\mathbb{Z}[i]$); $\bar{h}$ alternating (doubly-even), nondegenerate — $(V, \bar{h})$ symplectic over $\mathbb{F}_2$ with explicit Darboux basis. Core identity: $|H_y| = 7$, pair intersections 3; $BB^{\mathsf{T}} = 4I + 3J$ exact, characteristic polynomial $(x-7)(x-2)^9(x+2)^5$ — the multiplicities $(1, 9, 5)$ are the symplectic signature (honest control: $BB^{\mathsf{T}} = 4I + 3J$ alone holds for any nondegenerate form). Channel: $K^2 = (4/49)I + (45/49)\Pi_0$ exactly ($K$ itself not psd). NS/R as geometry: $\chi_{\text{NSR}} = \bar{h}(\cdot, y_0)$ for the unique $\sigma$-fixed $y_0 =$ FSUM; $248 = 120 + 128$. The v737 resolution: the seam-48 distribution derived exactly (16× the $3 \times 5$ label table), and the $5 \times 48$ coordinatization exists via exactly one $\bar{u}$-invariant spread (kill explained, not revised). Companion Lean module ProjectiveHamming.lean (32 theorems). No marker move. Machine-checked: [verification/v752_projective_hamming_incidence.py]. **[E]** (exact finite lemma; Lean companion)

**The Fano–discriminant bridge (QGEO.FANODISC7.01).** The seven-correlation, permanently typed (26 checks, ∼1 s, one preregistered break; verdict FANO-DISC7-DECORATIVE). The preregistered four-condition test breaks exactly at condition 2: 2 *splits* in $\mathbb{Q}(\sqrt{-7})$ ($-7 \equiv 1 \bmod 8$), so every 2-adic ideal quotient of $\mathcal{O}_{-7}$ has 2-rank $\leq 2 < 3$ (complete norm-8/16 census) — the code $\mathbb{F}_2$-structures are arithmetically unrealizable, the NS/R parity bit has no arithmetic carrier. The three-cycle half is real and exact: equivariant lifts exist for both canonical difference-set Fano models, the canonical order-3 action is $x \to 2x$ (the difference-set multiplier), every lift sends the $\bar{\sigma}$ fixed point to the C corner (norm $14 = \dim G_2$). Controls fire ($x \to 3x$: zero lifts; broken incidence: zero lifts). The seven-correlation is decorative for good. No marker move. Machine-checked: [verification/v757_fano_disc7_bridge.py]. **[E]** (exact machinery; preregistered break — honest negative)

**The Arf/spinor compiler (ARF.SPINORCOMPILER.01).** 46/46 checks, ∼2 s; verdict ARF-SPINOR-EXACT, zero kills, both must-fail controls fire. On the symplectic four-bit quotient $V = L/(1+i)L$ of the unimodular hermitian $E_8$ lattice: the parity lift $\iota\colon V \cong C_{\text{even}}(5)$ carries $\beta = \bar{h}$ exactly in all 256 cells; brute force over all $2^{16}$ functions yields exactly 16 quadratic refinements (Arf census $6 + 10$; $|Sp(4,2)| = 720$ with a faithful+surjective $S_6$ action); the frozen selector ($\sigma$-invariance census $4 \to q(A){=}1$: $2 \to q(F_\Sigma){=}0$: 1) picks a *unique* $q^*$, and the weight form *is* $q^*$; $\text{Stab}(q^*) \cong S_5$ gives $16 = 1 + \bar{5} + 10$ as Arf geometry on refinements and words; the $K_6$ duad model is an equivariant bijection onto $E(K_6)$; $B = I + A_{KG(6,2)}$ entrywise with the exact ovoid eigenvector; all six isotropic spreads carry signature $(1, 2)$; the hypercharge table $X = (-2, -2, -2, 3, 3)$ reproduces the v310/v14 dictionary; moments $(16, 0, 120, 0)$ with the chain 40/41/240/248; Pati–Salam $8 + 8$ with $B{-}L = 1 - 2n_c/3$; $\chi_{\text{NSR}} =$ the anchor bit. Naming discipline: the same $\sigma$ is the family 3-cycle $A_3^{\text{Fam}}$ on message bits and the 3+2 slot split $A_3^{\text{PS}}$ on carrier legs — two registers, never conflated. Lean companion: TfptCarrier/ArfSpinorCompiler.lean (23 kernel theorems). The

physical code→matter reading is *not* claimed here (fence: next block). No marker move, no RH claim. Machine-checked: [verification/v774_arf_spinor_compiler.py]. [E] (exact finite algebra; Lean companion)

**The fired matter kill (ARF.ROOTCLASS.01).** 18/18 checks; verdict ROOTCLASS-MIXED, all three must-fail controls fire; the prediction table SHA-frozen *before* any root data. No $D_5 \oplus A_3/SU(5)$ convention makes even one full set of 15 Gaussian classes pure: the embedding-independent counting bound (every embedding has exactly 128 spinor-side roots; $240 > 128$) forces $\geq 7$ adjoint-side classes for *every* embedding, coordinate or not; the exhaustive scan (56 splits $\times$ 240 assignments $\times$ 2 chiralities) finds zero passing conventions; all 8 spinor classes mix 8 distinct 16-weight pairs (best dictionary match 8/128; $X$-purity 0/15). Binding fence: the code→matter reading — including the "empty mailbox = right-handed neutrino" reading — must not enter any main claim; the Arf mathematics of v774 is untouched. No marker move, no RH claim. Machine-checked: [verification/v775_gaussian_class_d5_purity.py]. [E] (exact census kill; preregistered)

**Boundary code uniqueness (ARF.BOUNDARY.CODE.01).** 31/31 checks; verdict BOUNDARY-CODE-UNIQUE. The non-circular chain boundary → code → $E_8$ → $\mathbb{F}_2^4$ → five slots exists with $g_{\mathrm{car}} = 5$ as *output*: census $135 \to 30 \to 14 \to 2 \to 1$ rebuilt without Hamming/$E_8$ input; the flag datum is one deck-invariant $\mathbb{Z}_2$ (the product of the four side orientations; all three predeclared formalizations agree); the boundary gauge group (order 12) makes the flagless twofold pure gauge; Construction A of the selected code *is* $E_8$ (240/2160 theta, Cartan det = 1, arms 1-2-4); the Arf selector is unique; packets $1 + 4$. Honest residuals as recorded then: R1 the doubly-even self-dual *universe premise* (the circularity relocated and shrunk, not removed), R2 deck-placement blindness (semantic, typed), R3 anchor naming ($\bar{b}(F_\Sigma, A) = 1$). *All three moved in round 21 (2026-08-06, markers unchanged):* R1 is replaced by R1′ — the three physical axioms A1 mutual locality, A2 integer conformal spin, A3 holomorphy/index one, from which Type II is forced exhaustively ([verification/v799_seam_code_typeii.py]); R2 is placed — the deck is the metaplectic anomaly of the total modular $S$-lift at $\tau = i$ — and R3 is named through the $\sigma$-orbit Arf defects whose XOR is the anchor itself ([verification/v798_seam_clifford_modular_s.py]). No P2-removal claim; the contract form is now the sharpened SEAM.CODE.TYPEII.01 residual R1′ [C] (the seam satisfies A1–A3). No marker move, no RH claim. Machine-checked: [verification/v776_boundary_hamming_uniqueness.py]. [E] (exact selection chain) + [O] (the named universe premise)

**The executed clock contract (FTRANSFER.CLOCKS.01 + CONTRACT.F.01).** One module from two probes ($24 + 14$ checks, 0 failures; verdicts NO-COMMON-CONNECTION/FTRANSFER-FIBERED-CARRIES; prereg YAML and both data tables byte-frozen *before* the first run). All four dictionaries are construction-valid (Koide scheme check 571×; washout drift $1.15 \times 10^{-4}$; QCD overlay 0.83%; relic slope 0.0602) and the one-class prediction dies sympy-exactly: transported Schwarzians $\{-\Delta^2/2, -\Delta^2/2, 0\}$, $\Delta = 6\log(3/2)$ — the K1 kill fires exactly as the frozen prior expected; the pass-enabling reparametrization is the forbidden non-affine Möbius clock. The surviving architecture is the *fibered* functor: one shared discrete $\mathrm{PGL}_2$ base (anchor cross-ratio 4/3 exact; unit parabolic deck translation) with the constant seam obstruction cocycle, values exactly $\{0, \pm\Delta^2/2\}$; thermal/proper-time clocks $\{\tau, \beta\}$ vs the RG clock $\{\log \mu\}$ split the cosets; the relic channel stays degenerate, unclassified. The jet route of CONTRACT.F.01 closes at the physical-clock level; the named next step — derive the coset assignment from KMS/modular clock typing — is now a theorem ([verification/v792_ftransfer_kms_coset.py], KMS-COSET-THEOREM). No marker move, no RH claim. Machine-checked: [verification/v777_ftransfer_clock_jets.py]. [E] (sympy-exact kill and cocycle) + (MEASURED) (dictionary legs) + [X] (executed preregistered kill test)

**The grid-supremum closure (QGEO.GRIDSUP.01).** One module from two probes ($19/19 + 19/19$; verdicts GRIDSUP-SPLIT-CLOSED-LOG-LAW/FOURIER-LOGSUM-CLOSED). Remainder (a) of the v745 (L1) majorant lemma is fully closed in the corrected $(1 + \ln M)$ form: (a1) the continuum entrywise supremum, (a2) the configuration-$S_1$ closure with explicit $B_{\mathrm{conf}}(\varepsilon)$ (packing + $\zeta(2)/\zeta(3)$), and (a3) the refinement remainder with RHO($\varepsilon$) via exact $\mathrm{csc}^3$ integrals close elementarily with zero census violations; (a4) the leading part is honestly log-divergent (jump-driven; taper-controlled),

so the literal `v715` clause "one $B(\varepsilon)$ for all grid refinements" is false at $r_1 = 1$ and is repaired to $\|D_N^{(M)}\|_1 \le [B_0+(J/2)\ln(M+3)]/N$, certified at theorem level by the grid Fourier log-sum inequality $\|F\|_1 \le c_0 + (J/4)\ln(M+3)$, $J = 2|f_q'(\varepsilon)|$ (162-case census to $M = 6144$, zero violations; all 252 transfer cases verified; Araki/Powers–Størmer summability explicit). Honest: the measured slope is $J/\pi^2$ — the bound is valid, not sharp. Residues (b) L2 and (c) L3 stay open; `GATE.QGEO` does not move. No marker move, no RH claim. Machine-checked: [`verification/v778_qgeo_gridsup_logsum.py`]. **[E]** (theorem-level closure with explicit constants)

**The $G_{\text{net}}$ continuum strand (`GNET.GNSLIMIT.01`).** One module from two probes (27/27 + 27/27; verdicts GNS-PRECURSORS-COHERENT/GNET-ARF-NO-CORRESPONDENCE). The inductive system is exactly coherent: Haar scaling isometry tower intertwines at $\le 2.2 \times 10^{-16}$; the index-4 expectation squares commute exactly (the finite half of the Longo Q-system identification); the canonical state is Cauchy with geometric contraction $\sim 1/4$ per doubling; the index census is machine-exact at every level with zero degradation; the twisted Haag-duality defect is 0 with $\dim B'/\dim A' = 4 =$ the Watatani index. The exact dim law $d_c(\ell) = 4^{\ell-1} + 2^{\ell-1}\cos(\pi c/2)$ makes the $\mathbb{C}[\mathbb{Z}_4]$ quasi-basis obstruction exactly $2^{-\ell}$ (restored only asymptotically; it explains the $\ell = 1$ three-sector anomaly). The honest negative, exhaustive: the $16 = 1+5+10$ counting echo with the Arf geometry is exact but non-geometric ($\mathbb{F}_2$-rank 3 vs 4; $GL(4,2)$ census 0/20160; no internal order-3 carrier), and $K_{\text{matter}}$ does not govern the tower transport — the two fours stay in their registers. All five controls fire. The five `v746` continuum theorems stay named, not claimed (the limit-existence hypotheses are measured in round 21, [`verification/v802_gnet_martingale.py`]). No marker move, no RH claim. Machine-checked: [`verification/v779_gnet_gns_arf.py`]. **[E]** (exact coherence data; dim law) + (MEASURED) (precursors; honest negative)

**The $d = 4$ forcing census and the clock kill (`E8.ONEOBJECT.01`).** One module from two probes (21/21 + 23/23; verdicts ONE-OBJECT-PARTIAL/CLOCK-FIBER-DEAD). The TFPT glue pattern (cyclic isotropic glue, evenness $q \equiv 0 \bmod 2\mathbb{Z}$, unimodularity $|H|^2 = 4d$) admits an even unimodular *diagonal* glue of $A_{d-1} \oplus D_{d+1}$ iff $d = 4$ (exact Fraction census, $d = 2..12$): at $d = 4$ exactly the two `v92` Lagrangians ($k \mapsto k$, $k \mapsto 3k$) with the explicit $E_8$ certificate (det = 1, 240 norm-2 vectors); $d = 9$ dies at isotropy; both relaxation controls fire. Corollaries at the forced $d$: $g_{\text{car}} = d+1 = 5$, $N_{\text{fam}} = d-1 = 3$, rank $2d = 8$, index $4 = d = |\mu_4|$. Honesty gate: $c_3 = 1/(2\pi d)|_{d=4} = 1/(8\pi)$ is a corpus-legitimate rewriting (`v23` anchor), *not* a new derivation of P1. The clock kill: the fiber product $\mathbb{Z}_{12} \times_{\mathbb{Z}_6} \mathbb{Z}_{30}$ cannot act on the 60 Gaussian lines — the full line-compatible group has order 23040 with maximal element order 12 (orders 15/30/60 absent for every realization); the fiber product is cyclic $\mathbb{Z}_{60}$ (exact SNF). Round-22 addendum: a second, independent $d = 4$ mechanism — the null selector $|v_d|^2 = \det M_d = d(4-d)$ on the RR/Serre functor vector, with the bridge $g^2 - N^2 = 4d = |H|^2$ tying null to glue index on the same invariant $4d$ ([`verification/v807_lorentz_nullselector.py`]). No marker move, no RH claim. Machine-checked: [`verification/v781_one_object_clock_census.py`]. **[E]** (exact census + exact kill; $c_3$ link typed interpretation)

**The transition-bus torsor (`E8.TRANSITIONBUS.01`).** One probe (35/35; verdict TRANSITION-BUS-TORSOR). The 240 roots are 15 disjoint coordinate 8-frames over $V = L/(1+i)L$; on $L/2L$ the 120 root pairs are exactly the 120 $q = 1$ points of the plus-type quadratic space, and each class fiber is a canonical torsor under $v^\perp$ (affine $\mathbb{F}_2^3$, no distinguished origin); $J$ acts as translation by $v$. The obstruction census is proof-grade: no deck-equivariant root-level labeling $R(E_8) = V \times (V \setminus 0)$ exists. The frozen symplectic predicate reproduces all 57600 ordered inner products (sign law included; $h$-refinement); spread layer $240 = 5 \times 3 \times 16$ with the $q^*$ selector; the $A_{15}$ comparison is quantified (a strict bus *with* a start state; $E_8$ provably has none; cross-class non-orthogonality exactly doubled). Controls fire (the C3 control-design correction disclosed). Round-22 follow-up: the no-origin obstruction is now an algebraic theorem — $W = L/2L$ is free over the dual numbers $\mathbb{F}_2[\epsilon]$, the deck is the jet unit $J = 1+\epsilon$, and the exhaustive splitting census is empty ([`verification/v803_e8_jetcode_affine_nsr.py`]); the torsor is the non-split self-extension, not merely a census fact. No marker move, no RH claim. Machine-

checked: [verification/v782_e8_transition_bus.py]. [E] (exact structure theorem; bus language stays interpretation)

**The two-qubit Clifford identification (E8.CLIFFORD2Q.01).** One probe (32/32; verdict CLIFFORD-PARTIAL(H4)). Under the single predeclared chart the 60 Gaussian lines *are* the 60 pure two-qubit stabilizer rays, entry-exactly over $\mathbb{Z}[i]$ (60/60; support census $4+24+32$); $UG_{31}U^{-1}$ is an index-2 subgroup of the Clifford matrix group ($92160 = 2 \times 46080$, coset $\zeta_8 G_{31}$; projectively equal); $46080 = 4 \times 16 \times 720$ carries the semantics $\mu_4 \times \mathrm{Pauli}_{16} \times Sp(4,2)$, classes = Pauli contexts, spreads = MUB pentads. H4 fails as predeclared: the index bit is the Galois $\sqrt{2}$/Hadamard bit, not $J \leftrightarrow -J$ (that is the separate antiunitary outer $\mathbb{Z}_2$). All controls fire (T-gate twist exactly the predeclared 16/60). Round-21 follow-up: the missing coset is identified — the per-factor Hadamard is the metaplectic lift of modular $S$ at $\tau = i$ ([verification/v798_seam_clifford_modular_s.py]). Carrier mathematics only; the fired ROOTCLASS-MIXED kill is unaffected. No marker move, no RH claim. Machine-checked: [verification/v783_two_qubit_clifford.py]. [E] (H1–H3 exact) + (MEASURED) (H4 honest partial)

**The curve/code 2-torsion decider (CURVE.CODE.2TORSION.01).** One probe (33/33; verdict CURVE-CODE-PARTIAL). The frozen CM route through $y^3 = x^4 - 1$: every generator period certified as an integral $\mathbb{Z}[\zeta_{12}]$ recognition (dps 120, HNF index 1); $\tau = \zeta^4$ derived as the only order-3 class; minimal odd $\tau$-invariant polarization type $(1,3)$ (no principal form; unit-sign obstruction); $q_\Theta$ the unique $\tau$-fixed refinement. The correspondence dies at two exact layers: $\tau$ acts freely (cycle type $[3,3]$ on the six odd theta characteristics, syntheme side) vs $\bar{\sigma}$ fixing a 2-plane ($[3,1,1,1]$, duad side) — exchanged only by the outer $S_6$ automorphism; and $q_\Theta$ is even (Arf 0, 1+6+9) while $q^*$ is odd (Arf 1, 1+5+10). Joint census 0 with positive control 6; all controls fire. Round-21 follow-up: the outer twist is not a dead end — the canonical outer bridge exists and resolves the Arf mismatch by the anchor shift ([verification/v796_duad_syntheme_bridge.py]). No marker move, no RH claim. Machine-checked: [verification/v784_curve_code_2torsion.py]. [E] (exact finite algebra; certified periods; both kill layers named)

**The KMS coset theorem (FTRANSFER.KMSCOSET.01).** One probe (18 checks, 0 failures; verdict KMS-COSET-THEOREM). The coset carried by a clock chart of the executed $F_{\mathrm{transfer}}$ contract is a *function* of its KMS/modular typing datum alone: $S = -(\ln\lambda)^2/2$ with $\lambda$ the $\mathrm{PGL}_2$ multiplier of the flow's time-1 holonomy per clock e-fold. The classifier uses no Schwarzian and no derivatives (non-circularity gated on its own source): functional-identity holonomy witness, $\mathrm{PGL}_2$ class, detailed-balance multiplier cross-ratio. Blind assignment: $F_{\mathrm{pole}}/F_{\mathrm{Boltzmann}}$ modular with $|\ln\lambda| = \Delta = 6\log(3/2)$ exactly, $F_{\mathrm{QCD}}$ autonomous (parabolic), $F_{\mathrm{relic}}$ degenerate; unblinding exact (sympy zero). Controls: fake KMS caught; affine invariance; the v578 Möbius clock as the positive case. The executor → groupoid → coset chain is whole. No marker move, no RH claim. Machine-checked: [verification/v792_ftransfer_kms_coset.py]. [E] (sympy-exact theorem; blind protocol)

**The outer bridge (CURVE.CODE.OUTERSPIN.01).** One probe (24/24; verdict BRIDGE-CANONICAL). The Sylvester duality $\beta$: code odd forms → curve spreads with $\beta\sigma = \tau\beta$ and $\beta(q^*) = S^*$ (the unique $\langle\tau,\rho\rangle$-fixed spread) exists and is unique up to the order-6 code joint stabilizer (census $18+18 \to 6+6$, one orbit incl. the $\tau/\tau^2$ swap by an inner code symmetry); the outer witness is exact and uniform. It resolves the v784 Arf mismatch canonically: $q_\Theta$ pulls back constantly to $q^*_{\mathrm{even}}$ with $q^* = q^*_{\mathrm{even}} + \bar{h}(\cdot, A)$ (the anchor shift); $\bar{5}/10$ and 3+2 transport exactly onto the $S^*$-line geometry; $t_A \mapsto \rho = J$ on every bridge; $\chi_{\mathrm{NSR}}$ transports as an honest exact two-variant torsor (pinned by the v752 NS/R parity). The inner identification stays dead; no matter semantics. Controls fire. No marker move, no RH claim. Machine-checked: [verification/v796_duad_syntheme_bridge.py]. [E] (finite census theorem; structure labels only)

**The metaplectic $S$-lift (SEAM.CLIFFORD.MODULAR_S.01).** One probe (27/27; verdict MODULAR-S-PARTIAL). The per-binary-factor Hadamard $K_1 = H \otimes I$ *is* the metaplectic/Weil lift of modular $S$ at its fixed point $\tau = i$ and lands exactly in the missing coset $\zeta_8 G_{31}$ of v783 ($S^2 = I$, $T^2 =$ Pauli parity, $(TK)^3 = \zeta_8 I$ — the Weil constant); the total Fourier $H \otimes H$ lies inside $G_{31}$ with

metaplectic anomaly $(T_3K_3)^3 = iI = J$ — the deck is the anomaly of the total modular lift (residual R2 placed); $\zeta_8 = (1+J)/\sqrt{2}$ is the real half-deck swapping the seam pair $\{L, \zeta_8 L\}$. The strict uniqueness census over all 46080 coset elements is *empty* (no $\Theta$-real metaplectic involution fixes $q^*$: the Arf defect is unavoidable); the retyped census has exactly one $G_{31}$-conjugacy class, and the $\sigma$-orbit of lifts has Arf defects exactly $\{A + F_\Sigma + F_i\}$ whose XOR is the anchor $A$ itself (residual R3 named, not fixed). Controls fire (CZ inside $G_{31}$; wrong phase breaks $S^2$). No marker move, no RH claim. Machine-checked: [verification/v798_seam_clifford_modular_s.py]. **[E]** (exact coset/metaplectic identities; the strict census empty by its own frozen gate)

**Type II from the seam axioms (SEAM.CODE.TYPEII.01).** One probe (19/19; verdict TYPEII-FORCED). Exhaustively over all 308,993 subspaces of $\mathbb{F}_2^8$ (dim 0..4; dim $> 4$ by rank-nullity): mutual locality of the Construction-A boundary vertex words $\iff$ $C$ self-orthogonal; integer conformal spin (trivial $T$ sector phases $i^{\mathrm{wt}}$) $\iff$ $C$ doubly even; holomorphy/index one ($|\mathrm{disc}| = 1$, $S$-closure of the single character) $\iff$ $C$ self-dual — the survivor set is exactly the 30 $\hat{e}_8$ copies; no boundary-legal non-Type-II code exists; every non-self-dual doubly-even code has a fractional-weight discriminant sector (exhaustive; $[8, 3]$ witness with exact coset thetas). Residual R1 of ARF.BOUNDARY.CODE.01 is replaced by R1′: the physical seam satisfies A1 mutual locality, A2 integer conformal spin, A3 holomorphy/index one — each a standard physical demand, not a coding-theory premise. Imported lattice-VOA/theta ingredients cited and machine-verified in every instance used. Controls fire. No P2 marker move, no RH claim. Machine-checked: [verification/v799_seam_code_typeii.py]. **[E]** (exhaustive equivalence chain) + **[C]** (R1′: the seam satisfies A1–A3)

**The torsor Fourier modes (E8.TORSOR.FOURIER.01).** One probe (30/30; verdict TORSOR-FOURIER-PARTIAL). The $120 = 15 \times 8$ projective Fourier rays of the origin-free root fibers are canonical (an origin shift is a global sign; phase law exact on all 960 shifts), orthogonal (integer Gram $= 16I$), deck/Pauli-diagonal by characters and $G_{31}$-monomial with label action $Sp(4, 2)$ (kernel $64 = \mu_4 \times \mathrm{Pauli}_{16}$). The carrier block — the 8 R-fibers' 64 rays — factorizes as $(1{+}5{+}10) \boxtimes (1{+}3)$ under $W(D_5) \times W(A_3)$ (Clebsch charpoly $(x{-}5)(x{-}1)^{10}(x{+}3)^5$ exact; $S_4$-natural tetrad with $\sigma =$ family 3-cycle + anchor) — the carrier Pascal at *character* level, impossible at point level (v775); the honest cap: the weight and character bases are mutually unbiased ($|\langle \mathrm{pair}|\mathrm{mode}\rangle|^2 = 1/8$ exactly), so ROOTCLASS-MIXED stands. The 45 deck-even modes are the 45 (context, Pauli) incidences under the certified chart. Controls fire. No marker move, no RH claim. Machine-checked: [verification/v800_e8_torsor_fourier.py]. **[E]** (character theory exact; the predeclared P3 failure typed)

**The martingale limit hypotheses (GNET.MARTINGALE.LIMIT.01).** One probe (15/15; verdict MARTINGALE-HYPOTHESES-MET). The Haar ladder is an exact filtration (isometry with support coherence, leak 0; Watatani quasi-basis 185/3041 at $l = 2/3$; one index-4 expectation for the whole tower); the state defects are summable at measured rate $\sim$0.25 per doubling — one power better than the $2^{-l}$ dim-law envelope (reported, not forced); the unique limit follows by elementary telescoping and is verified on every rung; the split/normality precursors are uniform (determinant witness $D_k \geq 0.985$, stabilizing at 0.9928). Steps 5–7 (GNS reconstruction, local normality, split + $\mu_4$ simple-current extension) stay typed analytic remainder. The open QGEO majorant residues (b)/(c) are bypassed for limit *existence* and relocated into the typed normality step — a reallocation, not a proof. Controls fire (decimation converges to the wrong degenerate limit: the coherent bond is load-bearing). No marker move, no RH claim. Machine-checked: [verification/v802_gnet_martingale.py]. (MEASURED) (hypotheses met) + **[O]** (steps 5–7 typed)

**The jet-code normal form and the affine NS/R geometry (E8.RAMIFIED.JETCODE.01 + E8.AFFINE.NSR.01).** One module from two probes (18/18 + 15/15; verdicts JETCODE-EXACT/AFFINE-NSR-EXACT). Because 2 ramifies in $\mathbb{Z}[i]$, $\mathbb{Z}[i]/(2) \cong \mathbb{F}_2[\epsilon]/(\epsilon^2)$ with $\epsilon = 1{+}i$ (unique iso; the split ring $\mathbb{F}_2 \times \mathbb{F}_2$ fails all 24 bijections), and $W = L/2L$ is *free* of rank 4 over the dual numbers (Hermite-certified $\mathbb{Z}[i]$-basis): the mod-2 root system is a first-order jet space, the exact self-extension $0 \to \epsilon W \to W \to V \to 0$ with $\ker \epsilon = \mathrm{im}\, \epsilon$. The deck *is* the jet unit $J = 1{+}\epsilon$ — translation by the error part re-derives the v782 transition-bus torsor on all 240

roots — and among all 65536 sections there are *zero* $\mu_4$-equivariant splittings: non-splitness is the no-canonical-origin obstruction, now an algebraic theorem rather than a census fact. The 120 root pairs are the $q$=1 unit shell ($240 = 15 \times 8 \times 2$: root = address × error jet × orientation). Part 2: $\mathrm{Stab}(\chi_{\mathrm{NSR}}) = \mathrm{AGL}(3,2)$ of order $1344 = |\mathrm{Aut}(H_8)|$ with the *same* natural 8-point action, and affine functions on the NS/R torsor reconstruct $\mathrm{RM}(1,3) = H_8$ with exactly 1344 equivalences — a typed internal reconstruction that sharpens the `ARF.BOUNDARY.CODE.01` attack surface, *not* a P2 removal. Controls fire. No marker move. Machine-checked: [`verification/v803_e8_jetcode_affine_nsr.py`]. [E] (ring-theoretic normal form; exhaustive splitting census; reconstruction fenced)

**The syndrome-algebra kill (`E8.SYNDROME.ALGEBRA.01`).** One probe (13/13 incl. the 3 must-fire controls; verdict SYNDROME-DEAD; the kill fired exactly as preregistered). The sharpest error-correction-hull hypothesis — that the 240 root operators $U_r$ plus the 16 neutral kernels generate a twisted group algebra presenting $E_8$ as the syndrome hull of $H_8$ inside $\mathrm{End}(S_+)$ — is *dead*: max rank 222/256 over all four frozen phase conventions, the $\pm r$ proportionality obstruction is structural for the whole diagonal-phase family, and both post-hoc escape directions score worse (215/160). What survives is exact: $16 + 240 = 256$ at dimension level, the 16 `v112` neutral kernels = the diagonal MASA, and same-class products are $A_0$-valued with $\mu_4$ coefficients. The successor criterion is named (root-individual operator data beyond diagonal phases). The Construction-A Hamming theorem is untouched. Controls fire. No marker move. Machine-checked: [`verification/v805_e8_syndrome_algebra.py`]. Honest negative on record + [E] (the surviving identities)

**The Lorentz spinor coordinates and the null selector (`E8.ONEOBJECT.01` addendum).** The scoped TFPT statement is the exact RR/Serre algebraic selector. The compiler triple is a rank-one spinor square: $M(5,-3,4) = 2\,(1,2)^T(1,2)$ exact, Euclid $(2,-1) \mapsto (5,-3,4)$. Part 2: the null selector $|v_d|^2 = \det M_d = d(4-d)$ forces $d = 4$ on the RR/Serre functor vector with anchor identities $i{+}j = N_{\mathrm{fam}}$, $ij = |\mathbb{Z}_2|$, $i^2{+}j^2 = g_{\mathrm{car}}$, $2ij = |\mu_4|$; the bridge $g^2 - N^2 = 4d = |H|^2$ makes null $\iff$ glue index $= d$ — a second, independent $d = 4$ mechanism on the same invariant $4d$ as the `v781` census (honest downgrade recorded: the seam family has no $d$-dependence to vary; the missing theorem typed). Controls fire. No marker move. Machine-checked: [`verification/v807_lorentz_nullselector.py`]. [E] (spinor identities; algebraic selector) + [O] (seam-family identification)

**The Petersen carrier geometry and the sixth root (`CARRIER.PETERSEN.RADIAL.01` + `TRANSPORT.SIXTHROOT.01`).** One module from two probes (19/19 + 16/16; verdicts PETERSEN-RADIAL-EXACT/SIXTHROOT-SPECTRAL-ONLY). The $K_5$ edge machine: charges $(-4,+1,+6)$ match the certified `v774` $SU(5)$ decuple; the edge graph is Petersen $\mathrm{SRG}(10,3,0,1)$ with $|\mathrm{Aut}| = 120$ by census; the distance shells $(1,3,6)$ are charge-pure; the flavor hexagon *is* the distance-2 shell edge-by-edge; the equitable quotient carries the hypercharge eigenvector $(6,-4,1) \to -2$; and $\mathrm{spec}(P^6) = \{1, 64/729, 1/729\}$ *is* the deployed transport spectrum exactly — the $\{1, (2/3)^6, (1/3)^6\}$ triple acquires a graph-geometric origin. The 10-dim multiplicity prediction ($\times 5 = g_{\mathrm{car}}$ fast, $\times 4 = |\mu_4|$ slow) is typed [H neu]. Part 2, the honest middle: $B = \frac{1}{18}[[13,1,4],[1,13,4],[4,4,10]]$ is the *unique* symmetric doubly stochastic PSD sixth root in the corpus basis, and $B^6$ is $GL(3,\mathbb{Q})$-conjugate to `v486`'s $M^6$ and to the cusp-diagonal form but bit-equal to *neither* — the corpus freezes the spectrum, not the basis (the basis-freeze question named). *[2026-08-06 round-23 dated correction: the census had missed the deployed `v221` evaluation — $T_{v221} = B^6$ bit-exactly, so the verdict upgrades to BITEXACT-vs-v221 and the basis-freeze question is answered on the deployed route; the [H neu] 10-dim reading gains the canonical proposal $T_{10}$ (TRANSPORT10-PROPOSED, no deploy claim); see the six-step block below ([`verification/v814_k5_sixstep_transport.py`]).]* Controls fire. No marker move. Machine-checked: [`verification/v808_petersen_sixthroot.py`]. [E] (census identities; spectrum equality) + [H neu] (multiplicity reading) + [O] (basis freeze — answered bit-exactly in round 23, dated correction above)

**The doily incidence identity (`CURVE.CODE.DOILY.01`).** One probe (21/21 incl. the 3 must-fire controls; verdict DOILY-EXACT). The certified outer bridge sharpens into an incidence-operator

statement with a spectral payoff: the $K_6$ doily incidence $N$ (15 duads × 15 synthemes on the six odd Arf refinements) satisfies $NN^T = B + 2I$ entrywise with spec $= \{9, 4^9, 0^5\}$, and $N/3$ has singular values $\{1, 2/3, 0\}$ — the recovery base rate $2/3$ is the unique nontrivial singular value of the canonical duad-to-syntheme incidence (the $(2/3)^6$ identification typed *pending the six-step composition* — decided in round 23: SIXSTEP-CLOCK-ONLY, the deployed six is the clock exponent, [verification/v814_k5_sixstep_transport.py], see the six-step block below). Petersen edges = synthemes 15/15; every census bridge factors exactly $N_{\text{code}} = P_\Delta^T N_{\text{curve}}^T P_\Gamma$ — a transpose duality, not an isomorphism; the Arf switch set $T_6$ = the six within-block duads. Controls fire. No marker move. Machine-checked: [verification/v809_curve_code_doily.py]. **[E]** (incidence identities) + **[O]** (the six-step target — decided clock-only in round 23, dated note above)

**The cut code and the master moment (`CARRIER.CUTCODE.01` + `CARRIER.MOMENT.INCIDENCE.01`).** One module from two probes (17/17 + 16/16; verdicts CUTCODE-EXACT/MOMENT-INCIDENCE-PARTIAL). The cut map is injective-isometric on all 256 cells and the Arf split $16 = 1 + \bar{5} + 10$ *is* the cut classification $(0|5) + (1|4) + (2|3)$ of $K_5$ — an Arf geometry, exact but non-matter; Clebsch $40 = 5 + 20 + 15$ over the Arf shells with within-10 = Petersen (census 120); roots/lines grade $80 + 160$ / $20 + 40$ — Arf/cut grading, *not* matter semantics. The master moment is one symbolic identity: $\mathrm{Tr}_{S^+} X^2 = 4|X|^2 = 4h(E_8) = 120$. The frozen 210 decision: the $210 = \binom{10}{4}$ map is *decorative* per the frozen rule — the equivariant bijection is killed by the exhaustive fixed-point obstruction; the partial $80 \to 5 \times 16$ miss-slot map and the Newton tie survive as audit. Controls fire. No marker move. Machine-checked: [verification/v810_cutcode_moment.py]. **[E]** (cut classification; master moment) + honest negative (the 210 bijection)

**The flavor-filtration fingerprint (`FLAVOR.GRAPH.FILTRATION.01`).** One probe (16/16 incl. the 4 must-fire controls; verdict FILTRATION-FINGERPRINT-ONLY). The flavor-matrix filtration $Q \to K \to L$ is an incidence-compiler fingerprint: the row budgets are exact, order-exact counts of the $K_{3,2}$/prism/$K_5$/$H_8$ incidence chain with level-wise additivity mirroring $L = K + Q$; *no* entry-level reconstruction succeeds from any frozen candidate; the exhaustive false-positive rates (0.005–0.75%) make the misses informative. Typed audit fingerprint — kept beside the audit identities, not in any derivation. Controls fire. No marker move. Machine-checked: [verification/v812_flavor_graph_filtration.py]. **[E]** (budget identities) + honest negative (entry reconstruction)

**The P1 index-KMS identity (`P1.INDEX.KMS.01`).** One probe (13/13 incl. the 2 must-fire controls; verdict P1-IDENTITY-CLOSED). The closed normalization identity $I \cdot \beta \cdot c_3 = 4 \cdot 2\pi \cdot \frac{1}{8\pi} = 1$ is exact in formal arithmetic, with each factor independently sourced: the Jones index $I = 4 = |\mu_4|$ is a $c_3$-free theorem (zero occurrences of $c_3$ in all four sources, gated), the KMS angle $\beta_{\text{angle}} = 2\pi$ independently measured (`v526`; the radian pin is the one shared convention and cancels), and $c_3$ stays the P1 axiom that the identity *explains* — it does not derive it. The finite evenness precursor is measured ($w_1 = w_3 = \frac{1}{4}$ exactly at every level, symmetric zero-sum offset $\pm 2^{-(l+1)}$ — the no-anomaly shape); the sea state does *not* equidistribute (dev 0.188), so the open bridge must normalize the modular response over the $I$ deck sectors (kill: any asymmetric sector offset). Control: index 2 gives $1/(4\pi)$, booked elsewhere by the corpus. No marker move; P1 stays a declared input. Machine-checked: [verification/v813_p1_index_kms.py]. **[E]** (the identity; the convention audit) + **[O]** (the even modular-response bridge)

**The six-step composition and the $T_{10}$ proposal (`K5.SIXSTEP.TRANSPORT.01`).** One probe (23/23 incl. the 3 must-fire controls; verdicts SIXSTEP-CLOCK-ONLY + TRANSPORT10-PROPOSED). The rate $(2/3)^6$ arises canonically on three spatial routes (doily alternation, hexagon compression, hexagon circulation), but each carries a typed obstruction (no unit mode — the norm obstruction; the bipartite sign $-2/3$; the wrong space), and sixth-power blindness ($C_6$ vs $2K_3$: identical sixth powers) makes per-step data mandatory — the deployed six is the *clock* exponent. The dated correction: $T_{v221}$ (the exact spectral sum of `v221`'s transfer $(1-w)^6$) $= B^6$ *bit-exactly* — the `v808` sixth-root verdict upgrades from SPECTRAL-ONLY to BITEXACT-vs-v221. The canonical proposal $T_{10} = P_{10}^6$ in the $\Lambda^2$ basis passes all gates (spectrum $\{1, (1/3)^6 \times 5, (2/3)^6 \times 4\}$, equitable `v774`

quotient = $(Q_{\mathrm{Pet}}/3)^6$ bit-exact) — a proposal, no deploy claim. Controls fire. No marker move, no RH claim. Machine-checked: [verification/v814_k5_sixstep_transport.py]. [E] (operator identities; the bit-exact location) + [C] (the $T_{10}$ reading, proposal)

**The positive-descent master theorem (TFPT.POSITIVE_DESCENT.MASTER.01).** One probe (14/14 incl. the 4 must-fire controls; verdict MASTER-BOTH-INSTANCES; kernel-checked Lean core `PositiveDescentMaster.lean`). Summable state defects over character sectors force a unique positive limit state (telescoping + Cauchy tail; positivity closed; sectors commute) — and both instances measure their hypotheses: $G_{\mathrm{net}}$ at 0.267/doubling, and the prime packet state at 0.701/doubling $\approx 2^{-1/2}$ with $\hat{m}_2(p) = -\frac{1}{15}$ exact for every odd prime. Control K4: positivity is a separate hypothesis. No marker move, no RH claim. Machine-checked: [verification/v817_positive_descent_master.py]. [E] (the Lean core) + (MEASURED) (both instances)

**The anchor affine normal form and the flavor bidirectional checksum (ANCHOR.AFFINE.01 / FLAVOR.BIDIR.01).** One module from two probes (18/18 + 15/15; verdicts ANCHOR-AFFINE-EXACT + FLAVOR-BIDIR-CHECKSUM (selective)). Part A: the anchor power sums obey $p_{n+1} = 2p_n - 2$ identically (sympy); $T(x) = 2x - 2$ has the unique fixed point $2 = |\mathbb{Z}_2|$ and the $T$-orbit of 4 is the compiler quintet $(4, 6, 10, 18, 34)$ — the whole budget (240, 248, 30, 40, 48, 41) from one recursion. Part B: $Ra = (4, 10, 13)$ and $R^{\mathsf{T}}a = (6, 18, 8)$ — the same anchor decodes rows (odd power sums $+N(3{+}2i)$) and columns (even power sums $+h(D_5)$); of the 12 down-row candidates exactly one — the accepted $(1, 5, 2)$ — passes the full identity set, with the anchor contraction $(Ra)_2 = 10 = p_3$ as the one-number kill (siblings give 11 and 12). Controls fire (wrong map, wrong anchor, the sibling reproducing the corpus failure numbers verbatim, scrambled ordering). A compression / normal form, not a new claim. No marker move. Machine-checked: [verification/v832_anchor_flavor_checksum.py]. [E] (exact sympy/integer identities throughout)

**The $(1{+}i)$ ramification ladder (GAUSSIAN.RAMLADDER.01).** One probe (33/33; verdict RAMIFICATION-LADDER-EXACT, $\sim$ 5 s). The four documented roles of the ramified Gaussian prime $\pi_2 = 1{+}i$ are one machine-checked object: norm doubling ($(1{+}J)^{\mathsf{T}}(1{+}J) = 2I_8$ exactly; zero class empty 0/240; $240 = 15 \times 16$; elementary divisors $(1^4, 2^4)$); the 4-bit address $(L/(1{+}i)L \cong \mathbb{F}_2^4$, deck trivial); the non-split jet $(\mathbb{Z}[i]/(2) \cong \mathbb{F}_2[\epsilon]/(\epsilon^2)$ unique among 24 bijections, deck $= 1{+}\epsilon$, exactly 0 of 65536 sections deck-equivariant, the 120 root pairs the $q{=}1$ shell); the metaplectic lift ($\zeta_8 = (1{+}i)/\sqrt{2}$, $(SH)^3 = \zeta_8 I$, $(T_3K_3)^3 = iI$ = the deck; exact $\mathbb{Z}[\zeta_8]$ BFS census $|C_2/\mu_8| = 11520$, zero Galois-mixed classes, $C_2 = K \sqcup \zeta_8 K$; $\chi(H{\otimes}I) = 1$ the RM–CSS phase bit). Controls fire (the split ring admits 0 isos — 2 must ramify; the split extension splits 65536/65536). A compression of v689/v803/v783/v798. No marker move. Machine-checked: [verification/v833_gaussian_ramification_ladder.py]. [E] (exact integer/sympy arithmetic, exhaustive censuses)

**The PG(3,2) flag normal form and the incidence split (PG32.FLAGS.01).** One probe (26/26; verdict PG32-FLAGS-EXACT + INCIDENCE-SPLIT, $<$ 1 s). The counting normal form $\binom{4}{2}_2 = 35$, $140 = 35 + 105$, $105 = 35 \times 3$ flags $\cong$ point pairs; the RM layer rebuilt as set equalities ($[15, 10, 4]$ enumerator exact; Grams $22I{+}6J$ / $6I{+}J$ / $28I{+}7J$); the vacuum coordinate is origin restoration. The kill typed as a split: $105 = 45 + 60$ is the $\mathrm{Sp}(4, 2)$-orbit decomposition on both sides; the polarity dictionary on the 60 non-isotropic rungs is bijective, equivariant ($60 \times 720$) and incidence-preserving; the 45 doily rungs are obstructed ($(t{+}U) \cap U = \emptyset$ for 45/45; flag Gram $18I{+}3J \neq 22I{+}6J$) — "105 checks = 105 flags" is a deck-equivariant bijection, *not* an incidence isomorphism. The zero class carries no root (0/240 rebuilt). Controls fire. No marker move. Machine-checked: [verification/v834_pg32_flag_completion.py]. [E] (exact integer censuses, orbit decompositions, Grams)

**The message ladder and the Doily–Pascal rank theorem (MESSAGE.LADDER.01 + DOILY.PASCAL.RANK.01).** One module from two probes (23/23 + 16/16, verdicts MESSAGE-LADDER-EXACT and DOILY-PASCAL-RANK-EXACT, $\sim$2 s, exact integer/Fraction/sympy). The budget lines are one formula: $M_n = 15 \cdot 2^n = (15, 30, 60, 120, 240)$ with one named structure

per rung — $h(E_8) = 30$ derived from the rebuilt root system itself (highest root height 29, marks $\{2,2,3,3,4,4,5,6\}$), 60 Gaussian lines, $120 = |R^+|$, the $240 = 15 \times 16$ census, coda $248 = 15 \cdot 16 + 8$; wrong anchor/multiplier controls fire. And the doily incidence is a rank theorem: $NN^{\mathsf{T}} = B + 2I$ entrywise under the $K_6$ duad model, charpoly $= (x-9)(x-4)^9x^5$ exact — rank 10 $= A_\Lambda$, kernel 5 $= g_{\mathrm{car}}$ (spanned by the six ovoid indicators), multiplicity $9 = N_{\mathrm{fam}}^2$, recovery $2/3 = (N_{\mathrm{fam}}-1)/N_{\mathrm{fam}}$, six-step $(2/3)^6 = 64/729$ exact; no $(1/3)^6$ doily mode (typed, no upgrade); wrong-pairing controls fire. No marker move. Machine-checked: [verification/v844_message_doily_rank.py]. **[E]** (exact identities against the rebuilt objects)

**The finite compiler normal form (NORMALFORM.CFIN.01).** One probe (28/28, SPEC v2 declared, verdict NORMAL-FORM-ASSEMBLED, ∼4 s, exact integer/sympy). $C_{\mathrm{fin}} = (V, \hbar, q^*, \sigma, \iota)$ read off the Gaussian quotient as *one* object: the unique selector-picked refinement (16 refinements, Arf $6+10$), the order-3 family cycle, the parity lift onto $C_{\mathrm{even}}(5)$; derived counting $16 = 1+5+10$ with $|Sp(4,2)| = 720$ and $\mathrm{Stab}(q^*) = 120$ faithful $S_5$, the Pascal reading $(1,10,5)$, the ovoid projective reading, the budget $240 = 16 \cdot 15$ / $248 = 16 \cdot 15 + 8$ against the actual census. The kill is built in: the 128-spinor counting bound ($\geq 7$ of 15 classes always adjoint-side; saturated $8+7$, zero mixed classes) and the weight-level kill re-verify ROOTCLASS-MIXED — not overturned. P2 narrowed to the v799 residual R1′ ([C]), not eliminated. Controls fire. No marker move. Machine-checked: [verification/v845_cfin_normal_form.py]. **[E]** (the assembled object and its counting corollaries) + **[C]** (the typed P2 residual, per v799)

**The Arf–vacuum syndrome and the ovoid decoder (ARF.VACUUM.SYNDROME.01 + OVOID.DECODER.01).** One module from two probes (24/24 + 23/23, verdicts ARF-VACUUM-SYNDROME-EXACT and OVOID-DECODER-EXACT, <1 s, exact $\mathbb{F}_2/\mathbb{Z}/\mathbb{Q}$ algebra). The Arf bit is position-independent ($a(t+U) = \hbar(u,v)$ on all 140 flats); the two-bit syndrome table splits the flats exactly into 15/20/45/60; the code pencil $[15,11,3] \supset \{[15,10,4], 2\times[15,10,3]\} \supset [15,9,4]$ carries the canonicity lemma (one Arf functional on $H$) and the vacuum bit is the $\mathrm{RM}(2,4)$ parity bit. The doily kernel gains its integral inverse: $\mathrm{SNF}(N) = \mathrm{diag}(1^{10}, 0^5)$ torsion-free, $N^+ = N^{\mathsf{T}}/4 - J/36$ (Penrose exact in $\mathbb{Q}$), $P_5 = I/2 - B/4 + J/12$, the decoder exact, the kernel the $S_6$ standard representation (11/11 classes), and the three-ring correspondence ($\mathbb{R}$ ovoid span / $\mathbb{Z}$ torsion-free / $\mathbb{F}_2$ code $A_q = [15,5,6]$). Controls fire. No marker move. Machine-checked: [verification/v852_arf_vacuum_ovoid.py]. **[E]** (exact identities and set equalities throughout)

**The simplex-Fourier character theorem and the winding quadratic (E8.SIMPLEX.FOURIER.01 + FLAVOR.WINDING.QUADRATIC.01).** One module from two probes (26/26 + 20/20, zero fails, verdicts SIMPLEX-FOURIER-EXACT + INTERTWINER-IDENTIFIED and WINDING-QUADRATIC-EXACT; exact integer/Fraction/sympy). The census $240 = 15 \times 16$ is a Fourier law ($\hat{r} = (240, -16^{15})$; $P = (J-I)/15$, spectrum $\{1, (-1/15)^{15}\}$). The character-law extension to all odd shells is retained in E8.SHELL.GLOBAL.01 (the v875 card). The winding line is one adjugate quadratic $q_{\mathrm{wind}} = t^2 - g_{\mathrm{car}}t + |\mathbb{Z}_2|$; triple lock $s = 6$; $\Delta(6) = 39200 = 2 \cdot 140^2$; threshold $s^* \approx 2.8250$. Seven controls fire, two invariances hold. No marker move, no RH claim. Machine-checked: [verification/v857_simplex_fourier_winding.py]. **[E]** (exact identities and censuses throughout)

**The G31 clock alphabet (E8.G31.CLOCK_ALPHABET.01).** One module from one probe (19/19, zero fails, verdict CLOCK-ALPHABET-SELECTIVE). $\mathrm{Deg}(G_{31})/4 = \{2,3,5,6\} = \{|Z_2|, N_{\mathrm{fam}}, g_{\mathrm{car}}, |R^+(A_3)|\}$ as a normal form (gcd $4 = |\mu_4|$, lcm $120 = |R^+(E_8)|$, lcm/gcd $30 = h(E_8)$; $16 \times 15 = 240$ realized on the roots); $(8,12,20,24)$ the unique product-46080/sum-64 quadruple; the 607-group rank-4 kill scan leaves $G_{31}$ the sole full-battery passer (impostors typed by name; honesty line: degree-only audits cannot see the reducible torus); the $W(D_5) \times W(A_3)$ fence re-killed by computed centers 4 vs 1 (v634 cited). Controls fire. No marker move, no RH claim. Machine-checked: [verification/v858_g31_clock_alphabet.py]. **[E]** (exact group closure, censuses, computed centers)

**The feedback normal form and the winding decoder (FLAVOR.FEEDBACK.NORMALFORM.01 + E8.WINDING.DECODER.01).** Two exact compiler theorems from one probe pair (14/14, FEEDBACK-NORMALFORM-EXACT; 11/11, WINDING-DECODER-EXACT; all decisions exact sympy/integer). The

basis $P = [\,2\cdot\mathbf{1} \mid e_3 \mid Re_3\,]$ (det $-4$, SNF diag$(1,2,2)$) conjugates the whole winding line, symbolic in $s$, to the gain-plus-companion form $[[s{+}4,0,3],[0,0,-2],[4,1,5]]$: gain channel $s+|\mu_4|$, companion block of $q_{\text{wind}}$, coupling pair $(3,4)$ with loop product $12 = \dim g_{\text{SM}}$; $\chi_{R_s} = (t-s-|\mu_4|)\,q_{\text{wind}} - 12t$; the $(15,40,20)$ coefficients at $s=6$ are the gain-10 readouts. All three sibling-direction controls fire. The honest typing load-bearing: $\mathbb{Z}^3/P\mathbb{Z}^3 = \mathbb{Z}_2^2$ — the cardinality of $\mu_4$, *not* the group. The decoder: $q_{\text{wind}}(\{2,3,5,6\}) = (-4,-4,2,8)$ with $q(2)=q(3)$ forced by the palindrome axis; $D(y) = (y{+}4)^2(y{-}2)(y{-}8)$ with budget coefficients $(2,48,32,256)$; the bridge $|\mu_4|\,q(|Z_2|)/|R(E_8)| = -1/15$ hits the Walsh message eigenvalue exactly. Typed: exact *audit* theorem, not a functor. Lean mirror `FlavorFeedback.lean` (11 theorems, standard axioms). No marker move, no RH claim. Machine-checked: [`verification/v874_feedback_decoder_lean.py`]. **[E]** (both theorems, the controls, the Lean mirror) — the $\mu_4$ reading *not* claimed (typed honestly)

**The global Gaussian shell theorem (`E8.SHELL.GLOBAL.01`).** One probe (15/15, SHELL-GLOBAL-THEOREM): the $-1/15$ census law is a theorem for *all* shells — $\Theta_v(n) = (\Theta_L(n) - \Theta_0(n))/15$ for $v \neq 0$ at every $n$, $\hat{\Theta}_u(2m{+}1) = -16\,\sigma_3(2m{+}1)$, ratio $-1/15$ exactly at every odd $n$. Inputs warded: $\mathrm{Sp}(4,2)$ transitivity (the 60 transvections; identical to the shell-module action), $(1{+}i)$ doubling, $\Theta_L = 240\,\sigma_3$ (Eisenstein $E_4$; the 16500 census retained as ward). Controls: a non-transitive subgroup breaks the law (Walsh multiset $\{-48^4, -16^7, 16^4\}$); the even-shell formula differs correctly (7/135 at $n=2$). Consequence chain kernel-checked from the named hypothesis bundle (`GaussianShells.lean`, 7 theorems, no division anywhere). No marker move, no RH claim. Machine-checked: [`verification/v875_shell_global_lean.py`]. **[E]** (the theorem, the wards, the controls, the Lean skeleton — inputs named with typed provenance)

**The ledger integrity guard (`GATE.LEDGER.01`).** A CI sentinel, no new physics (10/10): round 36 found and fixed the ledger's first primary-key violation (`FORM.QGEO.03` double-booked 2026-06-15/18; the SeamDeckClosure row renamed `FORM.QGEO.04` with a dated QA note, every flat-closure reference updated). The guard enforces: the 10-column schema; globally unique claim ids (the fix pinned); existing supersession targets with an acyclic supersession graph; unique active supersession targets modulo a frozen remove-only allowlist (two legitimate 2026-06 joint closures); and the dependency cross-reference census frozen remove-only (13 SCCs / 265 ids — acyclicity enforced as *no new cycles*). Machine-checked: [`verification/v878_ledger_integrity.py`]. **[E]** (CSV integrity, frozen baselines remove-only)

**The four finite anchor theorems (`E8.FINITE.ANCHOR.01`).** One probe (26/26, FINITE-ANCHOR-CLOSED; embedded byte-exact). The census-free normal form: $q^*(x) = \sum_{i<j} x_i x_j + \sum_i x_i = \binom{|x|+1}{2} \bmod 2$ (weight-only, value table $(0,1,1,0,0)$); the `v845` selector *forces* the linear part $(1,1,1,1)$ — the $2^{16}$ search replaced by a theorem. The exclusive classification of the $PG(3,2)$ lines: all 15 doily lines carry weight multiset $\{2,1,1\}$ = the anchor $(1,1,2)$, per-line charpoly $(t-2)(t-1)^2 = t^3 - 4t^2 + 5t - 2$, Vieta $(e_1,e_2,e_3) = (4,5,2)$ with $e_2 = g_{\text{car}}$, $e_3 = |Z_2|$; the 20 non-isotropic lines split exactly 10 secants (= the *wrong* anchor $(1,2,2)$, Vieta $(5,8,4)$) + 10 external ($\{1,1,1\}$); budget $15 = M_0$, $30 = M_1 = h(E_8)$ (counting only). The doily *is* the multiplication table $\Lambda^2 E \times \Lambda^2 E \to \Lambda^4 E$ of the five-slot carrier (support partition $\{i\} \cup \{j,k\} \cup \{l,m\}$ per line, $b_{jk} \wedge b_{lm} = \pm f_i$ exact, bijection onto all $5 \times 3 = 15$ configurations, wedge nonzero *exactly* on doily lines). The exhaustion: $\Lambda^2 \wedge \Lambda^2 = \Lambda^{g-1}$ forces $g_{\text{car}} = 5$ uniquely among odd $g \in \{3..11\}$. Anchor–Vieta report: $p_2(a) = 6$, $\chi_L - \chi_R = -p_2(a)(t^2 - e_2 t + e_3)$, $M_n = 15(1, e_3, e_1, e_1 e_3, e_1^2)$. No marker move, no RH claim. Machine-checked: [`verification/v880_finite_anchor_theorems.py`]. **[E]** (the four theorems — exact) + **[O]** (the physical Calderón premise of the exhaustion — P2 narrowed, not eliminated)

**The finite Pfaffian/$K_6$ signature (`DOILY.PLUCKER.GR25.01`; `CFIN.AUT.FLAVOR.C6.01`; `DOILY.PFAFFIAN.WICK.01`).** Three probes (27/27 + 19/19 + 42/42; S6PLUCKER-CLOSED, C6FLAVOR-MEASURED + INTERTWINED, K6PFAFFIAN-CLOSED; embedded byte-exact; exact integer/Fraction/sympy in every decision). The 15 doily lines are the 15 Plücker monomials of $Gr(2,5)$ with perfect sign match ($(+,-,+)$ = the `v880` Hodge table); the $S_6$ Specht channels act by $(1,0,2/3)$ exactly; the $q^*$ zero set is a $(16,6,2)$ Hadamard difference set with anchor signature $(16,6,2,10)$; the vacuum selection: $\mathrm{Stab}(q^*) = 120$ acts as $S_5$ with message

orbits $[1,5,10] = (\Lambda^0, \Lambda^4, \Lambda^2)$; the flavor hexagon is one $\mathrm{Aut}(C_{\mathrm{fin}}) \cong C_6$ orbit ($g^2 = \sigma$ exact; $(2/3)^6 = 64/729$ the frozen recovery value; 0/600 control); the $K_6$ motif theorem (35 = 15 matchings +20 triangles; $q^*$ = vacuum incidence); the Pfaffian signs gauge-matched via the canonical $\chi(i) = (-1)^{i+1}$ (triple-identified: Laplace/volume-form/$C_6$); the self-hosting counting theorem $\binom{g+1}{2} = g!!$ forcing $g = 5$ uniquely nontrivial with $(g-2)!! = 3 = N_{\mathrm{fam}}$; the mod-30 clock CLOCK-DISTINGUISHED. The counting theorem is [E]; the Wick-compiler premise stays [O] — P2 narrowed to one premise; the operative-Coxeter-30 reading separately dead (sandbox). No marker move, no RH claim. Machine-checked: [verification/v888_finite_pfaffian_signature.py]. **[E]** (the bijections, the sign theorem, the counting theorem — exact) + **[O]** (the Wick-compiler premise — the one remaining P2 premise)

**The Wick functor arc (SEAM.CFIN.WICKFUNCTOR.01 [O]).** Three probes (27/27 + 33/33 + 26/26; WICKFUNCTOR-PARTIAL, WICKMIX-MEASURED (WICKMIX-OBSTRUCTED), WICKBLOCK-MEASURED (BLOCKFUNCTOR-CONSTRUCTED / SEAM-DIAGONAL); embedded byte-exact; exact integer/rational/symbolic in every decision typed PROVED or SYMBOLIC). The six compiler roles exist in the deployed seam ($D_5$ slot pairs + the $A_3$ boundary block; $\mathfrak{so}(16)$ sector law; $C_6$ intertwining exact) but the deployed vacuum kernel is channel-diagonal; the scalar obstruction is *proved*: the $C_6$ 2-cycle $\{4,5\}$ forces a duad zero — the scalar route reaches at most 14/15 duads (isotypic multiplicities $[3,0,1,1,1,0]$ re-derived by hand and by exact root-of-unity sums); the block functor is constructed exactly (commutant dimension 33; 15/15 cross-blocks; 15/15 monomials with the canonical $\chi(i) = (-1)^{i+1}$ sign law; interleaving character +1 symbolic); and the deployed seam is SEAM-DIAGONAL (zero cross-blocks) — the physical channel-mixing mechanism is the demand; the [O] premise unmoved. *Candidate note (round CXV):* the demanded state now has a constructed candidate ([verification/v898_kms_schur_mixing.py], next card); the physical realization remains the demand. No marker move, no RH claim. Machine-checked: [verification/v896_wick_block_functor.py]. **[E]** (the obstruction theorem and the block construction — exact) + **[O]** (the physical mixing mechanism — the one remaining premise)

**The channel-mixing candidate state (SEAM.CFIN.KMSMIX.01 + SEAM.CFIN.SCHURMIX.01).** One probe (30/30; KMSMIX-MEASURED [KMSMIX-FOUND(U=1, T=1/8, BETA=1), SCHURMIX-GENERATES(10/10), RP-THETA-OPEN]; embedded byte-exact). The $C_6$-covariant mixing KMS state exists (all 15 blocks at the floor, grading 5/5 + 10/10, canonical $\mathrm{Pf}_4$ signs; robust at $\beta = 2$); the exact sympy Schur identity $A_{\mathrm{eff}} = \kappa A_{CC} + \frac{t^2 m}{1-m^2} V J_3 V^{\mathsf{T}}$ generates all 10 carrier duads from the bare diagonal state — on both fronts the real correlation lives in the eliminated boundary; the frozen 15-signature table is the registered prediction; RP-THETA-OPEN is the named open check (dated note 2026-08-10: measured exclusionary at family level — see the seam RP/modular exclusion card below, [verification/v903_seam_rp_exclusion.py]). No marker move, no RH claim. Machine-checked: [verification/v898_kms_schur_mixing.py]. **[E]** (the commutant walks and the Schur generation identity — exact) + (MEASURED) (the KMS gates and the frozen signatures) + **[O]** (the physical realization — the premise unmoved)

**The seam RP/modular exclusion and the two dead readings (SEAM.STATE.DERIVATION.01; SEAM.CFIN.TWISTED.RP.01; SEAM.CFIN.RP.DILATION.01; SEAM.CFIN.GAP.PENCIL.01; SEAM.CFIN.MIXING.NORMALIZATION.01; SEAM.CFIN.MINIMAL.MEDIATOR.01).** Six probes (25/25 + 23/23 + 20/20 + 20/20 + 23/23 + 18/18; SEAMSTATE-MEASURED, SEAMNORM-MEASURED, MEDIATOR-MEASURED, TWISTEDRP-MEASURED, DILATION-SPLIT, GAPPENCIL-MEASURED; embedded byte-exact). Strict reflection positivity forces $t = 0$ under both deployable spatial reflections (OS positivity $\iff$ the v898 mixing floor falls: mutually exclusive on this family); $u \geq t$ forced; the $2\pi$-KMS locus is the point ($t$=0, $\beta$=$2\pi$); the complete twisted census is exclusionary (0/6); the dilation route splits (marginal cone-boundary witness with the exact identity $t^2 \cdot 3m/(1-m^2) = 1/200$; strict collar obstructed by an exact linear $2t$ law); the gap pencil is exact ($\mathrm{Pf} = -(t-1)(3t^2-1)(9t^3+21t^2-t-1)$, Sturm uniqueness on $(0, 1/4]$, kernel dimension exactly 2; no root is 1/8). The two dead readings booked as first-class negatives: '$t = 1/8$ is a compiler value' is a decoration (PIN-ABSENT; allowed set unbounded); '$N_{\mathrm{fam}} = 3$ as minimal mediation

rank' is refuted (rank 1 suffices; deployed rank exactly 2; purity = dimension bookkeeping). The RP-vs-mixing exclusivity is since typed *ray- and reflection-relative*: on the full 33-dimensional covariant mixing census $\theta_{ab\mathsf{T}}$ kills exactly one coordinate while $\theta_S$ leaves a 17-dimensional neutral subspace with strict witnesses ([verification/v972_seam_interaction_front.py], toy level). No marker move, no RH claim. Machine-checked: [verification/v903_seam_rp_exclusion.py]. **[E]** (the $\pm|a_J|$, Schur, Pfaffian and Sturm identities — exact) + (MEASURED) (the censuses, the exclusion and the two registered negatives) + **[O]** (the physical realization — the premise unmoved)

**The seam equilibrium wiring (SEAM.CFIN.MARGINAL.STABILITY.01; SEAM.CFIN.NESS.PARENT.01, closed not-needed).** Two probes (22/22 + 25/25; MARGSTAB-MEASURED (INTERIOR-EMPTY / FLOOR-EXCHANGE), NESSPARENT-MEASURED (PRICE-ZERO / TWO-SEAT-LAW / FINITE-NESS-NOGO / NESS-NOT-FORCED); exact arithmetic; embedded byte-exact). Strict 2-cycle reflection positivity is impossible on the whole $C_6$-covariant class (the seat law reduces exactly to $\mathrm{diag}(a_J, -a_J)$; the only escape is covariance-breaking at exact linear price $2\varepsilon$; floor exchange $\mathrm{Pf}_4(\varepsilon) = (\varepsilon - \frac{1}{200})(\varepsilon + \frac{1}{200})$ — crossover exactly at the 1/200 floor); the strict-collar obstruction is a two-seat linear law with kernel $\{J, Z\}$, the deployed wiring is PURE-I (maximally obstructed *in the deployed $\theta_S$ collar frame* — the frame qualifier of rounds 69–71), and *equilibrium* kernel witnesses carry the full 1/200 mixing at zero entropy production — no NESS is needed (stationary $\Rightarrow \sigma \equiv 0$ exactly; drive buys no RP); the registered successor contract — is PURE-I compiler-forced? (SEAM.STATE.WIRING.SELECTOR.01) — is closed in the next block. The v898 [O] premise unmoved. No marker move, no RH claim. Machine-checked: [verification/v908_seam_equilibrium_wiring.py]. **[E]** (the seat laws, witnesses and floor-exchange identities — exact) + (MEASURED) (the stratum and drive censuses) + **[O]** (the physical realization — the premise unmoved)

**The wiring freedom theorem (SEAM.STATE.WIRING.SELECTOR.01, closed: deployment choice; SEAM.STATE.THETA.SELECTOR.01; SEAM.STATE.WIRING.GAUGE.RP.01).** Three probes (30/30 + 26/26 + 21/21; WIRING-DEGENERATE, THETA-CONVENTIONAL, RP-FRAME-COVARIANT; exact integer/rational sympy, embedded byte-exact). The v908-registered successor contract is answered as a *freedom theorem*: the exact Gröbner census of the orbit/edge constraint classes leaves exactly *one* admissible three-dimensional wiring component modulo the rule gauge (the rotation cell); $Z/X$ wirings are excluded gauge-invariantly by orientation propagation ($\det u < 0$ identically); PURE-I is an *interior* point, connected to the pure-$J$ ray by an integer rule-gauge element — a deployment representative, *not* a compiler theorem. No compiler-side demand distinguishes the $\theta_S$ sheet-swap frame within the 9-dim rule-gauge algebra (five demand classes, all silent on the angle), and the strict-collar RP Grams transform exactly covariantly under the rule gauge: RP of a (frame, wiring) pair depends only on the relative angle — the deployed parent passes strict collar RP *exactly* in the integer-conjugated frame, so the rounds-58/59 obstruction is a frame statement, not a wiring no-go. Canonical sentence frozen; "PURE-I is compiler-forced" is forbidden. Honest scope: five enumerated demand classes; a demand outside the list is not excluded; the physical collar frame remains underived; the v898/v903 [O] premise unmoved. No marker move, no RH claim. Machine-checked: [verification/v911_wiring_freedom.py]. **[E]** (the census, gauge and covariance identities — exact) + (MEASURED) (the float wards and spectra) + **[O]** (the physical collar frame — underived)

**The deck-sector trace identity (QGEO.DECKSECTOR.01, scoped physical lift).** For the specified 48-site antiperiodic lift, the three deck charges are $\{1/6, 1/2, 5/6\}$, with 16 modes in each sector. On the selected half-turn character $m \equiv 1 \pmod 4$, its three tower channels are the explicit geometric series

$$T_r(t) = \sum_{k\geq 0} e^{-(6k+r/2)t} = \frac{e^{-rt/2}}{1-e^{-6t}}, \qquad r \in \{1,5,9\}, \quad t > 0, \qquad \sum_r T_r(t) = \frac{e^{-t/2}}{1-e^{-2t}}.$$

The common normalization is 1, fixed by the large-$t$ asymptote. This is the elementary trace identity of this specified lift. Selecting that half-turn character, identifying a physical response and controlling the continuum limit remain separate hypotheses; the identity alone supplies none

of those identifications. The retained module is scoped to this NS-circle and deck-series algebra. [`verification/v705_chain_deck_sector.py`]. **[E]** (specified-lift trace algebra).

## 6 Further audit lemmas (machine-verified this round)

Five more exact identities tie the flavor matrices and the $E_8$ exponents to the compiler. All checked with exact integer arithmetic.

**(i) Full-length Frobenius norm is three $E_6$ blocks.** $\|L\|_F^2 = 234 = N_{\text{fam}}\,\|R\|_F^2 = 3\dim E_6$, and $\|L\|_F^2 = \|R\|_F^2 + 12\langle R, W\rangle + 36\|W\|_F^2 = 78+48+108$, with $\langle R, W\rangle = 4 = |\mu_4|$ and $\|W\|_F^2 = 3 = N_{\text{fam}}$. The full word-length load is exactly the residue $E_6$ norm + the $A_3$-winding over $\mu_4$ + the family-winding square. **[E]**

**(ii) $A_3$-winding determinant lemma.** The rank-one winding $L = R + 6\,\mathbf{1}e_1^\top$ and the matrix-determinant lemma with $e_1^\top R^{-1}\mathbf{1} = \frac{1}{4}$ give $\det L = \det R(1 + \frac{|R^+(A_3)|}{|\mu_4|}) = 8(1+\frac{6}{4}) = 20 = 2\mathcal{A}_\Lambda$. So $h(D_5) = 8 \xrightarrow{A_3^+/\mu_4} 20 = |R(A_4)|$: the winding injects the carrier factor 5 into the cokernel ($\operatorname{coker} R \simeq \mathbb{Z}_8$, $\operatorname{coker} L \simeq \mathbb{Z}_{20}$). **[E]**

**(iii) The scalaron 7 is the $E_8$ exponent variance per decuple.** Centring the exponents at $h/2 = 15$, $\sum_m (m-15)^2 = 560$ and $\text{Var} = \frac{560}{8} = 70 = 7\cdot\mathcal{A}_\Lambda =$ (scalaron 7) $\cdot$ (decuple 10). The halved positive offsets $\{1,2,4,7\}$ satisfy $\sum = 14 = 2\cdot 7$, $\sum^2 = 70$, $\prod = 56 = \dim \mathbf{56}_{E_7}$ — a micro-code carrying the scalaron, the decuple and the $E_7$ minuscule at once. Also $\sum_{\text{pairs}} m(30-m) = 620 = \frac{h\dim E_8}{12} = 2\cdot 10\cdot 31$. **[E]**

**(iv) Exponents + degrees give the 120+128 split.** The invariant degrees $d_i = m_i + 1 = \{2,8,12,14,18,20,24,30\}$ obey $\sum_i m_i = 120$, $\sum_i d_i = 128, 248 = 120+128$, $\prod_i d_i = |W(E_8)|$, matching the $D_8$ branching $248 = 120 + 128$; the degrees are a TFPT dictionary $(2, 8, 12, 14, 18, 20, 24, 30$ = sheet, $h(D_5)$, $|R(A_3)|$, $\dim G_2$, $p_4$, $\det L$, $\dim A_4$, $h(E_8))$. **[E]**

**(v) Root shell = local $E_7 \times A_1$.** Around any $E_8$ root the inner-product multiplicities are $\{+2{:}1,\ +1{:}56,\ 0{:}126,\ -1{:}56,\ -2{:}1\}$, i.e. the local $248 = (133,1)+(1,3)+(56,2)$. A distinguished root (seam/Higgs direction) leaves an orthogonal $E_7$ and an active coupling shell 56+56. **[E]**

## 7 What this changes, and what stays open

$E_8$ is one container whose maximal subalgebras each project a different TFPT module:

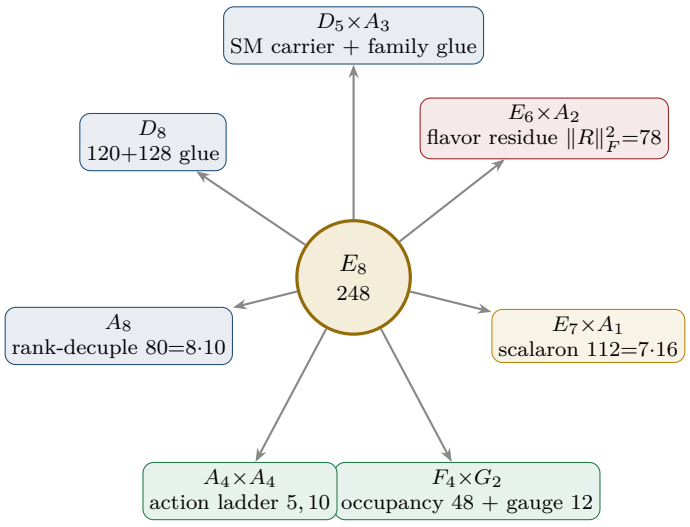



Figure 3: $E_8$ as an audit container: each maximal subalgebra projects one TFPT module. The discipline rule — every load-bearing number appears in $\geq 1$ projection — makes the structure falsifiable.

**Net.** The atlas turns isolated fingerprints (e.g. $\|R\|_F^2 = 78 = \dim E_6$, the scalaron 7) into *branching projections* of one container. The discipline rule — "every load-bearing number must appear in $\geq 1$ projection" — makes the structure falsifiable: a number in no projection is suspect. This is a *program*, not a proof of new physics: all readings are [O]. The two genuinely load-bearing, derived results remain the $\mu_4$ glue ($E_8 = (D_5 \oplus A_3) + \mu_4$, [E] [verification/v1_e8_glue.py]) and the spectral selector of the flavor matrix ($\det R = h(D_5)$, $\mathrm{PrinMin}_2 = (2, 3, 5)$, [E] [verification/v4_flavor_matrix.py]). The remaining flavor-geometry step (historically "H2") is sharpened in the parabolic note to: *the $D_4$-equivariant stable parabolic structure on $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ realises the unique branch with* $\det R = 8$ *and* $\mathrm{PrinMin}_2(R) = (2, 3, 5)$ — no longer "find the matrix", but "prove this one geometric realisation". (Status: since v118–v122 the dictionary *values* are exact theorems — hexagon = sign-twisted family spectrum, addresses pinned, margins forced; what remains is exactly the realisation statement above, the selector establishment R4′.)

# Part II
# The $E_8$ cascade bridge

**The connection in one sentence.** The old $E_8$ cascade (Paper V1.06: the nilpotent-orbit chain $D_n = 60 - 2n$, $n = 0 \dots 26$) and the new compiler/atlas are *the same even-integer $E_8$ spine*, read three ways: the cascade *orders* the numbers $\{8, 10, \dots, 60\}$ as a scale ladder, the atlas *reads* them as $E_8$ subalgebra branchings, and the compiler *generates* them from the $(1, 1, 2)$ anchor. The cascade *starts* at $D_{\text{start}} = 60 = p_2 p_3$ and *ends* at $8 = h(D_5) = \det R$ — the flavor selector.

## 8 The old cascade, recalled

Paper V1.06 builds a deterministic scale ladder from the nilpotent orbits of $E_8$. With centralizer dimension $D = 248 - \dim \mathcal{O}$ and the Hasse chain restricted to $\Delta D = 2$, a unique maximal chain runs from $A_4{+}A_1$ ($D_0 = 60$) down to the regular orbit ($D_{26} = 8$):

$$\boxed{D_n = 60 - 2n, \quad n = 0, \dots, 26; \qquad \varphi_{n+1} = \varphi_n e^{-\gamma(n)}, \quad \gamma(n) = \lambda \left[\ln D_n - \ln D_{n+1}\right]}$$

The cascade arithmetic (endpoints, exponent rungs, IR-tail product, variance) is machine-checked. [verification/v5_e8_cascade.py] with the single structural normalisation $\lambda = \gamma(0)/(\ln 248 - \ln 60) = 0.58770$, fixed by a curvature-extremum on the chain (no fit). The $E$-windows of the two-loop flow anchor it: $\alpha_3 = 1/(8\pi) = c_3$ (the $E_8$ window) and $\alpha_3 = 1/(6\pi)$ (the $E_6$ window, near $\varphi_0^{\text{ret}}$).

## 9 The spine: every rung is a current number

Reading $D = 60 - 2n$ against today's compiler atoms, the secondary-branching atlas and the $(1, 1, 2)$ anchor power sums $p_n = 2 + 2^n$, **25 of the 27 rungs are load-bearing numbers** (machine-checked):

| $n$ | $D$ | **current meaning** | $n$ | $D$ | **current meaning** |
|---|---|---|---|---|---|
| 0 | 60 | $D_{\text{start}} = p_2 p_3 = \mathcal{A}_\Lambda \lvert R^+(A_3)\rvert$ **(start)** | 13 | 34 | $p_5$ anchor power $= 2 \cdot 17$ |
| 1 | 58 | $2 \cdot 29$ ($E_8$ exponent) | 14 | 32 | $2^5 = 2 \dim S^+$ |
| 2 | 56 | $\dim \mathbf{56}_{E_7} = a^\top L \mathbf{1} = \sum \lvert m{-}15\rvert$ | 15 | 30 | $h(E_8) = 2{\cdot}3{\cdot}5$ **(Coxeter rung)** |
| 3 | 54 | $2 \cdot 27$, $27 = \mathbf{1}^\top R a = 3^3$ | 16 | 28 | $4 \cdot 7$ (half of 56) |
| 4 | 52 | $\dim F_4 = \lvert R(D_5)\rvert + \lvert R(A_3)\rvert$ | 17 | 26 | $2 \cdot 13$ (old hadronic block) |
| 5 | 50 | $A_4 \times A_4$ block $= 5{\cdot}10$ | 18 | 24 | $\dim A_4$; 24+24=48 |
| 6 | 48 | $\Omega_{\text{adm}} = \lvert R(F_4)\rvert = 2 p_1 p_2$ | 19 | 22 | $\sum R = \mathbf{1}^\top R \mathbf{1}$ (residue sum) |
| 7 | 46 | $2 \cdot 23$ ($E_8$ exponent) | 20 | 20 | $\det L = \lvert R(A_4)\rvert = 2\mathcal{A}_\Lambda$ |
| 8 | 44 | $4 \cdot 11$ ($E_8$ exponent) | 21 | 18 | $p_4 = N_{\text{fam}} \lvert R^+(A_3)\rvert$ |
| 9 | 42 | $6 \cdot 7 = \lvert R^+(A_3)\rvert \cdot 7$ | 22 | 16 | $\dim S^+$ (one generation) |
| 10 | 40 | $\lvert R(D_5)\rvert = \sum L = \mathbf{1}^\top L \mathbf{1}$ | 23 | 14 | $\dim G_2 = \sum L_d = 8{+}6$ |
| 11 | 38 | $2 \cdot 19$ ($E_8$ exponent) | 24 | 12 | $\lvert R(A_3)\rvert = \dim \mathfrak{g}_{\text{SM}}$ |
| 12 | 36 | $6^2$ (old EW block) | 25 | 10 | $\mathcal{A}_\Lambda = \binom{5}{2}$ (old cosmo block) |
| | | | 26 | 8 | $h(D_5) = \det R = 2p_1$ **(end: flavor selector)** |

Three structural facts jump out (all **[E]** arithmetic):

- **Endpoints are the compiler boundary.** The cascade *starts* at $60 = p_2 p_3 = D_{\text{start}}$ and *ends* at $8 = h(D_5) = \det R$ — the flavor spectral selector is literally the deepest orbit of $E_8$.
- **Coxeter rung is $h(E_8)$.** At $n = 15$, $D = 30 = h(E_8)$ — this is the *Coxeter rung*, not the arithmetic centre; the median node is $n = 13$, $D = 34 = p_5(a)$ (below). It matches the exponent-centering audit $\sum_m \lvert m - 15\rvert = 56 = \dim \mathbf{56}_{E_7}$ (which is itself rung $n = 2$).
- **Doubled $E_8$ exponents are rungs.** $\{58, 46, 38, 34, 26, 22, 14\} = 2 \times \{29, 23, 19, 17, 13, 11, 7\}$ are exactly the cascade rungs at $n = 1, 7, 11, 13, 17, 19, 23$; three of them coincide with anchor

numbers ($34 = p_5$, $22 = \sum R$, $14 = \dim G_2$). Only $n = 8$ ($D = 44 = 4 \cdot 11$) is not independently labelled.

**Cascade arithmetic — endpoints, centre and length (verified)**

The 27 rungs encode $E_8$ in their endpoints, and their centre is the anchor:

$$\tfrac{D_{\text{start}}\,D_{\text{end}}}{2} = \tfrac{60\cdot 8}{2} = 240 = |R(E_8)|, \qquad 240 + D_{\text{end}} = 248 = \dim E_8, \qquad D_{\text{start}} = 2\,h(E_8) = 60;$$
$$\text{arithmetic median} = \tfrac{60+8}{2} = 34 = p_5(a) = h(E_8) + |\mu_4|, \qquad \textstyle\sum_{n=0}^{26}(D_n - 30) = 27 \cdot 4 = 27|\mu_4|;$$
$$\text{width} = D_{\text{start}} - D_{\text{end}} = 52 = \dim F_4, \qquad 60 \to 30 = h(E_8), \quad 30 \to 8 = \textstyle\sum R = 22;$$
$$\#\text{nodes} = 27 = 3^3 = \mathbf{1}^\top R\, a \ (E_6 \text{ cubic block}), \qquad \#\text{steps} = 26 = \dim \mathbf{26}_{F_4}.$$

So the cascade is an $E_8$ Coxeter ladder centred on $p_5(a)$ with a per-node $|\mu_4|$ glue shift; its length (27 nodes, 26 steps) carries the $E_6$ cube and the $F_4$ fundamental. **[E]**

**The IR tail $n = 24 \ldots 30$: the compiler atoms**

Continuing $D = 60 - 2n$ past the last orbit ($D$=8, $n$=26) is no longer a nilpotent-orbit chain but the *algebraic compiler tail* of the Coxeter clock:

$$D_{24\ldots30} = \underbrace{12}_{|R(A_3)|=\dim \mathfrak{g}_{\text{SM}}}, \underbrace{10}_{\mathcal{A}_\Lambda}, \underbrace{8}_{h(D_5)}, \underbrace{6}_{|R^+(A_3)|}, \underbrace{4}_{|\mu_4|}, \underbrace{2}_{\text{sheet}}, \underbrace{0}_{\text{closure}}.$$

This is exactly what the old "$n = 0 \ldots 30$ clock" was scanning: $n$=0 … 26 the $E_8$ orbit spine, $n$=27 … 30 the compiler atoms. (Mark the tail as the *compiler tail*, not orbits.) **[E]**

**Three further cascade lemmas (all machine-verified)**

The clock carries more $E_8$ data than the endpoints:

- **Primitive rungs sum to the root count.** Evaluating $D_n = 60 - 2n$ at the *primitive* indices $n \in (\mathbb{Z}/30)^\times = \{1, 7, 11, 13, 17, 19, 23, 29\} = \mathrm{Exp}(E_8)$ gives $\{58, 46, 38, 34, 26, 22, 14, 2\}$ with

$$\sum_{n \in \mathrm{Exp}(E_8)} (60 - 2n) = 240 = |R(E_8)|,$$

  and since $30 - n$ is primitive when $n$ is, the clock indexes its own $E_8$ exponents.
- **The IR tail product is the main Weyl stabiliser.** The nonzero tail $\{12, 10, 8, 6, 4, 2\}$ has

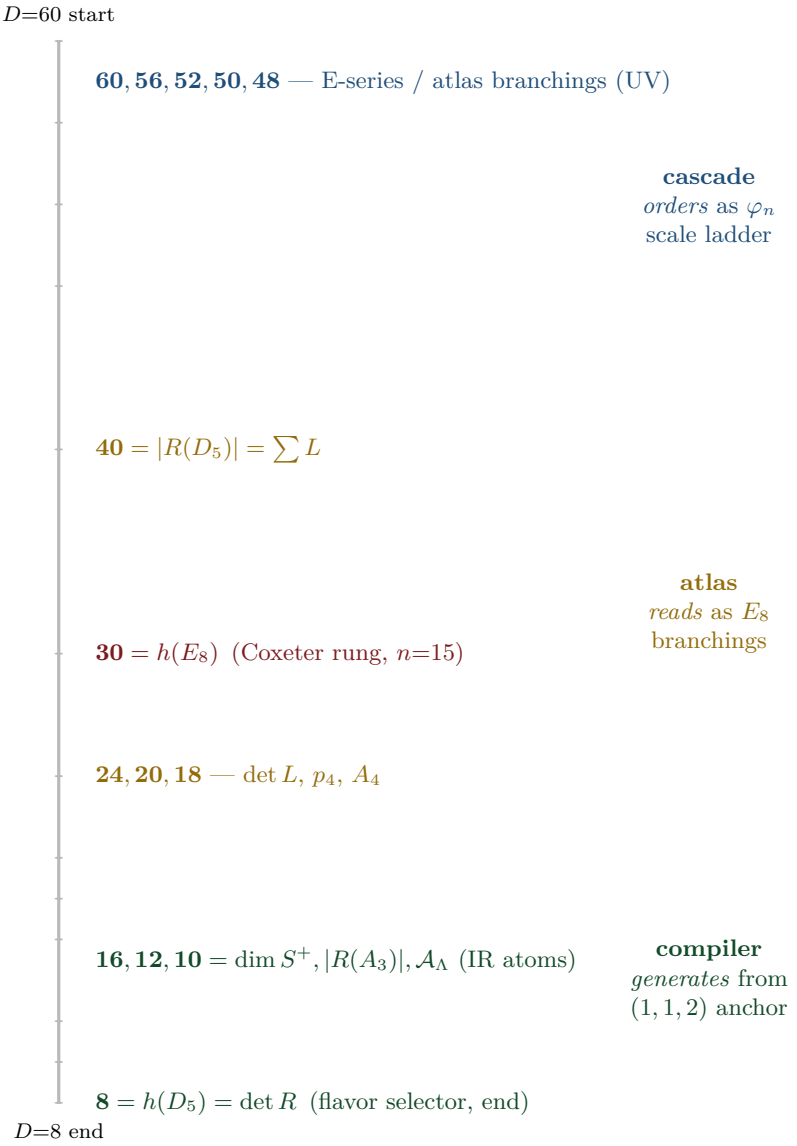
$$\prod = 46080 = |W(D_5)|\,|W(A_3)| = 1920 \cdot 24, \qquad \sum = 42 = |R^+(A_3)| \cdot 7.$$

- **Cascade variance $= E_6$ flavour norm times the scalaron-gauge block.** The 27 rungs $60, \ldots, 8$ have mean $34 = p_5(a)$ and

$$\sum_{n=0}^{26} (D_n - 34)^2 = 6552 = 78 \cdot 84 = \dim E_6 \cdot (7 \dim \mathfrak{g}_{\mathrm{SM}}),$$

  tying the cascade simultaneously to the $E_6 \times A_2$ flavour shadow and the $A_8$ scalaron-gauge shadow. **[E]** (audit level)

# 10 Three views of one spine




**The unification.** The same numbers $\{8, 10, \ldots, 60\}$ are: (a) the *scale ladder* $\varphi_n$ of the old cascade (orbit dimensions, ordered by $e^{-\gamma}$); (b) the *branching blocks* of the secondary-atlas ($\dim F_4$=52, $\dim \mathbf{56}_{E_7}$, 5·10, …); (c) the *anchor power sums and matrix invariants* of the compiler ($p_n = 2 + 2^n$, $\det R$=8, $\sum L$=40). The cascade is the *UV→IR ordering*; the compiler is its *IR endpoint*, where the small numbers 8, 10, 12, 16 are the flavor/family/carrier data.

**The spine tetrahedron: the microkernel as *one* finite object [E]** ([`verification/v91_spine_tetrahedron.py`])

The spine $T = \{2, 3, 4, 5\} = \{|\mathbb{Z}_2|, N_{\text{fam}}, |\mu_4|, g_{\text{car}}\}$ equals $\{e_3(a), p_0(a), e_1(a), e_2(a)\}$ for the anchor $a = (1, 1, 2)$ — the tetrahedron on these four nodes is the anchor plus its zeroth moment, not a second structure. Its combinatorics *are* the central integer grammar: edge products $\{6, 8, 10, 12, 15, 20\}$ ($|R^+(A_3)|$; rank $E_8$=$h(D_5)$= $\det R$; $\mathcal{A}_\Lambda$=$|E(K_5)|$; $|R(A_3)|$= $\dim \mathfrak{g}_{\text{SM}}$; $\dim A_3$; $\det L$), face products $\{24, 30, 40, 60\}$ ($|W(A_3)|$; $h(E_8)$; $|R(D_5)|$= $\sum L$; $D_{\text{start}}$), volume $2{\cdot}3{\cdot}4{\cdot}5 = 120 = |R^+(E_8)| = 5!$; and the graph reading $|R(E_8)| = 240 = |\mu_4| \cdot |E(K_4)| \cdot |E(K_5)|$ with $|E(K_4)| = p_2 = 6$, $|E(K_5)| = p_3 = 10$ — including the honest negative control that the identification *breaks* at $K_6$ ($|E(K_6)| = 15 \neq p_4 = 18$), so it is specific, not generic. Two disciplined caveats: the "dual cuts" (node × opposite face = 120, etc.) are tautologies of any complementary 4-set partition and are typed as presentation only; and the tetrahedron grammar is a *sub*-grammar — the load-bearing 7, 16, 41, 48, 240, 248 are *not* subset products and still require the

anchor power sums / Pascal / glue layers.

## 11 Direct $\varphi_0^{\rm ret}$ relations: old vs current

The old cascade's "fundamental relations near $n = 0$" connect directly to the current note — one is *identical*, two are early forms of current refinements:

| Quantity | old (V1.06) | current | relation |
|---|---|---|---|
| $\Omega_b$ | $\varphi_0^{\rm ret}(1-2c_3) = 0.04894$ | $(1-\frac{1}{4\pi})\varphi_0^{\rm ret} = 0.04894$ | **identical** ($2c_3 = \frac{1}{4\pi}$) [E] |
| $V_{us}/V_{ud}$ | $\sqrt{\varphi_0^{\rm ret}} = 0.2306$ | $\lambda_C = \sqrt{\varphi_0^{\rm ret}(1-\varphi_0^{\rm ret})} = 0.2244$ | refined (the $(1-\varphi_0^{\rm ret})$ factor) [E] |
| $r$ (tensor) | $(\varphi_0^{\rm ret})^2 = 0.00283$ | $12/N_\star^2 = 0.00397$ | two branches, same order [C] |

*Marker key:* ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`.

So $\Omega_b = \varphi_0^{\rm ret}(1-2c_3)$ was already the exact current formula in V1.06; the Cabibbo ratio gained its $(1-\varphi_0^{\rm ret})$ factor; and the old direct $r = (\varphi_0^{\rm ret})^2$ and the current Starobinsky $r = 12/N_\star^2$ agree in order ($\sim 0.003$) but are distinct predictions — a clean, testable discriminant, stated honestly.

## 12 Sector inventory: have all the old-cascade sectors been computed?

The old cascade also carried *calibrated absolute scales*: per block, one unit $\zeta_B$ was fixed on a single reference and every other scale in the block followed fit-free from the ratio law $\varphi_m/\varphi_n = (D_m/D_n)^\lambda$ (Paper V1.06, App. C). These block assignments are the place where the proton mass, the EW masses and the CMB temperature lived. The honest question is: *has the new compiler re-derived each of these, or only some?* The full inventory, with the current status spelled out:

| Old block (rung) | Old observable & calibration | Current-model status | Mark |
|---|---|---|---|
| near $n$=0 | $\Omega_b = \varphi_0^{\rm ret}(1-2c_3)$, $V_{us}/V_{ud} = \sqrt{\varphi_0^{\rm ret}}$, $r = (\varphi_0^{\rm ret})^2$ | **covered** — dimensionless, derived from $\{c_3, \varphi_0^{\rm ret}\}$; see the $\varphi_0^{\rm ret}$-table above. $\Omega_b$ is *identical*. | [E] |
| EW, $n$=12 | $v_H = \zeta_{\rm EW} M_{\rm Pl}\varphi_{12}$, $M_W = \frac{1}{2}g_2 v_H$, $M_Z$ | **scale covered, absolute scheme-level** — the EW scale enters the current note via the action-grammar exponent and the two-loop flow; $M_W, M_Z$ are dimensionful and remain RG/scheme outputs (one $\zeta_{\rm EW}$), *not* new compiler powers. | [C] |
| hadronic, $n$=15, 17 | $\mathbf{m_p}$=$\zeta_p M_{\rm Pl}\varphi_{15}$, $m_b$=$\zeta_b M_{\rm Pl}\varphi_{15}$, $m_u$=$\zeta_u M_{\rm Pl}\varphi_{17}$ | **ratios closed, absolute scale an anchor.** The quark mass *ratios* sit on the $\varphi_0^{\rm ret}$-word-length ladder and are closed (integer Plücker readouts, v49/v71); the *absolute* values need the per-sector $\Lambda/U_f^\star$ normalisation, an anchor ([O]). The **proton mass** is QCD-confinement, $m_p \sim \Lambda_{\rm QCD}$ by dimensional transmutation — a *cross-sector* ratio, listed as a genuine frontier item ($m_p/m_e$, tfpt_4). | [O] |
| cosmo, $n$=25 | $T_{\gamma 0}$=$\zeta_\gamma M_{\rm Pl}\varphi_{25}$, $T_\nu$=$(\frac{4}{11})^{1/3}T_{\gamma 0}$, $\rho_\Lambda$ | **partly.** $\rho_\Lambda/\Lambda$ is *typed* by the action grammar and the seam (Λ-typing, tfpt_2); $T_\nu/T_{\gamma 0}$ is standard QFT (not cascade). The **absolute CMB temperature** $T_{\gamma 0}$ is a downstream readout of the reheating pipeline (the old cosmology/CMB layers; today the scalaron-reheating chain, v86), conditional — not re-derived as a compiler number. | [C] |

*Marker key:* ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`.

**Direct answer: which sectors are closed, which are not**

**Closed (derived from $\{c_3, g_{\rm car}, \varphi_0^{\rm ret}\}$):** the dimensionless near-$n$=0 anchors ($\Omega_b$ exactly, Cabibbo, tensor branch) and all the *dimension/rung* structure of the spine. **Scale-level only (one $\zeta_B$ each, scheme/RG):** the EW masses $M_W, M_Z$ and the cosmological absolute scales $T_{\gamma 0}$ — exactly as in the current SM/cosmology notes; these were *never* pure predictions, not even in V1.06 (each needed a block unit). **Genuinely open:** the **proton mass** $m_p$ (and $m_p/m_e$) is QCD-confinement across two sectors and is *not* claimed as a compiler power — it stays a frontier item ([O], tfpt_4); the quark mass *ratios* are closed (integer Plücker, v49/v71) and only the *absolute* quark scale $m_b, m_u$ needs the per-sector $\Lambda/U_f^\star$ normalisation, an anchor ([O]). So: yes, the *spine* is fully audited and the *dimensionless* sectors are derived; the *dimensionful* hadronic/cosmo absolute scales are honestly still scheme-level, with the proton mass the one explicitly open cross-sector number — nothing is silently claimed.

# 13 What is new, what is old, what stays open

**New structures found via the bridge**

1. The flavor selector $\det R = h(D_5) = 8$ is the *terminal orbit* of the $E_8$ cascade; the compiler is the cascade's IR endpoint. [E]
2. The atlas branching blocks $(52, 56, 60)$ are the cascade's *UV head* ($n = 0, 2, 4$); the secondary atlas and the orbit cascade are the same spine read top-down vs. as subalgebras. [E]
3. The anchor power sums sit on doubled-exponent rungs: $p_5 = 34$ ($n$=13), $\sum R = 22$ ($n$=19), $\dim G_2 = 14$ ($n$=23). [E]
4. The cascade Coxeter rung $D = 30 = h(E_8)$ ($n$=15) ties to the exponent-centering audit $\sum |m-15| = 56$. [E]

**Honest separation**

The *dimension/rung coincidences* are exact arithmetic [E] — they show the old and new constructions live on one $E_8$ spine. The old cascade's *scale assignments* $\varphi_n$ (EW at $n$=12, hadronic at $n$=15, 17, cosmo at $n$=25) are a *different* mechanism from the current $\varphi_0^{\rm ret}$-power mass ladder: the cascade orders *orders of magnitude* (which sector sits where) via $e^{-\gamma}$, while the $\varphi_0^{\rm ret}$-ladder gives *values within* a sector. Both are real, complementary orderings; neither is forced onto the other. The per-sector $\zeta$-calibrated absolute scales (proton mass, EW masses, CMB temperature) remain scheme-level — audited one by one in §12, with the proton mass the single explicitly open cross-sector number.

**Bottom line.** The old $E_8$ cascade was an early, correct intuition: the physically relevant numbers are exactly the even-integer $E_8$ orbit spine $\{8, \ldots, 60\}$. The current work *explains why* — they are the $(D_5 \times A_3) + \mu_4$ compiler atoms and their $E_8$ branching blocks — and identifies the spine's two ends as the compiler boundary ($D_{\rm start} = 60$, $\det R = 8$). Old and new are the same structure at different resolution.

# Part III
# The self-consistency loop

**The question, and the honest answer.** The theory rests on two inputs, $c_3 = \frac{1}{8\pi}$ and $g_{\text{car}} = 5$. The old theory had a Möbius seam and the seed was the *start* of the $E_8$ cascade. Run it *backwards*: can the cascade/$E_8$ *return* $c_3$ and $g_{\text{car}}$, closing a self-consistent loop? **Answer:** the *discrete* content is a closed, overdetermined fixed point — $g_{\text{car}} = 5$ and the "8" in $c_3$ are forced by the $E_8$ closure. The *only* irreducible primitive left is the geometric $\pi$ (the Möbius/boundary Gauss–Bonnet). So "two axioms" collapses to "*one* continuous primitive $\pi$ + *one* discrete self-consistency fixed point."

## 14 The loop for $g_{\text{car}}$ — an overdetermined fixed point

Two *independent* structural conditions, both pure Lie-theory, force $g_{\text{car}} = 5$ once $E_8$ is the closure object and $A_3$ is the four-puncture family geometry:

**$g_{\text{car}} = 5$ is forced twice over**

(A) rank fill.
The glue $D_{g_{\text{car}}} \oplus A_3 \hookrightarrow E_8$ must fill the rank: $\operatorname{rank} D_{g_{\text{car}}} + \operatorname{rank} A_3 = \operatorname{rank} E_8$, i.e. $g_{\text{car}} + 3 = 8 \Rightarrow g_{\text{car}} = 5$.

(B) Coxeter match.
The carrier Coxeter number must equal the $E_8$ rank, $h(D_{g_{\text{car}}}) = 2g_{\text{car}} - 2 = 8 \Rightarrow g_{\text{car}} = 5$.

Both give $g_{\text{car}} = 5$, and the loop *closes*: $g_{\text{car}} = 5 \Rightarrow D_5 \oplus A_3 \xrightarrow{\mu_4} E_8 \Rightarrow h(E_8) = 30 = 2 \cdot 3 \cdot 5 \Rightarrow$ largest prime $= 5 = g_{\text{car}}$. **[E]**

**Reverse glue selection — $D_5$ *and* $A_3$ from unimodular closure**

The closure even *selects the two factors*, not just the rank. The discriminant of $A_{\mu-1}$ is $\mathbb{Z}_\mu$ and of $D_g$ (odd $g$) is $\mathbb{Z}_4$; a common cyclic glue forces $\mu = 4$, hence $A_{\mu-1} = A_3$. The even-glue norm $q(D_g) + q(A_3) = \frac{g}{4} + \frac{3}{4} = 2$ then forces $g = 5$, hence $D_5$. A second, independent confirmation of $\mu = 4$ is the *mirror identity* $\dim D_{\mu+1} + \dim A_{\mu-1} = \dim V_{D_{\mu+1}} \cdot |R^+(A_{\mu-1})|$, i.e. $3\mu(\mu+1) = \mu(\mu+1)(\mu-1) \Rightarrow \mu - 1 = 3 \Rightarrow \mu = 4$ (the $45 + 15 = 10 \cdot 6$ identity). So

$$\boxed{\text{common } \mathbb{Z}_4 \text{ glue } + \text{ root norm } 2 \implies D_5 + A_3}$$

— $g_{\text{car}} = 5$ is recovered from the closure, not just posited. **[E]**

**Familyful $E_8$-closure fixed-point theorem**

*Among rank-filling pairs $D_g \oplus A_m$ $(g + m = 8)$ admitting an integer glue index $\sqrt{\det D_g \det A_m} = \sqrt{4(m+1)}$, exactly two close to an even unimodular lattice: $D_8 \oplus A_0$ (no family) and $D_5 \oplus A_3$ (familyful). The unique familyful solution is $D_5 \oplus A_3$, with lattice $\cong E_8$ and $g_{\text{car}} = 5$.*

| $D_g + A_m$ | $g+m$ | glue idx$^2$=$4(m+1)$ | integer? | status |
|---|---|---|---|---|
| $D_8 + A_0$ | 8 | 4 | 2 | valid, *no family* |
| $D_7 + A_1$ | 8 | 8 | — | no integer glue |
| $D_6 + A_2$ | 8 | 12 | — | no integer glue |
| **$D_5 + A_3$** | 8 | 16 | **4**=$\lvert\mu_4\rvert$ | **valid, familyful,** $E_8$ |
| $D_4 + A_4$ | 8 | 20 | — | no integer glue |

Equivalently, with $q(A_{\mu-1}) = \frac{\mu-1}{\mu}$ and $g = 9 - \mu$ the norm closure $q(D_g) + q(A_{\mu-1}) = 2$ becomes $\mu^2 - 5\mu + 4 = 0$, whose nontrivial root is $\mu = 4$ ($A_3$, $g = 5$). So $g_{\text{car}} = 5$ is forced *three* ways: rank-fill, Coxeter-match, and integer-glue/norm closure. **[E]** [`verification/v6_bootstrap.py`]

**Bootstrap classification: $D_5 \oplus A_3$ is the *unique* familyful $E_8$ glue [E]**

$E_8$ is the unique even unimodular rank-8 lattice. Its two-factor root-lattice gluings $L_1 \oplus L_2$ (rank $= 8$) that admit a *cyclic* glue are exactly those whose discriminant groups are isomorphic and cyclic:

$$D_5 \oplus A_3\ (\mathbb{Z}_4), \quad E_6 \oplus A_2\ (\mathbb{Z}_3), \quad E_7 \oplus A_1\ (\mathbb{Z}_2), \quad A_4 \oplus A_4\ (\mathbb{Z}_5).$$

Imposing the two TFPT data — one factor with a 16-dimensional half-spinor ($\dim S^+_{D_g} = 2^{g-1} = 16 \Leftrightarrow g = 5$) and one factor giving exactly $N_{\text{fam}} = 3$ families ($\operatorname{rank} A_m = m = 3$) — selects

$$\boxed{D_5 \oplus A_3 \text{ (uniquely), glue group } \mathbb{Z}_4 = |\mu_4|}.$$

$E_6 \oplus A_2$ gives only 2 families, $E_7 \oplus A_1$ only 1; $D_8$ has non-cyclic glue $(\mathbb{Z}_2)^2$ and $A_8$ is a single rank-8 factor. So the carrier+family decomposition is forced. [`verification/v15_bootstrap_classification.py`]

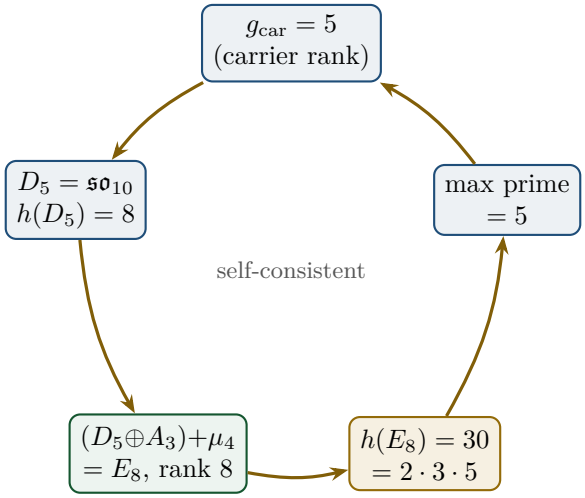



Figure 4: The closed loop: $g_{\text{car}} = 5$ builds $E_8$, whose Coxeter number $30 = 2 \cdot 3 \cdot 5$ returns the carrier prime 5. The value $g_{\text{car}} = 5$ is the unique fixed point (overdetermined by rank-fill *and* Coxeter-match).

The carrier split $3 + 2$ and the one-generation count $\dim S^+ = 2^{g_{\text{car}}-1} = 16$ are then automatic, as is the glue norm $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2$. The family rank is $\operatorname{rank} A_3 = 3 = N_{\text{fam}}$ and the glue index is $h(A_3) = 4 = |\mu_4|$ — the Coxeter numbers of the three players are $(h(A_3), h(D_5), h(E_8)) = (4, 8, 30) = (|\mu_4|,\ \det R,\ \text{compiler})$.

## 15 The loop for $c_3$ — the "8" is the $E_8$ rank

The boundary normalisation is $c_3 = 1/(8\pi)$. Its integer is exactly the carrier Coxeter number, which the loop above equals to the $E_8$ rank:

$$\boxed{c_3 = \frac{1}{h(D_5)\,\pi} = \frac{1}{(2g_{\text{car}} - 2)\,\pi} = \frac{1}{8\pi}, \qquad 8 = h(D_5) = \operatorname{rank} E_8 = \det R = \varphi(30)}$$

where $\varphi(30) = 8$ is Euler's totient (the number of primitive residues mod 30 = the $E_8$ exponents) — so the "8" has *five* concordant readings. The tree seed $\varphi_{\text{tree}} = 1/(6\pi)$ has $6 = |R^+(A_3)| = h(E_8)/g_{\text{car}}$, the three Gauss–Bonnet boundary cycles ($6\pi = 3 \times 2\pi$). So the *discrete* parts of both $c_3$ and $\varphi_{\text{tree}}$ are $E_8$/carrier invariants; only the factor $\pi$ is genuinely continuous.

**Honest caveat on $8\pi$**

$8\pi$ is *also* the standard Einstein normalisation ($G_{\mu\nu} = 8\pi G\, T_{\mu\nu}$). The loop *identifies* this gravitational 8 with rank $E_8 = h(D_5)$ — a non-coincidence *if* $E_8$ is the closure, but one should be clear that $8\pi$ has an independent GR reading. The loop promotes the two 8's to the same number; it does not derive $\pi$.

## 16 The Möbius origin of the irreducible $\pi$

What the loop does *not* remove is $\pi$. In the old theory this is exactly the Möbius datum: the orientable double cover of the Möbius fibre is a cylinder with two boundaries plus one $\mathbb{Z}_2$ seam $\Gamma$; Gauss–Bonnet gives $\oint k\, ds = 2\pi + 2\pi + 2\pi = 6\pi$ (hence $\varphi_{\text{tree}} = 1/(6\pi)$), and the $\mathbb{Z}_2$ identification *is* the sheet "2" of $30 = 2 \cdot 3 \cdot 5$. So the Möbius seam supplies precisely the two things the discrete loop cannot: the continuous $\pi$ and the sheet parity $\mathbb{Z}_2$.

**Why "Möbius" is not incidental: it is the boundary Lorentz structure.** The Möbius group of the Riemann sphere is *exactly* the proper orthochronous Lorentz group: $\text{Möb}(\hat{\mathbb{C}}) \cong \text{PSL}(2,\mathbb{C}) \cong \text{SO}^+(3,1)$, with the sphere realised as the celestial sphere of null directions via stereographic projection (Penrose & Rindler, *Spinors and Space-time*, Vol. 1, CUP 1984, §1.2–1.4). This is the spacetime meaning of the self-consistency loop: the seam's boundary-2-sphere carries the Lorentz symmetry, the carrier clock $z \mapsto iz$ is an elliptic $\text{PSL}(2,\mathbb{C})$ rotation of that sphere (the order-4 Möbius map, [verification/v180_clock_is_mobius.py]), and the RP → OS-reconstructed causal cone (the shared $v_\gamma = v_g = c$ of `tfpt_horizon_readouts`) is the same Lorentz structure read off the seam. So "Möbius origin of $\pi$" and "one Lorentzian cone" are two faces of one boundary $\text{PSL}(2,\mathbb{C})$ — not a coincidence of names. [O] ($\pi$ stays primitive; the identification is a structural reading). The *celestial* reading of this sphere has since been pinned at the exact-arithmetic level and has grown into a programme of its own (the research contract `CELEST.SEAM.01`); it is presented as a dedicated section below — the celestial and twistor continuum route (Section 17).

**Net: two axioms → one primitive + one fixed point.**

$$\{c_3, g_{\text{car}}\} \quad \longrightarrow \quad \underbrace{\pi\ (\text{Möbius/Gauss–Bonnet})}_{\text{one continuous primitive}} + \underbrace{E_8\ \text{closure}}_{\text{one discrete fixed point}}$$

The discrete data $\{2, 3, 4, 5, 6, 8, 30\}$ are *all* self-consistent outputs of the $E_8$ closure ($g_{\text{car}} = 5$, $h(D_5) = 8$, $h(A_3) = 4 = |\mu_4|$, $h(E_8) = 30 = 2{\cdot}3{\cdot}5$, $|R^+(A_3)| = 6$). The only irreducible input on the *dimensionless* axis is $\pi$. [E] (discrete loop) / [O] ($\pi$ is primitive)

On the *dimensionful* axis there is exactly *one* more irreducible: a single mass scale. But even that is no longer a *free* constant — fixing the physical $\Lambda$ branch pins $\bar{M}_{\text{Pl}}$, so $G_N$ is a $\Lambda$-metrology *output* (`origin_theory`, $\Lambda$-typing); the one genuinely irreducible scale is the induced-gravity (Sakharov) anchor. So the complete honest reduction is "$\pi$ + the $E_8$-closure fixed point + *one* dimensionful scale".

*One sharpening of the "8".* Both $\pi$-coefficients are $2\pi$ times a compiler integer: $\varphi_{\text{tree}} = 1/(6\pi)$ with $6 = 2N_{\text{fam}}$ (three Gauss–Bonnet boundary circles) and $c_3 = 1/(8\pi)$ with $8 = 2|\mu_4|$ (seam winding), so $\varphi_{\text{tree}} = (|\mu_4|/N_{\text{fam}})c_3 = \frac{4}{3}c_3$. The *same* integer 8 is the Hawking/Einstein coefficient ($G_{\mu\nu} = 8\pi T_{\mu\nu}$, Appendix H): the seam normaliser, the $E_8$ rank and the universal horizon factor are one number, overdetermined ($8 = 2|\mu_4| = \text{rank}\, E_8 = h(D_5) = \varphi(30) = \det R$). [E] [verification/v54_seam_horizon_keystones.py]

**A cyclic element *inside* the hull: the order-30 Coxeter rotation [E] ([verification/v55_coxeter_cycle.py])**

The compiler hull carries a *literal* cyclic symmetry. Built from the $E_8$ Cartan matrix, the Coxeter element $c = s_1 \cdots s_8$ has **order exactly** $h(E_8) = 30 = |\mathbb{Z}_2| \cdot N_{\text{fam}} \cdot g_{\text{car}} = 2{\cdot}3{\cdot}5$ (verified by direct matrix powers), and its eigenvalues are the primitive 30th roots $e^{2\pi i\, m/30}$ with $m$ the $E_8$ *exponents* $\{1, 7, 11, 13, 17, 19, 23, 29\}$ — which are precisely the $\varphi(30) = 8$ *totatives* of 30. Hence

$$\text{rank}\, E_8 = 8 = \varphi(30) = \#\{\text{exponents}\} = \#\{\text{live phases of the order-30 cycle}\}$$

So the 8 in $c_3 = \frac{1}{8\pi}$ is the count of live phases of a primitive order-30 cyclic rotation of the hull, whose period $30 = |\mathbb{Z}_2| \cdot N_{\text{fam}} \cdot g_{\text{car}}$ is the product of the three core integers. The cyclic structure is *present in the mathematics*, not interposed.

**The compiler atoms *are* the $E_8$ Casimir degrees [E] ( [`verification/v66_e8_casimir_degrees.py`] )**

The load-bearing integers are not an ad-hoc list: they are the fundamental *invariant degrees* of $E_8$ (exponents +1),

$$\deg(E_8) = \{2,\ 8,\ 12,\ 14,\ 18,\ 20,\ 24,\ 30\},$$

with the clean (non-trivial) readings $2{=}|\mathbb{Z}_2|$, $8{=}\operatorname{rank} E_8$, $14{=}\dim G_2$ (the $[K,Q]$ commutator $G_2$, flavor sector), $20{=}\det L$, $24{=}|W(A_3)|{=}4!$, $30{=}h(E_8)$ (and the fittable $12{=}|R(A_3)|$, $18{=}p_4(a){=}2{+}2^4$). Two exact sums close it:

$$\sum \deg(E_8) = 128 = 2^7 \quad \text{(the de Sitter prefactor } 128c_3^4;\ 7{=}\text{scalaron exp)},$$
$$\sum \exp(E_8) = 120 = |R^+(E_8)|.$$

Since $E_8$ is the audit hull, its invariant degrees being the theory's integers is *structurally expected* — an organizing simplification that confirms the $E_8$ choice rather than a coincidence.

**The dual-degree modular checksum 744 — audit-only [E] ( [`verification/v532_e8_degree_modular_checksum.py`] )**

The degrees pair dually, $d_i + d_{9-i} = 32 = h + 2$, and the four dual products $(2{\cdot}30,\ 8{\cdot}24,\ 12{\cdot}20,\ 14{\cdot}18) = (60, 192, 240, 252)$ satisfy *simultaneously*

$$\sum = 744 = 3 \cdot 248, \qquad \prod = 696\,729\,600 = |W(E_8)|, \qquad \gcd = 12 = \dim \mathfrak{g}_{\text{SM}},$$
$$\tfrac{1}{12}(60, 192, 240, 252) = (g_{\text{car}}, \dim S^+, \det L, 7N_{\text{fam}}),$$

where 744 is the constant coefficient of the modular $j$-function ($j = E_4^3/\Delta = q^{-1} + 744 + 196884\,q + \ldots$, recomputed from the $q$-series; $\chi^3_{(E_8)_1} = j$, v496) — a finite-to-affine checksum. *Mandatory fence:* the checksum is *not* $E_8$-exclusive — $D_{16}$ (same $h = 30$, rank 16) passes with $1488 = 3 \cdot 496 = 2 \cdot 744$, exactly the heterotic dimension equality; the non-heterotic controls all fail the 3 dim rule ($A_8$: 100/240, $D_8$: 200/360, $E_6$: 117/234, $E_7$: 316/399, $F_4$: 72/156, $G_2$: 12/42). *Prime-totative fingerprint (audit):* $\{2,3,5\} \cup (\text{Exp}(E_8) \setminus \{1\}) = \{2,3,5,7,11,13,17,19,23,29\}$ — all primes below 30; every totative of 30 except 1 is prime, and 30 is the *largest* integer with this property (exact scan; full list $\{1,2,3,4,6,8,12,18,24,30\}$): $h_{E_8}$ is the maximal prime-totative Coxeter number. Under the no-free-pattern rule none of this is load-bearing — it is recorded as a checksum with its own negative controls.

**Two-level foundation, and the parameter audit**

**Local foundation:** $c_3, g_{\text{car}}$ are the two operational inputs (the two axioms). **Global bootstrap foundation:** $g_{\text{car}}$ and the integer in $c_3$ are $E_8$-closure fixed points; only $\pi$ is primitive. The audit:

| former input | status now | recovered by |
|---|---|---|
| $g_{\text{car}} = 5$ | discrete fixed point | rank-fill, Coxeter-match, integer-glue/norm, $h(E_8)$ |
| 8 in $c_3$ | discrete fixed point | $h(D_5)$, rank $E_8$, $\det R$, $\varphi(30)$, $2\lvert\mu_4\rvert$ (seam winding) |
| 6 in $\varphi_{\text{tree}}$ | discrete fixed point | $\lvert R^+(A_3)\rvert = h(E_8)/g_{\text{car}}$ |
| $\pi$ | geometric primitive | Möbius / Gauss–Bonnet (*not* derived) |
| $\varphi_0^{\text{ret}}$ | derived readout | $\frac{4}{3}c_3 + 48c_3^4$ |

**Correct one-liner:** *TFPT has no freely tunable fundamental numbers left; the only remaining continuous origin is $\pi$ from the Möbius boundary geometry.* (Not "zero axioms".) [E]/[O]

## 17 The celestial and twistor continuum route

The previous section identified the seam's boundary 2-sphere with the celestial sphere of null directions ($\text{Möb}(\hat{\mathbb{C}}) \cong \text{PSL}(2,\mathbb{C}) \cong \text{SO}^+(3,1)$, clock = order-4 Möbius map [verification/v180_clock_is_mobius.py]). Taking that identification seriously and reading the seam through the celestial-holography lens (the celestial-chiral-algebra construction of Bittleston–Homans–Sharma on ALE spaces; Costello's twistorial SDYM) produces a coherent thirty-two-step story — the narrative arc from the $\mu_4$ clock to the $(E_8)_1$ boundary shadow — executed as work packages WP1–WP5d (both WP5d stages) plus the WP5e $\alpha$, $\beta$, $\gamma$, $\delta_1$, $\delta_2$, $\varepsilon_2$ and $\varepsilon_1$ stages, the three back-reaction milestones M1–M3, the measure decision (Step 21) and the constructive $w_m$ derivation (Step 24) of the research contract `CELEST.SEAM.01`, plus the RP-blindness decision on the alignment bit (Step 22) with its twist-state completion (Step 26) with its twist-class definition (Step 29) and the interacting FK-toy scoreboard extension (Step 30) with the straddle-cone selection law (Step 31, since refined to the odd-split law — the refinement note after Step 31), the thermal-seam third leg of $c_3$ (Step 27) with the silver-eigenvalue demystification (Step 28), and the $\alpha$, $\beta_1$, $\beta_2$ and $\beta_3$ stages of the OS twistor bridge `WOIT.OS.TWISTOR.01` (Steps 20, 23, 25 and 32), whose full technical statement, kill tests and typing fence live in `tfpt_research_contracts`. Every step below is exact machine arithmetic [E] plus explicitly named identifications [C]; the route is a *programme* [O], not a claim, and nothing in it moves `SEAM.EQUIV.01`. The name of the route is deliberately *structural* ("celestial and twistor continuum route", matching the contract heading): what the route supplies is a candidate *continuum* construction on the twistor side, and the object diagram, the exact group-extension definition ($\Gamma_{\text{ALE}}$ vs the normalising clock $\langle\rho\rangle$) and the $A_3$ role table that fence its semantics live in the contract (`tfpt_research_contracts`, `CELEST.SEAM.01`/`WOIT.OS.TWISTOR.01`).

**Step 1 — the setup: the $\mu_4$ glue is a flat $\mathbb{Z}_4$ monodromy on the $A_3$ ALE space [E].** The $E_8$ $\mu_4$-glue grading 240 = 52+64+60+64 (v128) is *inner*: $h = (2,2,2,2,2;0^4)$ reads the glue class mod 4 on all 240 roots, so the glue is $\text{Ad}(\exp(2\pi i h/4))$ — a flat $\mathbb{Z}_4$ monodromy in the Kronheimer–Nakajima sense on the $A_3$ ALE orbifold $\mathbb{C}^2/\mathbb{Z}_4$ (which *is* the $A_3$ singularity $XY = Z^4$, exactly). The glue-equivariant SDYM($E_8$) sector closes as a Lie algebra with graded dimensions $60(d{+}1)/64(d{+}1)$ — possible *only* because $\dim \mathfrak{g}_j = (60,64,60,64)$ — with zero modes = the carrier $D_5{\oplus}A_3{+}\text{Cartan} = 60$, and the four glue-sector characters sum *exactly* to the $(E_8)_1$ character $\chi = 1{+}248q{+}4124q^2{+}34752q^3$ (v377). The critical correction: the $A_3$ deck induces on the celestial sphere only the sheet flip $z \mapsto -z$; the order-4 clock $z \mapsto iz$ is realised as the Kähler $U(2)$ phase $\text{diag}(i,1)$, and **clock$^2$ = deck** exactly (spin bridge $\mathbb{Z}_8$, $8 = 2|\mu_4|$ — the $c_3 = 1/(8\pi)$ seam winding integer). [verification/v492_celestial_z4_orbifold.py] That spin bookkeeping is now *forced* on the seam fermions: the NS implementer of the deck satisfies $U^2 = (-1)^F$ exactly (nonsplit $\mathbb{Z}_4$ extension, no phase choice escapes it), and the quarter-shift lifts

to a canonical $\mathbb{Z}_8$ tower with $V^2 \propto U$, $V^4 \propto (-1)^F$ — "$8 = 2|\mu_4|$" fermionically explained; the R-sector control splits ($U_R^2 = +1$), so the forcing rests exactly on the established bounding spin structure. [verification/v506_seam_clock_rigidity.py]

**Step 2 — the deformation: $a_0$ is a pure scale and the clock is the monodromy [E].** Clock-invariance selects from the versal $A_3$ family exactly the one-parameter slice $XY = Z^4 + a_0$, and $a_0$ carries *no* shape modulus: the binary-quartic invariants are $I = 12a_0$, $J = 0$ *identically*, so $j = 1728$ and the $\tau{=}i$ pillowcase (v168/v214) is frozen for every $a_0$. The clock acts on $H_2$ as a *Coxeter element* of $W(A_3)$ (eigenvalues $\{i, -1, -i\}$ = the $A_3$ exponents, $h(A_3){=}4{=}|\mu_4|$) and *is* the Picard–Lefschetz monodromy of the family; the deformed-algebra pattern transfers $\mathbb{Z}_2 \to \mathbb{Z}_4$ grade-conserving (no $\mu_4$-sector leak). [verification/v493_celestial_wp2_clock_deformation.py] The converse is now tested *light*: the free field alone does *not* select $\tau = i$ (the global det′ winner is the hexagonal torus, whose $\mathbb{Z}_6$ carries no order-4 element) — the square is pinned by the clock's *rigidity* (the raw DtN is mod-4 block-diagonal *iff* $\tau = i$), and this order-4 clock is the *same* carrier datum as the P2 cusp class (v499): one common residual, not two ([verification/v503_qgeo_emergence_light.py]; the v215 deferred emergence test, light version — no marker moves, `QGEO.REALIZE.01` stays [C]/[O]). That common residual is now reduced to *one bit*: the deck $z \mapsto -z$ has exactly two Möbius square roots ($z \mapsto \pm iz$), both automatically of order 4, and a marks-preserving root exists *iff* the deck is the *central* involution of the mark-stabiliser $D_4$ — given that alignment bit, order, uniqueness up to $\mathrm{Aut}(\mathbb{Z}_4)$, the free 4-cycle and $\tau = i$ (scale free) are all theorems; harmonicity alone is *not* enough on bare mark data (exact counterexample — though on free-deck circles it becomes sufficient, see below), and the fermionic $\mathbb{Z}_4$ forces the *order*, not the alignment ([verification/v506_seam_clock_rigidity.py], `SEAM.CLOCK.RIGIDITY.01`; markers unchanged). The bit itself has now survived its tautology attack: marks and deck *do* share one torus origin — the marks are the four half-periods $\Lambda/2\Lambda$ of the seam double, and *every* mark-free collar deck is a half-period translation (in all three stabiliser types the freely-acting involutions are exactly the three $\sigma_c$), so the deck's *class* is derived — yet the origin does not force the alignment: the marks-preserving root exists iff the deck pairs separate harmonically ($\lambda \in \{-1, 2, \frac{1}{2}\}$, one value per half-period), the aligned half-period is the CM-fixed one $c^* = (1{+}\tau)/2$ (the clock's curve lift needs the unit $i$ of $\mathbb{Z}[i]$, while half-period decks lift rationally), and the silver counterexample sits *in-family* as "$\tau = i$ with the deck a non-CM-fixed half-period". The bit is thereby an equivalence *tetrad* — clock $\iff$ ($\tau{=}i$ and deck = CM-fixed half-period) $\iff$ (Stab = $D_4$ and deck central) $\iff$ harmonic deck pairs $\iff$ *nonsplit* NS Fock lift — with a physical face: the central arrangement carries the nontrivial Fidkowski–Kitaev-type extension class ($U^2 = (-1)^F$), the edge arrangement splits ($U_e^2 = +1$, zero roots, no $\mathbb{Z}_8$ tower); only the *position* (which half-period) stays the carrier input ([verification/v507_seam_bit_origin.py], `SEAM.BIT.ORIGIN.01`; no marker moves). And that position half is now *topology*: the collar deck is the deck transformation of a *covering* (the $\mathbb{Z}_2$ seam of the Möbius double), and covering decks act freely — on the seam circle the unique free involution is the antipode ($\mathrm{Aut}(C_{16}) = D_{16}$, complete census; every reflection has two circle fixed points, and its quotient is an *interval*, breaking the closed-circle $6\pi$ budget), the NS Fock lift is nonsplit *iff* the deck is free (all 17 involutions in exact Cl(16)), and the edge/silver arrangement — whose mark-fixing real structure has its fix circle *through* the deck poles — is the restriction of the *branched* pillowcase involution, not a covering deck: excluded. On free-deck circles "harmonic" already solves to $b = \pm i$, so the alignment bit reduces from "harmonic *and* central" to the square-modulus datum $\tau = i$ alone — the position half is topology, the modulus half stays the one carrier input ([verification/v510_seam_bit_freedom.py], `SEAM.BIT.FREEDOM.01`; markers unchanged). The modulus half now has a sharp *discrete* face: on the free seam circle the deck-invariant 4-mark configurations form the one-parameter family $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$, and *bare* mark-transitivity is automatic for every $\delta$ (the pair-exchanging $V_4$; all undirected per-mark data are identical, so no local jet sees the bit — "transitivity from v216-locality" is falsified) — what selects $\delta = \pi/2 \iff \tau = i$ is exactly *flag* transitivity (marks *and* their two sides indistinguishable, the $V_4 \to D_4$ symmetry lift), welded into a 13-fold exact equivalence web (clock rotation, mark mirror, odd-doublet split $(2/\pi)\ln\cot(\delta/2)$, arc-Laplacian degeneracy, the v503 mod-4 indicator in closed form, $\rho$-twist existence, cusp-4-cycle implementability, harmonicity, $j = 1728$,

and — since Step 22 — the $D_4$ closure of the OS-reflection group); the carrier input thereby reduces from a continuous modulus to *one discrete symmetry-lift bit* whose negation is concretely measurable ([verification/v512_seam_tau_flag.py], SEAM.TAU.FLAG.01; markers unchanged).

**Step 3 — consistency, honestly demoted [E]/[C].** $E_8$ is on Costello's axionic-consistency list (no independent quartic Casimir; the Okubo coefficient is *derived*, not quoted), and the Green–Schwarz alignment is exact: $\tilde{\lambda} = 6$ and $(\kappa/c_3)^2 = 12 = |\mu_4| N_{\text{fam}}$. *But* the look-elsewhere battery shows the same alignment format passes for *all eight* algebras on the list (8/8): *alignment survives; selectivity does not* — the $c_3$-connection is compatibility plus genuine $\lambda$-arithmetic, not $\mathfrak{e}_8$-selective evidence. [verification/v495_celestial_wp3_gs_alignment.py]

**Step 4 — the type mismatch and the boundary limit [E].** The $(E_8)_1$ character is *not* a conformal block of the celestial S-algebra in its own jet grading — the jet Fock grows as $n^{2/3}$ against the character's $n^{1/2}$, and the level-2 truncation has no jet analogue. But the boundary/period reading holds *exactly* at the current stratum: one full $\mu_4$ period of loop energies sums to 248, and the glue-diagonal weights $(0, 1, 1, 1)$ are integers. The character is a *boundary-limit shadow* of the S-algebra — precisely the scaling-limit shape of SEAM.EQUIV.01 (MMST: the rational net exists in the limit of the lattice collar, not in the pre-limit algebra; v336/v449). [verification/v496_celestial_wp4_character_shadow.py]

**Step 5 — the null ideal, quantitatively [E].** The shadow is made a *precise coefficientwise limit*: the exact $\chi_w$ family contains the jet grading at $w$=2 and stabilises with threshold $w = n{+}1$. The deleted ideal is *derived* from root data (Freudenthal + Weyl + peeling): $\text{Sym}^2(248) = 27000{+}3875{+}1$ with $27000 = 30^3 = (h^\vee)^3$ exactly, and the level-2 quotient $31124{-}27000 = 4124$ equals the independent $\mu_4$ theta-split sector sum — two routes, one number. The $SO(16)_1$ negative control separates: four blocks, $5304 \neq 14^3$, half-integer weight breaking one-block fusion. [verification/v497_celestial_wp5a_null_ideal.py]

**Step 6 — the deleting object as an operator [E].** The level-2 singular vector is constructed explicitly, $|s\rangle = (E^\theta_{-1})^2|0\rangle$, in a machine-built Chevalley/Frenkel–Kac basis, and $J^a_1|s\rangle = 0$ is machine-verified for all 248 generators (case tally 190/57/1 — exactly one case sees the level). The *level dial* is the $F^\theta_1$ coefficient $2(1{-}k)$: only $k$=1 deletes — without the central extension the deletion operator does not exist. $\mu_4$ data: glue class $j(\theta) = 1$, clock phase $-1$ on $|s\rangle$. The $D_8$ contrast types the finding honestly: the singular-vector mechanism is level-1 *generic*; the *one-block closure* is the $E_8/\mu_4$-specific part. [verification/v498_celestial_wp5b_singular_vector.py]

**Step 7 — the limit state [E]/[C].** The state exists: the quasi-free family $\omega_w$ (affine $k$=1 vacuum on the currents + $x^{wr}$-contracted radial oscillators) stabilises exactly at the WP5a threshold $w = N{+}1$, and its limit carries the null ideal in its GNS kernel — the complete 9361-block exact level-2 Gram has rank 4124 with kernel $27000 = V(2\theta)$ weight by weight, $|s\rangle$ *is* the GNS zero vector, and a CCR obstruction shows no $w$-uniform state could do this (the family is *necessary*). [verification/v500_celestial_wp5c_gns_limit_state.py]

**Step 8 — the two-interval index [E]/[C].** The Haag-duality discriminator is measured on the seam lattice edge: the fermionic two-interval mutual information is extensive ($\mu$=1 reference) while the orbifold prescription pays exactly one classical bit (the $\ln 2$ plateau at machine precision), giving $\mu_{\text{gauged}} = 4 =$ the v490 parity census, and the condensation arithmetic $16 \to 4 \to 1$ (KLM/Longo–Rehren, $\theta_v = 1$ at $\nu = 16$) closes with $\mu = 1$ — the preregistered kill ("$\mu$-offset $\neq 0$ after condensation") does not fire. [verification/v501_celestial_wp5d_two_interval_index.py]

**Step 9 — the prefactor and the level, pinned on the CFT side [E]/[C].** The two exactly computable consistency anchors of the twistor uplift hold. The $q^{-1/3}$ prefactor of $E_4/\eta^8$ is exact $\mu_4$ vacuum-energy bookkeeping: the clock is *inner*, so the twist on the 8 torus bosons is $\theta = 0^8$ in every sector (a *shift* orbifold — not a rotation orbifold) and each sector carries the same $-c/24 = -\frac{1}{3}$ at $c = 8$; the sector weights $(0, 1, 1, 1)$ *are* Casimir energies, via spectral flow and exactly via the 16-Majorana seam carrier (R–NS shift $n/16$: $\frac{5}{8}, \frac{3}{8}, 1$); and $k = 1$ is forced three independent ways (current condition $h(J) = k$, conformal embedding $47(k{-}1)(k{+}266/47) = 0$, central charge $240(k{-}1) = 0$) plus the Step-6 singular-vector dial ($31124 - 27000 = 4124$ at $k = 1$ only). Honest

sharpening: glue-$h$ *integrality* holds at every level and fixes nothing — the naive integrality route is retired; the deck rotation reading fails ($\frac{3}{16} \neq \frac{3}{8}$). [verification/v502_celestial_wp5e_prefactor_level.py]

**Step 10 — the KLM completion on the lattice [E]/[C].** Complete rationality (Kawahigashi–Longo–Müger) has three ingredients — split property, strong additivity, finite two-interval $\mu$-index — and with Step 8 supplying the index, the two remaining legs are now witnessed for the same orbifold prescription. Strong additivity is *algebraically exact*: with the shared boundary Majorana, $\mathrm{Even}(A) \vee \mathrm{Even}(B) = \mathrm{Even}(A \cup B)$ exactly (GF2 spans full, matrix rank 32/32), while disjoint intervals give exactly index 2 — the Step-8 $\ln 2$ bit, localised at the split point; entropically the touching defect stays *bounded* by $\ln 2$ with the Ising $\frac{1}{4}$-exponent approach while the preregistered $U(1)$/Dirac control *diverges* (Klich–Levitov pinned): bounded-vs-divergent is the lattice discriminator (the naive "defect $\to 0$" expectation is false at a sharp lattice split — bounded $\Leftrightarrow$ finite index, Longo–Xu). The split property is quantitative: the coupling singular values fall at the elliptic-nome rate $\pi K(1-x)/K(x)$ to 1.3–2.0%, and the orbifold *inherits* the same ladder exactly ($C \to -C$; the $\Lambda^2 C$ compound — Longo heredity). The Pimsner–Popa bound is a one-line identity, $E(a) - a/2 = PaP/2$, attained in exact integers ($\lambda = \frac{1}{2} = 1/[F : F_{\mathrm{even}}]$, $\lambda_{E_4} = \frac{1}{4} = 1/\mu$), and the index is pinned by two independent routes ($e^{\Delta_\infty} = 2 = 1/\lambda_{\mathrm{PP}}$). All three KLM ingredients now carry lattice witnesses; the continuum uplift is honestly fenced — "finite-group orbifolds of completely rational nets are completely rational" is *Xu's theorem*, cited, not claimed. [verification/v504_celestial_wp5d_klm_completeness.py]

**Step 11 — the equivariant anomaly ledger on twistor space [E]/[C].** The twistor side of the uplift is pushed as far as exact arithmetic reaches: the Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop box anomaly on $\mathbb{C}^2/\mathbb{Z}_4$ with the $\mu_4$-glue monodromy is exact — denominators $(2, 4, 2)$ with Dedekind sum $5/4 = (|\mathbb{Z}_4|^2 - 1)/12$, equivariant characters $(248, 0, -8, 0)$ by two routes, invariant average $60 =$ the carrier. The honest sharpening: only the *invariant* sector is Okubo-quadratic ($36\langle x, x\rangle^2$, $36 = \tilde{\lambda}^2_{e_8}$); the twisted sectors are not, and the AB-weighted sum cancels the $D_5$ quartic exactly while leaving the *rigid* residual $32\,T_3$ — twisted-sector closed strings must carry the rest. The *index bridge* is the centrepiece: $f(m) = \mathrm{ch}_2(T_m)$ exactly (fixed-point ledger = McKay/Kronheimer intersection ledger), with the integral glue defect $-78$ by both routes; the level dials say $k = 1$ *geometrically* (lattice current count 240/0/0/0, embedding residual $(0, 360, 814, 1362)$, one scale $\Rightarrow$ one level — since Step 13 a *theorem*, no longer a heuristic). And an honest *refutation*: the clock-invariant modulus $a_0$ fills the BSS *graviton* slot $O(2)$, *not* the axion slot $O(-2)$ (weight mismatch $4 = |\mu_4|$) — "the theory brings its own GS axion as $a_0$" is false; instead the three $H^2(\mathrm{ALE})$ classes carry exactly the three twisted-sector Coxeter characters $\{i, -1, -i\}$ (the three twisted axion slots), and the bulk axion must come from the $O(-2)$ tower field itself. The preregistered kill ("inflow demands a level $\neq 1$") does *not* fire on the equivariant skeleton. [verification/v505_celestial_wp5e_beta_equivariant_ledger.py]

**Step 12 — the exchange no-go and the contact-term criterion [E]/[C].** Could the three sphere axions of Step 11 cancel the rigid $32\,T_3$ residual by Green–Schwarz-type *exchange* with sector-compatible quadratic vertices? No — and the refusal is a theorem, not a fit. The $W(D_5)\times W(A_3)$-invariant vertex space on the glue Cartan is computed exactly (quadratics $= \mathrm{span}\{s_5, s_3\}$, dim 2; quartics $= \mathrm{span}\{P_1, P_2, P_3, T_5, T_3\}$, dim 5 — complete by Weyl nullspace arithmetic over all bidegrees), and the *product theorem* kills the mechanism wholesale: the product of any two invariant quadratics has identically zero $T_5/T_3$ content, so *every* exchange image — any number of axions, any selection rule, any couplings — is annihilated by $\Phi_{T_3}$ while $\Phi_{T_3}(A_{\mathrm{fix}}) = 32 \neq 0$. The enumeration sharpens where it fails: the strict two-index $\mathbb{Z}_4$ rule collapses the sphere couplings entirely (only $m = 0$ survives); the generous twist-insertion channels give a rank-2 exchange matrix with $\mathrm{rank}([M \mid A_{\mathrm{fix}}]) = 3$ and annihilator certificate $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\mathrm{fix}}) = (0, 32, 72)$ — two independent obstructions, the second driven by the side discovery $K^{(0)} = -15\,K^{(2)}$ (the even-sector quadratics are parallel). Naturalness *dissolves*: $\mathrm{ch}_2$-natural and Atiyah–Bott-weight couplings both carry certificate $(0, 0)$ against the required $(32, 72)$ — the ansatz space misses the target identically, scale- and coupling-independently. Controls: $SO(16)$ has *no* sphere partners and uncancelled $T_5$ ($E_8$ is doubly special); $D_8$ has no $T_3$ structure at all. The slot bijection of Step 11 is untouched,

and the preregistered level-kill still does not fire — what dies is the exchange *realisation* of the pairing; the $32\,T_3$ burden moves to the twisted BCOV contact terms, with a *binary* criterion: the one-loop coefficient on $\mathbb{C}^2/\mathbb{Z}_4$ must come out exactly $(9, -30, -15, 0, 32)$, computed, not fitted. [verification/v508_celestial_wp5e_gamma_pairing.py]

**Step 13 — one level is a theorem; the sector counter pins $k = 1$ [E]/[C].** The Costello–Paquette–Sharma level-from-flux mechanism (Burns holography: "level = flux quantum × Dynkin index of the boundary matter") is executed on the lockstep spheres. The CPS skeleton is replicated exactly ($S^3$ period $(2\pi i)^2 N$; exceptional-sphere Kähler flux $2\pi N$; boundary level magnitude $2N$ by the symplectic-boson contraction, $T(\text{vector}) = 1$), and the geometric side is *pinned, not postulated*: the Step-11 $\mathrm{ch}_2$ ledger + integrality + unimodularity + effectivity force the pairing matrix $c_1(T_m).[\Sigma_i] = \delta_{mi}$ by complete enumeration (48 unimodular solutions, exactly 2 effective: identity + diagram flip), so the glue fluxes are $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$ with *exactly one* quantum per charged root current. The consistency test then does real work: "level = total open-string flux" is *killed by the lockstep test itself* (three unequal numbers — $F_i$ is the flavour multiplicity), while "level = flux per adjoint current" gives $(1, 1, 1)$, anchored by the lattice current count $(240, 0, 0, 0)$ and the embedding index 1. The ord-4-vs-level-1 tension resolves — the clock order counts *fractional flux sectors* ($\mu = 16 = \mathrm{ord}^2$, Lagrangian $\mu_4$ glue, KLM $16 \to 1$) — and a new dial pins the value: $\#\text{primaries}((E_8)_k) = (1, 3, 5, 10, 15, 27)$ for $k = 1..6$, exactly *one* sector iff $k = 1$. The centrepiece is the *lockstep theorem*, lifting Step 11's "one scale ⇒ one level" from heuristic to theorem: clock invariance forces the four branch points into one $\mu_4$ orbit, the period moduli $|\Pi_j| = \sqrt{2}\,t$ are equal on all three spheres, and $\det(A - 1) = -4$ leaves no invariant flux vector — $k_1 = k_2 = k_3$ *is a theorem* of clock invariance, with a new falsifier (the clock-forbidden family $Z^4 - Z$: zero lockstep chains over all 24 orderings) and an honest limit (uniform doubling keeps the lockstep: the theorem proves *equality*, the value 1 comes from the counters). Controls: $k = 2$ dies on the closure dials while the lockstep dial honestly does not fire; $SO(16)$ leaves two spheres flux-dark; $A_2$ admits no Lagrangian glue at all ($\mu = 12$ not a square). The CPS dictionary on $\mathbb{PT}/\mathbb{Z}_4$ itself stays [C] (their geometry is the $\mathbb{C}^2$ blow-up, not the $A_3$ ALE); the type-I B-model back-reaction stays [O]. [verification/v509_celestial_wp5e_eps2_level_from_flux.py]

**Step 14 — the full-tensor ledger: the collapse is real, and one cubic door opens [E]/[C].** Step 12's collapse was a *Cartan-restricted* statement — the named escape route ($\delta_2$) asks whether the full adjoint tensor structure of $\mathfrak{e}_8$ (non-Cartan external legs, higher-arity vertices) reopens the pairing. The answer has two halves, both exact. *The collapse is confirmed*: $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$ is semisimple with no $\mathfrak{u}(1)$ factor (class-0 roots 40+12, root span rank 8), the sectors factorise into minuscule Weyl orbits $\mathfrak{g}_1 = (16_s, 4)$, $\mathfrak{g}_2 = (10, 6)$, $\mathfrak{g}_3 = (\mathfrak{g}_1)^*$, and the *innerness theorem* decides everything wholesale: the grading element $h$ lies in the Cartan *of* $\mathfrak{g}_0$, so every $\mathfrak{g}_0$-invariant tensor carries total sector charge $0 \bmod 4$ — bilinear invariants exist only at $j' = -j$ (dims 2/1/1/1) and *all* 15 non-neutral trilinear sector triples have Hom = 0: the sphere axions have no invariant partner operator at any arity $\leq 3$, full-tensorially. *And one cubic door opens*: of the nine neutral trilinear structures (dims $(3, 2, 2, 1, 1)$) all cross-sector ones are pure brackets, but the *unique* totally symmetric survivor — the $\mathfrak{su}(4)$ $d$-symbol ($\mathfrak{so}(10)$ has no cubic Casimir, so $T_5$ stays protected) — carries an exchange quartic $Q_{dd} = \frac{1}{60}(T_3 - P_3/4)$ (two independent routes, Killing propagator $2h^\vee = 60$ from the roots) with $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$: Step 12's master kill covers *quadratic* vertices only and does not extend to cubic ones. The pairing nevertheless remains obstructed in the charge reading — $M = [E_{13}, E_{22}, E_{00}, Q_{dd}]$ has rank 3 against augmented rank 4 — but with the genuinely *weaker* certificate $\{\Phi_{T_5}, \psi = \Phi_P - \Phi_{T_3}/4\}$, $\psi(A_{\text{fix}}) = 72 - 8 = 64$; relaxed, it becomes *exactly solvable*: $A_{\text{fix}} = -u + 8v + 2w + 1920\,Q_{dd}$. Number fences typed [C] only: $64 = \dim \mathfrak{g}_1 = 2^6$, $1920 = |W(D_5)| = 8 \cdot 240$. Controls: $SO(16)/D_8$ has no symmetric cubic, no odd sectors and no $A_3$ block ($E_8/\mu_4$ doubly special); false $\mathfrak{g}_0$/sector assignments inflate or kill the tables. The burden on the $\delta_1$ contact terms is thereby *sharpened*: they must supply either the $\psi = 64$ slice or the selection-rule relaxation — a burden since *discharged* by Step 17 (the declared KS measure supplies the $\psi = 64$ slice exactly, so the cubic $d$-channel is not needed) — and the $\delta_1$ run, undecided at exploration level at the time (the strict holomorphic reading is

refuted at the $\mathbb{Z}_2$/Eguchi–Hanson anchor — a method boundary, not a kill; the contact term is the *modular completion* of the Atiyah–Bott data, a Harvey–Moore-type $\tau$-integral with the forced leading $(T_5, T_3)$ ratio 4:3), has since been *decided* in Step 19 (kill under the derived measure, v518). [verification/v511_celestial_wp5e_delta2_tensor_ledger.py] The 1920 fence itself has since been stress-tested and *fails as a derivation claim*: the $1920 = |W(D_5)|$ reading is look-elsewhere-loaded (11/924 catalog expressions hit 1920, vs 8/924 for the control target 1800) and convention-contingent (only 1 of 5 normalisations produces 1920; charge-*legal* flux pairings give 2048/1800 and miss; the twisted cubic index $32i$ clashes in class provenance with the true quartic 32) — the convention-stable **[E]** core is the factorisation $c_d = \Phi_{T_3}(A_{\text{fix}}) \times 2h^\vee(E_8) = 32 \times 60$, the fence stays **[C]**, and the physical generation of the 32 stays **[O]** ([verification/v513_celestial_dterm_nonderivation.py], `CELEST.DTERM.NONDERIV.01`; no marker moves).

**Step 15 — the bulk axion is a construction; $\tilde{\lambda} = 6$ is pinned three ways [E]/[C].** Step 11 refuted "$a_0$ is the GS axion" and left the bulk axion to the $O(-2)$ tower field *as a slot*; the $\varepsilon_1$ stage now *builds* it. On $\mathbb{PT}' = \text{Tot}(O(1)\oplus O(1) \to \mathbb{P}^1)$ the pushforward ledger $\pi_* O(-2)$ closes the Penrose accounting exactly ($(d+1)^2$ classes per degree = the exact wave-operator nullspaces, $d \le 6$), and *equivariantly*: the clock acts on the fibre coordinates as $(\mu_1, \mu_2) \mapsto (i\mu_1, i^{-1}\mu_2)$, the incidence relation forces the column weights $(+1, +1, -1, -1)$, and the bookkeeping closes *block by block* for all four characters. The character series are $P_0 = 1 + 3t^2 + 15t^4 + \ldots$, $P_1 = P_3 = 2t + 8t^3 + \ldots$, $P_2 = 6t^2 + 10t^4 + \ldots$; the $d = 0$ slot has multiplicity $(1, 0, 0, 0)$ — *the bulk axion survives the projection* — and the invariant fibre ring is the *hypersurface* $(1 - t^8)/((1 - t^4)^2(1 - t^2))$: $Z \in O(2)$, $X, Y \in O(4)$, one relation in degree 8 = the $a_0$ weight (the Step-1/Step-2 geometry re-emerging from the slot); the Step-11 bijection sharpens to a per-character statement (twisted minimal content $(2t, 6t^2, 2t)$ = the Coxeter eigenvalues, no degree-0 mode, $\det(A - 1) = -4$), and the graviton control separates $a_0$ cohomologically (the $O(+2)$ slot starts invariant only at fibre degree 4 with multiplicity $3 = \{X, Z^2, Y\}$). The Green–Schwarz residue is then pinned *three* independent ways: Okubo ($\text{Tr}_{\text{adj}}\, X^4 = (6\langle x, x\rangle)^2$ exactly on the 240 glue roots, $36 = h^\vee + 6$, $\tilde{\lambda} = 6 = |\mathbb{Z}_2| N_{\text{fam}}$), the measure chain on $\mathbb{PT}/\mathbb{Z}_4$ (the quotient factor $\mu = 1/4$ enters vertex$^2$/propagator/anomaly and cancels *exactly*; the wrong bookings are quantified and excluded: anomaly-only $\Rightarrow \tilde{\lambda} = 3$, missing propagator renormalisation $\Rightarrow \tilde{\lambda} = 12$), and the flux side (minimal CPS quantum $N = 1$, a single exchange channel exactly at $k = 1$, $(\kappa/c_3)^2 = 12$). The first honest *back-reaction* step follows in Gibbons–Hawking form: clock-invariant four-centre configurations form exactly two branches (four axis points = pure resolution, one free $\mu_4$ orbit = pure deformation), the free orbit re-derives the Step-2 family ($\prod_p (Z - i^p z_0) = Z^4 - z_0^4$, $a_0 = -z_0^4$, monodromy = one clock step), the charge-4 GH point is exactly the $S^3/\mathbb{Z}_4$ seam boundary, the periods $\Pi_j = 4\pi t_0 (i - 1)(1, i, i^2)$ confirm the Step-13 lockstep *from geometry* ($\Pi \cdot A = i\Pi$), the source ledger carries charge $4 = |\mu_4|$, and the honest sharpening: the Ricci-flat ALE has asymptotic log coefficient *exactly zero* (the CPS log is an exceptional-locus statement; Burns contrast $\det g \neq 1$), with the multipole selection rule $m \equiv 0 \bmod 4$ storing $-a_0$ in the first symmetry-breaking $(4, \pm 4)$ harmonic. Controls: $\mathfrak{so}_8$ ($\tilde{\lambda}^2 = 12$ irrational — perfect squares in the Deligne series only $\{9, 36\} = \{\mathfrak{sl}_3, \mathfrak{e}_8\}$), $\text{diag}(i, i)$ (Veronese cone, no hypersurface, the bijection collapses), $k = 2$ (three exchange channels). What stays honest: the chain is conditional on Costello's flat-$\mathbb{PT}$ matching **[C]**, and the *quantised* BCOV coefficient on $\mathbb{PT}/\mathbb{Z}_4$ plus the twisted channels ($32\, T_3$) stay **[O]** — preregistered as the M1–M3 back-reaction milestones (the $A_3$ $\Omega_N$, the twisted KS measure, the $a_0$ uplift), each with success and kill criteria; all three are executed in Steps 16–18. [verification/v514_celestial_wp5e_eps1_axion_slot.py]

**Step 16 — the back-reacted $\Omega_N$: closed form, integral periods, forced charge 4 [E]/[C].** The M1 milestone preregistered in Step 15 is now *executed*, with the preregistered SUCCESS criterion met and neither KILL fired. The residue 2-form of the family is *derived*, not assumed: on $F = XY - P(Z)$ all three ambient representatives satisfy $dF \wedge \omega_2 = -dX \wedge dY \wedge dZ$ and pull back to $\omega_2 = dX \wedge dZ/X$; at $a_0 = 0$ the orbifold-cover pullback is $\omega_2 = 4\, dz_1 \wedge dz_2 = |\mu_4| \times$(flat form) — the residue normalisation itself *carries* the source charge 4 — and the clock multiplies $\omega_2$ by $i$ (the Step-1 $\det = i$ replicated). The four $O(2)$ centre sections $q_p(\lambda) = i^p t_0 - i^{-p} t_0 \lambda^2$ close the twistor family in one line, $XY = Z^4 + 4t_0^2 \lambda^2 Z^2 - t_0^4 (1 - \lambda^4)^2$ ($e_1 = e_3 = 0$ identically; the

seam fibre $\lambda = 0$ is exactly the Step-2 family, the $a_2 \in O(4)$ slot opens only off-seam), with the CY-compatible clock lift $\gamma : (Z, \lambda) \mapsto (iZ, -i\lambda)$, $\gamma^4 = 1$, $\Omega \to +\Omega$. The periods reduce generically ($\int_{S_j} \omega_2 = 2\pi i(q_{j+1} - q_j)$ for arbitrary profile and path), giving the seam-fibre lockstep vector $2\pi i\, t_0(i-1)(1, i, i^2)$ with the exact covariance $\Pi_{j+1}(-i\lambda) = i\,\Pi_j(\lambda)$ for *all* $\lambda$, and the 12 collision nodes — honest conifold points (Hessian $\det = 512t_0^6$, quadric rank 4) — sit exactly on the 8 eighth roots of unity ($8 = 2|\mu_4|$, the Step-1 $\mathbb{Z}_8$ bridge), in clock orbits 4+4+4. The back-reacted 3-form is then *closed form*: $\Omega_N = \Omega_0 + \sum_p N_p K_p$ with the CPS/Bochner–Martinelli kernel on the four centre twistor lines ($\int_{S^3} K = (2\pi i)^2$ exactly, $dK = 0$ off-source, scale degree 0); the undeformed $\Omega_0$ carries *zero* quantised 3-flux (all flux is sourced), every 3-cycle period is $(2\pi i)^2$-integral, the flux vector $N(1, 1, 1, 1)$ is forced uniform *twice over* (clock orbit + the $K_4$ connectivity of the four lines), and the lens fundamental domain *forces* $N \in 4\mathbb{Z}$: the source charge $4 = |\mu_4|$ from quotient large-gauge quantisation, minimal invariant charge $\leftrightarrow$ the CPS quantum $N = 1 \leftrightarrow k = 1$ (Step 13). The honest fence has teeth: on the clock-forbidden family $Z^4 - Z$ the $(2\pi i)^2$ quantisation *also* holds — integrality alone is *not* the discriminator; the discriminator is the lockstep phase/modulus structure (0/24 orderings vs 8/24, node support on twelfth roots instead of $\mathbb{Z}_8$) together with the clock forcing of equal fluxes. Anchors and controls: EH/$\mathbb{Z}_2$ forces charge $2 = |\mathbb{Z}_2|$ from the same machinery, the CPS flat patch allows every integer $N$ (the unconstrained anchor), $\mathrm{diag}(i, i)$ collapses at step zero ($\Omega_0 \to -\Omega_0$), and fractional charges $N = 1, 2, 3$ are excluded. Global kernel patching, the Hitchin small resolution and the CPS brane dictionary stay [C]; M2–M3 are executed in Steps 17–18, and the quantised BCOV coefficient stays [O] only in its measure question (the $\delta_1$ chain itself is decided in Step 19). `[verification/v515_celestial_wp5e_m1_omega_n.py]`

**Step 17 — the twisted KS measure: every sector the same Okubo square [E]/[C].** The M2 milestone preregistered in Step 15 is now *executed*, with the preregistered SUCCESS wording met *on the declared completion measure* and the KILL (a leftover independent quartic in *any* sector) not fired. The measure ansatz is declared before computing, and honestly typed: Step 12 killed every *exchange* realisation, and the $\delta_1$ exploration identified the twisted contact term as the *modular completion* of the very Atiyah–Bott data that produced $A_{\mathrm{fix}}$; M2 declares the completion contact term per twisted box channel, $\mathrm{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$ — in the channel with the $g^j$ zero-mode normalisation $1/\det_j$, the twisted-sector *loop* contributes the *unphased* sector trace, of which the Atiyah–Bott skeleton kept only the phase-weighted insertion part. Everything downstream of the declaration is exact arithmetic, and it locks in place with no dial to turn: per glue class the contact weights obey the *completion-weight identity* $w_m = \sum_j (1 - i^{jm})/\det_j = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4| h_m = -4\,\mathrm{ch}_2(T_m)$ ($h_m = m(4-m)/8$ the Step-9 sector Casimir energies, $\mathrm{ch}_2(T_m)$ the McKay/Kronheimer charges of the Step-11 index bridge): the three sphere axions pair through their *own* $\mathrm{ch}_2$ charges — an exact identity, *no free scale, no fit.* The locks are parameter-free: $T_5 = 0$ for *any* scale $c$ (the $\mathrm{ch}_2$ pattern alone), the ratio $h_2{:}h_1 = 4{:}3$ *reproduces* the $\delta_1$-forced leading ratio, and the $T_3$ budget fixes $c = 4 = |\mu_4|$ uniquely (leading system det 256). The success table is then uniform: $\mathrm{skeleton}_j + \mathrm{contact}_j = Q^{(0)}/\det_j = 36\langle x, x\rangle^2/\det_j$ — a *perfect Okubo square* ($T_5 = T_3 = 0$, $\tilde{\lambda}^2 = 36$) in *every* twisted channel; the total $A_{\mathrm{fix}} + \mathrm{contact} = 45\langle x, x\rangle^2 = \frac{5}{4} \times 36\langle x, x\rangle^2 =$ Dedekind $\times$ Okubo is exactly the *unique* quartic-free weighting of the Step-11 rigidity theorem. Both Step-12 certificates are killed ($\Phi_{T_3} : 32 \to 0$, $\Phi_P : 72 \to 0$), and the Step-14 slice $\psi = 64$ is *supplied exactly* — the cubic $d$-channel is *not needed* ($c_d$ free $= 0$). The remainder sits inside the exchange span (the bulk axion cancels every channel with the *same* $\tilde{\lambda}^2 = 36$, the twisted propagator carrying the zero-mode factor $1/\det_j$), and the Step-15 measure chain stays $\mu$-blind. Controls with teeth: wrong scale $c \in \{1, 2, 3, 5\}$ leaves $T_3 = (24, 16, 8, -8)$; the $h_1 \leftrightarrow h_2$ shuffle breaks $T_5$ *and* $T_3$; $SO(16)$ keeps $T_5 = 20/T_3 = -40$ — the KILL *fires* there ($E_8$ doubly special); $\mathrm{diag}(i, i)$ degenerates to the $\mathbb{Z}_2$ target; and the $\mathbb{Z}_2$/Eguchi–Hanson anchor *passes* with the same mechanism ($9\langle x, x\rangle^2 = \frac{1}{4} \times 36\langle x, x\rangle^2$, coupling scale $2 = |\mathbb{Z}_2|$ = centre count). The mandatory fence: the completion reading (loop = unphased sector trace with the same zero-mode normalisation) is *declared* within this step, supported by the $\delta_1$ modular-completion finding, *not* derived from the BCOV integral here — the $\delta_1$ chain has since been decided in Step 19 (v518: the *derived* chiral measure fails all three testers), and the declared-vs-derived measure question has since

been decided in Step 21 (v520: the completion reading is now *derived at probe level* from two independent constructive sources — F-independence + the Quillen pairing — under the typed premises TP-1..TP-4); the residual [O] is the constructive BCOV derivation of the $w_m$ normalisation itself. [verification/v516_celestial_wp5e_m2_twisted_ks_measure.py]

**Step 18 — the $a_0$ uplift: four coupled centre-count scales [E]/[C].** The M3 milestone preregistered in Step 15 is now *executed*, with the preregistered SUCCESS wording ("a log-type correction whose coefficient is tied to the source charge $4 = |\mu_4|$") met and the KILL (decoupling from the centre count) not fired. The uplift object is the generalized-Legendre-transform kernel per centre, $G(\eta_p) = \eta_p \log \eta_p$ (the Lindström–Roček / Ivanov–Roček dictionary, typed [C]); its derivative kernel $\chi = \sum_p \log \eta_p = \log P_4$ is the *log of the Step-16 family polynomial*. The bridge is well-typed and exact: the $O(2)$ section is a *null* coordinate ($\Delta_{3d} G(\eta_p) = 0$ identically for an *arbitrary* kernel), and the residue identity $\partial_x[\text{transform}(\log \eta_p)] = 1/r_p$ holds exactly — the Gibbons–Hawking potential $V = \sum_p 1/r_p$ *is* the $x$-derivative of the contour transform of $\chi$ (flux $-4\pi$ per centre, source charge $4 = |\mu_4|$). The $a_0$ correction then arrives on *four coupled centre-count scales*: (i) the asymptotic kernel $\log \chi = 4 \log \eta + a_2/\eta^2 + (a_0 - a_2^2/2)/\eta^4 + \dots$ carries the log coefficient $4 = |\mu_4|$ = centre count (the CPS "$N \log$" analogue at kernel level); at the clock-fixed seam fibre the $a_2$ slot closes and the first correction is *exactly* $a_0/\eta^4$ = the $(4, \pm 4)$ multipole in kernel clothes, with exact $m$-grading (the $m = \pm 4$ pieces are *pure* $a_0$, the $m = 0$ piece is $\varphi_0$-free) and *no* log ×power terms — the Step-15 honest zero is preserved; (ii) the once-integrated GLT tower $\sum_p \eta_p \log \eta_p = 4\eta \log \eta + p_4/(12\eta^3) + \dots$ obeys the selection rule $p_n = 0$ unless $n \equiv 0 \bmod 4$ with $p_{4k} = 4(-a_0)^k$ — every correction carries the prefactor 4; (iii) the exceptional-locus log $\chi(0) = \log a_0 = 4 \log t_0 + i(4\varphi_0 + \pi)$ ($t_0$-coefficient $4 = |\mu_4|$, clock phase $4i$; the modulus response equals 1 *at* the locus and falls off as a power law at infinity — the log lives where Step 15 said it must); (iv) the period response is $d \log \Pi_j / d \log a_0 = 1/4 = 1/|\mu_4|$ *uniformly* (the periods are linear in $t_0$: lockstep and the phase ratios $(1, i, i^2)$ are preserved exactly at first order), and integrating the shift around the $a_0$ circle gives the monodromy $e^{2\pi i/4} = i$ = *one Coxeter clock step* — the Step-2 monodromy reproduced from perturbation theory; the $\mathbb{Z}_8$ node support is $a_0$-rigid, and the $(2\pi i)^2$ fluxes are topological (they cannot move). Controls: the $(4, 0)$ multipole is clock-invariant and breaks nothing; the $\mathbb{Z}_2$/Eguchi–Hanson analogue reads $2 = |\mathbb{Z}_2|$ on *every* dial (amplitude, kernel log, exceptional log, period response 1/2, monodromy $-1$); wrong centre counts $k = 3, 5$ move the coefficient to $k$ with modulus slots $O(6)/O(10) \neq O(8)$ — the observable *moves with the centre count*; the clock-forbidden family fails both dials ($p_3 = 3 \neq 0$ breaks the selection rule, $e_4 = 0$ kills the $a_0$-log). What stays [O]: the full nonlinear Kähler potential of the resolved $A_3$ ALE (no closed form exists). [verification/v517_celestial_wp5e_m3_a0_uplift.py]

**Step 19 — the derived measure decides — and disagrees with the declared one [E]/[C].** The $\delta_1$ question that Step 17 left open (*derive* the completion reading from the Harvey–Moore/BCOV $\tau$-integral) has now been executed as a three-stage probe chain and consolidated: instead of declaring or scanning a measure, the measure is *solved for* from blockwise $SL(2, \mathbb{Z})$ covariance of the dressed 16-component Weil system. Four exact results. (i) *The completion closes*: the discriminant module of $D_5 \oplus A_3$ is $\mathbb{Z}_4 \times \mathbb{Z}_4$ with $q = (5x^2 + 3y^2)/8$; the Gauss sums are $2\zeta_8^5 \times 2\zeta_8^3 = 4 = \sqrt{16}$ (signature 0 mod 8 = rank $E_8$), the Weil relations hold exactly in $\mathbb{Q}(\zeta_8)$, the diagonal and anti-diagonal Lagrangians are the two $E_8$ gluings, and the 16-vector theta $S$-covariance drops the naive 4-character-rule residual from $2.91 = O(1)$ to $\sim 10^{-39}$. (ii) *The obstruction is a finite $\mu_4$ character, identified*: the dressed gauge blocks carry a multiplier system whose $SL(2, \mathbb{Z})$ relation defects are $(1, 1, 1)$ exactly on all 15 sector pairs — a *character* of the orbit stabilisers $\Gamma_1(4)/\Gamma_0(2)^{\pm}$, not a genuine 2-cocycle — and on $\Gamma_1(4)$ it is $\lambda(\gamma) = i^{2B+C/4}$. (iii) *The cancellation exists and is the twisted fibre block*: $G[a, b] = f_1 f_3$ is an exact identity (the twisted $\mathbb{C}^2$ gauge block *is* the weight-$(1, 3)$ axion pair), the T-fix mechanism is one line of exact phase arithmetic ($(-1) \times e(-1/6) = e(1/3) = \chi_4(T)$), and the exact cancellation table reads: bare → order 4, the three sphere axions $f_1 f_2 f_3 \to 4$, $f_2 \to 6$, $f_1 f_3 \to 1$ — *only* the twistor-fibre content cancels the $\mu_4$ system, with the strict solutions exactly $\chi_4$ (dims $(3, 3)$) and $\chi_{10}$ (dims $(1, 1)$), certified on the dressed functions. (iv) *All three preregistered testers fail*: under both derived solutions the

integral misses $T_5 = 0$ (fractions 0.5/0.83), misses the forced 4:3 leading ratio ($W_{13} = 0$) and misses $-A_{\text{fix}}$/the $\psi = -64N$ slice (spreads 1.05/1.47), and the honest $(N_1, N_2)$ orbit rebalance rescues nothing in the positive cone — a genuine KILL on the derived surface. Controls: the $\mathbb{Z}_2$/Eguchi–Hanson anchor (residual order 2, cancelled to 1 by the same mechanism), $SO(16)$ (the order-4 supply structurally absent), wrong form/wrong signature (break exactly). Fences [C]: the $f_1 f_3$ = KS-weight identification (the deck weights $(1, 3)$ are the twistor fibre weights forced by the Step-15 incidence ledger — an exact match, *not* a complete BCOV derivation) and the kernel-family convention. TENSION, stated honestly: the *declared* completion reading (Step 17) delivers the $\psi = 64$ slice and cancels the $32\,T_3$; the *derived* chiral measure (this step) fails all three testers — both are exact; the sharp open question was which of the two is the true BCOV measure. Neither result is hidden behind the other — and Step 21 has since decided the question at probe level in favour of the declared reading, sharpening this kill. [`verification/v518_celestial_wp5e_delta1_derived_kill.py`]

**Step 20 — the real structure exists — and free reflection positivity picks the same family [E]/[O].** The $\alpha$ stage of the OS twistor bridge `WOIT.OS.TWISTOR.01` is executed (`WOIT.THETA.FREE.01`). The classification is complete: exactly *two* families of anti-linear structures on $\mathbb{C}^2$ normalise the clock $\rho = \mathrm{diag}(i, 1)$ — family D ($z \mapsto \mu\bar{z}$, $|\mu| = 1$) satisfies $\Theta\rho\Theta = \rho^{-1}$ *exactly* with $\Theta^2 = +1$ ($-1$ is *impossible*: $M\bar{M} = \mathrm{diag}(|\mu|^2, 1)$ — the clock-inverting family is Kramers-free), inverts the deck, and reflects the seam circle with two cut points; family A ($z \mapsto \mu/\bar{z}$) centralises the clock projectively and *never* inverts it, with $\Theta^2 = \pm 1$ per $\mu$. The role separation is sharp: family A *defines* the euclidean section — Woit's $\rho_{\text{tw}}$ ($\rho_{\text{tw}}^2 = -1$, no real points) is replicated exactly on $\mathbb{C}^4$ and is family A globally — while family D *reflects* it (the OS conjugation; $\sigma_{\text{std}}$ with real points $\mathbb{RP}^3$). Mark compatibility pins $\mu \in \mu_4$ (a 4-element torsor with $\rho\,\Theta_\mu\,\rho^{-1} = \Theta_{-\mu}$: two clock orbits); the $\mathbb{Z}_8$ spin plane has no phase leaks ($\overline{\zeta_8^k} = \zeta_8^{-k}$); and in exact Cl(16) the Fock implementer $\Theta_{\text{Fock}} = U_r \circ K$ has $\Theta_{\text{Fock}}^2 = 2^7 > 0$ (normalised $+1$), inverts the Fock clock tower ($V \mapsto 4096\,V^{-1}$, exact scalar) and normalises the deck — while the *deck*-induced candidate has $\Theta_t^2 = 256\,\gamma_1 \cdots \gamma_{16} = (-1)^F$: the Step-2/v510 split/nonsplit dichotomy *is* the $\Theta^2 = +1$ vs $(-1)^F$ dichotomy, so the deck does *not* furnish the OS $\Theta$ — the seam-circle *reflection* does. Free reflection positivity then holds for exactly that pinned $\Theta$: on the bond cut (marks at the bond midpoints of the 16-Majorana circle) the one-particle Gram is positive definite ($(8, 0, 0)$, min eigenvalue $1.888 \times 10^{-3}$ at 40 digits), the even deg $\leq 2$ sector is $(29, 0, 0)$, and at $N = 8$ the *complete* half-sided algebra is RP with no degree truncation; the twist $\eta = +i$ is forced ($\eta = 1$ non-Hermitian); the cut *through* the sites fails exactly (det $= 0$, inertia $(3, 3, 1)$ — a lattice-placement artifact, the continuum Cauchy–Stieltjes control is strictly positive), and the clock-centralising family-A structure fails RP *structurally* ($(4, 4, 0)$): RP and $\Theta\rho\Theta = \rho^{-1}$ select the *same* family. Bonus: the anti-chiral state flips the odd sector to negative definite — an exact free-level shadow of kill test 3, living entirely in the fermionic sector. **Kill test 1 of the contract does *not* fire at the free/equivariant level and stays live on the interacting algebra**; the interacting algebra, gauge-fixed RP, OS reconstruction, chirality and $\mu_4$ incidence remain open contract work (the $\beta/\gamma$ milestones named in the contract). No marker moves. [`verification/v519_woit_theta_rp_free.py`]

**Step 21 — the measure decision: single-valuedness is derived, the completion reading wins [E]/[C].** The declared-vs-derived tension that Steps 17 and 19 left as the named open face is now *decided at probe level* (ERFOLG-A on the preregistered decision layer; the $\delta_{1e}/\delta_{1f}$ consolidation). The centrepiece is exact linear algebra: the joint invariant subspace $\{v : \rho(T)v = v,\ \rho(S)v = v\}$ of the dressed 16-component Weil system has $\dim_{\mathbb{Q}} = 8 = 4 \times 2$ and *every* kernel vector lies in the $\mathbb{Q}(\zeta_8)$-span of the two Lagrangian gluings $\{e_H, e_{H'}\}$ — both with theta series $E_4$, both carrying the *unphased* sector trace $Q^{(0)} = (36, 72, 36, 0, 0)$. Single-valuedness of the physical one-loop lattice factor — Step 19's typed premise — is then *derived* from two independent constructive sources rather than postulated. *Source 1 (the fundamental-domain lemma):* every strict chiral closure at physical weight carries a nontrivial character ($\chi_4(T) = e(1/3)$, $\chi_{10}(T) = e(5/6)$, exact; zero strict trivial-character solutions across all eleven dressings); the canonical column-matched member's total block sum obeys $G(\gamma\tau) = \chi_4(\gamma)G(\tau)$ *pointwise* (certificate $3.9 \times 10^{-16}$, nonvacuous $|G| \geq 59.6$), and an exact change of variables turns fundamental-domain independence into $\int = 0$ for *every* chiral integrand: the Step-

19 route was never a well-defined nonvanishing moduli integral — the kill is *sharpened*, its unfolded values are convention bookkeeping. *Source 2 (the Quillen pairing):* BCOV's $F_1$ pairs hol × conj(hol), and the doubled transports satisfy $|t\bar{t} - 1| < 8.3 \times 10^{-40}$ on all 15 pairs ($|\chi_4|^2 = 1$ exactly, while the one-sided square $\chi_4^2 = e(2/3) \neq 1$ keeps defects on 15/15 pairs); the doubled 256-dimensional system closes *strictly* at trivial character and physical weight (nullspaces $(28, 17) \geq 11$), contains the unphased diagonal *and* the physical columns $w_b \otimes \bar{w}_b$, and the wrong pairing hol ⊗ hol is *empty* $(0, 0)$. The consistency circle then closes: the forced unphased numerator, inserted into the Step-19 scaffold, reproduces the Step-17 Okubo squares *levelwise* ($\sum_m V \propto \langle x, x\rangle^2$ exactly on every level $n \leq 8$; $J$-weighted spreads $\sim 5 \times 10^{-41}$, zero negative cells) with $\psi(\text{skeleton}) = +0.2302 = -\psi(\text{contact})$ — the $J$-weighted mirror image of $\psi(A_{\text{fix}}) = +64/\psi(\text{contact}) = -64$. The declared reading also wins the two decision-layer discriminators: the physical AB column $\sum_m i^{bm} e_{(m,m)}$ is contained *exactly* in the $\chi_4$ family ($\sigma \sim 1.8 \times 10^{-16}$, one scalar per orbit — canonicalising the $(N_1, N_2)$ freedom) yet the canonical member still fails every tester; and at the $\mathbb{Z}_2$/Eguchi–Hanson anchor only the declared reading hits ($9\langle x, x\rangle^2 = \frac{1}{4} \times 36$, $\psi(A_2) = -8$, scale tooth $c = 1/2 = |\mathbb{Z}_2| h^{A_1}$ unique) while all three derived instantiations fail. Typed premises [C] (the mandatory fence): TP-1 (F-independent moduli integral), TP-2 (nonvanishing — it must source $\psi = 64$), TP-3 (the Step-19 kernel convention), TP-4 (the Quillen structure of BCOV $F_1$). What stayed [O] — the constructive BCOV derivation of the $w_m$ normalisation itself — is since executed by Step 24, narrowing the [O] to the global BCOV integral beyond the fibre zero-mode factor. No marker moves. [verification/v520_celestial_wp5e_measure_decision.py]

**Step 22 — free OS positivity does not see the bit: the eighth side-blind test [E]/[C].** Could the Step-20 machinery *derive* the Step-2 alignment bit? The hypothesis — free RP or Θ existence breaks for $\delta \neq \pi/2$ on the deformed mark family $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$ — is now decided *negative*, exactly as preregistered (KILL branch). The census is exact and symbolic: for *every* $\delta$ the two mark-swapping reflections $\Theta_{\pm e^{i\delta}}$ exist (all 20 cut/mark incidence solvesets empty), while mark-*fixing* reflections exist iff $\delta = \pi/2$ (solveset $\cos\delta = 0$, lost immediately at $\pi/2 - \varepsilon$). Free RP is $\delta$-blind in the strongest sense: the bond-cut Gram inertias are $(8, 0, 0)$ PD for all tested $\delta$ with the full sorted spectra *identical to* 40 *digits* (deviation 0.0); the $N = 16$ odd-$m$ failure is a lattice-placement artifact (at $N = 32$ all $m = 1..7$ have mark-compatible bond axes with PD inertia $(16, 0, 0)$); the continuum OS kernel in axis-adapted coordinates is *exactly* $1/\sin((s+t)/2)$ — the axis position drops out identically — and the v510/v512 counterwitness passes free RP. Θ existence is $\delta$-blind too ($\Theta^2 = +1$, deck normalisation, Fock implementability $U^2 = +2^7/2^8 > 0$ for every $\delta$); the *only* $\delta$-sensitive clause of the Step-20 battery ($\Theta\rho\Theta = \rho^{-1}$) is for $\delta \neq \pi/2$ not violated but *not formulable* (a mark-compatible order-4 rotation exists iff $\delta = \pi/2$) — it presupposes the bit instead of deriving it. The battery has teeth at every $\delta$ (family A fails RP structurally $(4, 4, 0)$; the chirality–$\eta$ pinning persists $(0, 8, 0)$): RP separates the real-structure *families*, never the *sides*. Structure gained [C]: the group generated by the admissible OS reflections is $V_4$ for every $\delta < \pi/2$ and closes to $D_4$ exactly at the clock point ($\Theta_i \circ \Theta_1 =$ the clock) — face #13 of the Step-2 equivalence web ($12 \to 13$). The free RP/Θ battery is hereby the *eighth* side-blind test on the v512 scoreboard ($7 \to 8$); the honest [O] gap is the mark-decorated *state* class (defect/twist insertions, the interacting $\mathcal{A}_{\text{hol}}$) — the mark-decorated half of that gap is since decided side-blind too (Step 26: the *ninth* test, $8 \to 9$; the free-plus-twist class is exhausted, the residual gap narrows to the genuinely interacting $\mathcal{A}_{\text{hol}}$) — the alignment bit remains genuine discrete input. No marker moves. [verification/v521_seam_bit_rp_blind.py]

**Step 23 — the clock is time-like, GSO is the gauge datum [E]/[C]/[O].** The $\beta_1$ stage of the OS twistor bridge is executed as a *typed* result (verdict UNDECIDED per the frozen preregistration — the preregistered reading was the wrong operationalisation, and the module documents why, including its own first frozen run at 6/13 and a preregistration combinatorics error 30 vs 32). The mechanism: the one-step clock insertion $T_1 = \langle \theta(e_a), \alpha_S(e_b)\rangle$ violates Hermiticity *exactly* (witness entry $-i/(8\sin(5\pi/16))$ against $+i/(8\sin(5\pi/16))$; the pairing is complex-symmetric — 745 matching, 96 anti-matching, 0 violations — and "OS-symmetric" is strictly weaker than Hermitian), so "positivity after gauge fixing" is not well-posed for the clock reading. The typing is

a census: *all* 16 dihedral reflection axes of the NS seam circle invert the clock lift ($RSR^{-1} = S^{-1}$), none commutes — the $\mu_4$ clock *is* Woit's euclidean rotation, and the gaugeable ($\theta$-compatible) part of its $\mathbb{Z}_8$ Fock tower is exactly the 2-torsion $\{1, (-1)^F\}$ = the GSO/fermion-parity $\mathbb{Z}_2$ (the free-fermion shadow of the $E_8$ glue: $(E_8)_1$ = even NS of $SO(16)_1$ + R spinor). Under that corrected typing gauge-fixed RP *holds exactly*: $(29, 0, 0)$ PD at $N = 16$ (deg $\leq 2$, min eigenvalue $1.78 \times 10^{-6}$), $(8, 0, 0)$ PD on the complete $N = 8$ half algebra; the site-cut defect survives gauge fixing ($(7, 9, 6)$ — the bond placement is gauge-invariant information); family A stays indefinite ($(17, 12, 0)$); the Ramond control forces the NS/$\mathbb{Z}_8$ tower. Kill test (2)'s free shadow does *not* fire, kill test (1) stays discharged also gauge-invariantly; both stay live on $\mathcal{A}_{\text{hol}}$, and the clock-equivariant statement is re-routed through $\beta_2$ (OS quotient first, then the clock as reconstructed transfer operator — contract precision (iii)). No marker moves; `WOIT.OS.TWISTOR.01` stays **[O]**. [`verification/v522_woit_beta1_gso_gauge.py`]

**Step 24 — the $w_m$ normalisation derived: the fixed-point factor is computed, not declared [E]/[C].** The named remaining target of Steps 17–21 — the *constructive* BCOV derivation of the completion normalisation $1/\det_j$ — is now executed (verdict ERFOLG per the frozen preregistration), from three independent sources. *Route (i), Atiyah–Bott:* the equivariant mode ledger of the twistor fibre $\mathbb{C}^2$ — $c_n = \sum_{a+b=n} i^{j(a-b)}$ — equals the closed form $1/((1-q\,i^j)(1-q\,i^{-j}))$ exactly in $\mathbb{Q}(i)$ at every order $n \leq 120$; the closed form is *regular* at $q = 1$ (the twisted sectors carry no volume pole) with Abel value $(1/2, 1/4, 1/2) = 1/\det_j$, $\det_j = \det(1 - g^j) = (2, 4, 2)$; the $(C, 2)$ Cesàro means converge in exact arithmetic; and the fixed-point factor *splits off exactly at every truncation level* $d \in \{7, 12, 25\}$, for the phased (skeleton) *and* the Step-21-forced unphased (completion) trace alike — the Abel-limit contact$_j = (Q^{(0)} - Q^{(j)})/\det_j$ is the Step-17 contact vector *computed*, not declared. *Route (ii), zeta/Quillen:* the spectral determinant of the $g^j$-twisted circle Laplacian is $4\sin^2(\pi j/4) = \det_j$ exactly (Lerch + reflection formula symbolically; Hurwitz certificates $\sim 10^{-41}$); the Quillen split gives $\det \Delta = \det_j^2$ with $|\det \bar\partial| = \det_j$ and the real *positive* holomorphic section unique via the $SU(2)$ conjugation pairing; and the $\delta_{1f}$ modular block $f_1 f_3$ at the $(0, b)$ nodes carries the exact constant term $1/\det_b$ (deviations $\leq 5.2 \times 10^{-28}$) — the Step-19/21 transport machinery carries the fixed-point weight verbatim. *Route (iii), the consistency chain:* the derived weight reproduces the Step-17 chain number by number — $w_m = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4| h_m = -4\,\mathrm{ch}_2(T_m)$, $T_5 = 0$ at every scale, $h_2 : h_1 = 4 : 3$, the $T_3$ budget $32 - 8c$ forcing $c = 4 = |\mu_4|$ uniquely, per-sector Okubo squares $(18, 9, 18)\langle x, x\rangle^2$, total $45\langle x, x\rangle^2$, $\Phi/\psi$ certificates $\pm 64$. The negative controls have teeth: $1/\det^2$ leaves $(T_5, T_3) = (2, 12)$ and contradicts the Quillen split; $1/|1 - i^j|$ leaves $\mathbb{Q}$ ($w_1'' = 1 + \sqrt{2}$); the $\mathbb{Z}_2$/Eguchi–Hanson anchor *passes* with the same derivation ($\det = 4$, $w = 1/2 = |\mathbb{Z}_2| h^{A_1}$, $c = 2 = |\mathbb{Z}_2|$); the $SO(16)/D_8$ kill fires; $\mathrm{diag}(i, i)$ breaks the zeta = AB identity itself; and on $\mathbb{C}^2/\mathbb{Z}_k$ for $k = 3, 5$ the weights wander correctly ($w_m = m(k-m)/2 = k\,h_m$ exactly in cyclotomic arithmetic) — only $k = 4 = |\mu_4|$ carries the $E_8$-glue chain. Typed premises [C] (the mandatory fence): TP-REG (Abel/Cesàro/zeta regularisation *is* the definition of the equivariant character), TP-Q (the Quillen structure of BCOV $F_1$ = Step-21 TP-4), TP-NUM (the unphased numerator = the Step-21 decision layer), TP-CH (the insertion channel computes the box-channel weight). Nothing in $w_m$ is declared any more at this level; what stays **[O]** is the *global* BCOV integral on $\mathbb{PT}/\mathbb{Z}_4$ beyond the fibre zero-mode factor (and the resolved-ALE route not built). No marker moves. [`verification/v523_celestial_wp5e_wm_derived.py`]

**Step 25 — the OS quotient made explicit: the clock gets its spectral calculus [E]/[C].** The $\beta_2$ stage of the OS twistor bridge is executed (verdict SUCCESS per the frozen preregistration, [C]-typed per contract precision (iii)). $\mathcal{H}_{\text{phys}}$ is *explicit and nondegenerate*: at $N = 16$ (deg $\leq 2$) the $37\times37$ bond-cut OS Gram is exactly Hermitian and parity-block-diagonal with inertia $(37, 0, 0)$ PD (min eigenvalue $1.7801\times10^{-6}$ at 40 digits) and null space $\{0\}$, $\dim 37 = 29 \oplus 8$; at $N = 8$ the *complete* half algebra gives $(16, 0, 0)$ PD with $\dim 16 = 8 \oplus 8 = 4^2$ — compact euclidean time reconstructs a *thermal* (KMS) representation, with the exact certificate $\sin^2(3\pi/8) - \sin(\pi/8)\sin(5\pi/8) = 1/2$. The euclidean rotation becomes a Klein–Landau *local symmetric semigroup*: $\tau_k$ is exactly Hermitian on every shrinking domain ($k = 1..7$), the chain identities are exact, the vacuum is fixed, and the positivity pattern *is* the Step-20 site/bond dichotomy — even steps are PSD via the exact

square identity $T(2j) = A_j^* A_j$, odd steps are indefinite with exactly zero one-particle diagonal (the one-step transfer is *not* positive: the free chirality datum, kill-test-3 shadow sharpened). The clock is the quarter turn $T^{N/4}$: positive self-adjoint on its domain ($(11, 0, 0)$, min $6.7 \times 10^{-5}$; compressed spectrum $[0.0195, 1.6414]$ with vacuum eigenvalue exactly 1), with certified spectral projections ($\sim 10^{-40}$) — exactly the calculus the non-Hermitian pre-quotient average of Step 23 could not have — and at $N = 8$ the compressed clock spectrum is *exactly* $\{1, \sqrt{2} - 1\} = \{1, 1/\delta_S\}$ in both parity sectors: the silver axes of the Step-20 $\mu_4$ torsor return as the clock eigenvalue; the reconstructed rotation group $U(s) = e^{isH}$ is unitary with the group law — per precision (iii) the [C]-operationalisation of "the clock acting unitarily". The Step-23 non-Hermiticity is *resolved*: the census $(745, 96, 0)$ is reproduced and every anti-matching entry is a wrap overlap (every overlap-free entry matches) — the pre-quotient failure was exactly the domain/wrap artifact. The pre-declared KMS deviation arrives with exactly the declared witnesses: no contraction on the compact circle (the edge mode expands by $C(1)/C(3) = 1 + \sqrt{2} = \delta_S$ exactly; $\det(G - \tau_4) < 0$). GSO/$\Theta$: the grading survives, the *perpendicular* torsor mirror descends anti-unitarily with $\Theta^2_{\text{phys}} = +1$ on every sector (the quotient is Kramers-free), and $\theta_{\text{cut}} \circ \theta_\perp = \alpha_{N/2}$ exactly (the torsor closes onto the deck). Controls: the site cut is indefinite (the contract kill branch fires for the site placement), family A admits no quotient, the anti-chiral state flips to $(8, 8, 0)$ — quotient existence itself selects the chiral orientation. Kill tests (1)/(2) strengthened, (3) shadow sharpened, none fires; all seven stay live on $\mathcal{A}_{\text{hol}}$; $\beta_3$ is next. Transparency: run 1 scored 18/20 — a sympy simplify failure on a *true* $\pi/16$ product formula, fixed by a Laurent-polynomial certificate; no criterion changed. No marker moves; `WOIT.OS.TWISTOR.01` stays [O]. [`verification/v524_woit_beta2_os_quotient.py`]

**Step 26 — the twist-state kill: the ninth side-blind test, and the free-plus-twist class is exhausted [E]/[O].** Step 22 left one named door: marks decorating the *state* (defect/twist insertions). That door is now decided *negative*, exactly as preregistered (KILL) — the *ninth* side-blind test on the v512 scoreboard ($8 \to 9$). Every well-defined mark-decorated state functional in the free-plus-twist class is RP-blind on the always-existing cuts along the whole $M(\delta)$ ladder: the $\sigma$-gauge arc decoration satisfies $\sigma \circ r = \sigma$ on the swap axes, so the twisted Gram is a diagonal unitary congruence — spectra *identical* to 40 digits; the Kadanoff–Ceva 4-twist insertion $\omega(D\,\cdot)/\omega(D)$ is well-defined on the $N = 32$ ladder ($\omega(D)$ real nonzero; the $N = 16$ odd-$m$ degeneration is the checkerboard artifact) and its RP spectra are the *first* free-class data of the programme that depend on $\delta$ at all (pairwise up to 1.05) — yet the *inertia* stays $(16, 0, 0)$ PD with the same $\eta = +i$ for every member $m = 1..7$: the decoration sees $\delta$, the positivity does not; the $\mu_4$ defect at $\beta = \pi/2$ is pure plane gauge, and the genuine Bogoliubov defect at $\beta = \pi/4$ — where $\Theta$-compatibility selects exactly 2 of 16 sign patterns (the crossed pattern forced by the NS wrap signs) and the winner is genuinely non-monomial — is still PD with spectra again identical. The structure theorem behind the blindness is exact: the defect planes never straddle a cut bond, so every $\Theta$-compatible *quasi-free* mark decoration is RP-spectrally *invisible* (orthogonal congruence). The sub-results sharpen the verdict: the defect energy $E_{\text{def}} = 1.2752872$ is exactly $\delta$-independent (spread $4 \times 10^{-40}$); the 4-twist Casimir $\ln|\omega(D)|$ *measures* $\delta$ smoothly but is exactly mirror-symmetric under $m \leftrightarrow 8 - m$ — the $\delta \leftrightarrow \pi - \delta$ equivariance makes $\pi/2$ the symmetric point of every invariant readout: a family gauge, never a side selector; the twisted parity $\langle\Gamma\rangle_{\text{tw}} = +1$ for every member. The harvest: two genuinely twist-visible structures — the $\eta$ flip $+i \to -i$ on the mark-fixing axes ($\sigma \circ r = -\sigma$) and the twist frustration of the mark-fixing mirror ($(4, 4, 0)$ indefinite for both $\eta$: the lattice shadow of the Ising twist field's mirror-oddness) — both live *only* on the fixing axes existing iff $\delta = \pi/2$: they presuppose the bit (the v512 facet class). Controls: the twist-free limit reproduces Steps 20/22 exactly; the site cut keeps failing; the v512 counterwitness is arc-equivalent at $N = 16$ to the blind member $\pi/4$ — no state-level selector can exclude it; a single twist has $\omega(D) = 0$ identically (twists exist only in mark pairs). The free-plus-twist class — everything Wick-computable — is hereby *exhausted*: what could still see the bit is only a genuinely *interacting* $\mathcal{A}_{\text{hol}}$ whose OS data are not Pfaffian-reducible to the chiral vacuum — and that door is since probed by Step 30 (the tenth side-blind test $9 \to 10$, with Kill-Test 2 firing at toy level under a typed fence) — and answered *positively at selector level* by Step 31: reflection positivity, imposed on the interacting mark family,

keeps exactly one member alive, $\delta = \pi/2$. The alignment bit remains genuine discrete input. No marker moves. [verification/v525_seam_bit_twist_blind.py]

**Step 27 — the thermal seam: KMS closes onto the Nariai/BH chain as the third leg of $c_3$ [E]/[C].** The free OS quotient of Step 25 is not merely thermal in dimension $(4^2)$: its temperature is *measured*, not assumed. The circle kernel $C(d) = (2/N)/\sin(\pi d/N)$ admits exactly one finite thermal-mode representation with detailed balance — at $N = 8$ one real frequency $\omega = \operatorname{arcsinh}(2^{-3/4})$ with weight ratio $\kappa = q^{-8}$ exactly ($\beta = 8 = N$; Q-polynomial identity $Q^2 - (2+\sqrt{2})Q + 1 = 0$), at $N = 16$ both mode pairs real with pairing exponent $(N/2) + 1 = 9$ ($\beta = 16 = N$), and wrong candidates $\{N/2, N/4, 2N\}$ are excluded by quantified gaps. In angle units (one clock step $= 2\pi/N$, the v519 pin) one obtains $\beta_{\text{angle}} = 2\pi$ exact, hence $T_{\text{seam}} = 1/(2\pi) = 4c_3$ — $N$-independent, and exactly the Bisognano–Wichmann/Hawking modular normalisation that v208 carries into the horizon sector. The modular Hamiltonian is closed ($\nu_1 = \cos(\pi/24)$, $\nu_2 = \cos(7\pi/24)$; $\varepsilon_i = 2\ln\cot(\theta_i/2)$; Tomita KMS-$\beta = 1$ on all 256 monomial pairs). The anti-numerology audit selects the class-1 identity $r = 1$: temperature joins geometry and anomaly as the *third leg* of $c_3$. The Hawking/Nariai chain then reads in the same normalisation: $T_H = c_3/M = 1/(8\pi M)$ (the axiom *is* the Hawking coefficient, now from the seam KMS), SdS $1/(4\pi) = 2c_3$, Nariai $T_N = 4c_3\sqrt{\Lambda}$, and the v104 cross-link $2H/T_N = 4\pi = 1/(2c_3)$ exact. Entropy confirms the chiral $c = 1/2$ log-law ($\Delta S/\ln 2 \to 1/6$) but the v129 horizon fractions $\{1/3, 2/3\}$ and the v190 floor do *not* transfer — the temperature bridge closes, the entropy-fraction bridge honestly does not. Typed non-gating: the naive clock reading $\{1, \sqrt{2}-1\} = e^{-\Delta K}$ fails; the exact replacement is $\sqrt{2} - 1 = C(3)/C(1) = 1/(2\cosh 2\omega - 1)$ — the silver eigenvalue is a thermal kernel ratio (Step 28). Non-compact control: the Stieltjes half-line has $\kappa = 0$, $\beta = \infty$, $T = 0$ — the temperature hangs exactly on compactness of the euclidean circle. [C]-fence: the reading "seam euclidean circle = thermal circle of the reconstructed horizon dynamics". No marker moves. [verification/v526_seam_thermal_kms_nariai_bridge.py]

**Step 28 — the silver eigenvalue explained and demystified [E].** The compressed clock spectrum $\{1, \sqrt{2}-1\}$ of Step 25 is now closed-form: at $N = 8$ the Hankel pencil is diagonal, $\tan(\pi/8) = C(3)/C(1)$, the doubling is exact fermionic second quantisation, the geometric carrier is the silver-midpoint axis of the $\mu_4$ torsor, and the gap is $\operatorname{arcsinh}(1) = \mathrm{gd}^{-1}(\pi/4)$. $N$-scaling ($N = 8..24$) shows a pure $N = 8/16$ Majorana *lattice* datum — not a continuum limit; a positive $\mu_4$ clock exists only for $N \equiv 0 \pmod 8$; the golden family appears as the $N = 10$ control. The connection verdict is an honest KILL in the v100/v513 style: a preregistered 11466-candidate census over the 16 frozen registry numbers yields 0 hits above null expectation, with golden/plastic placebos comparable — no readout connection. Cross-link to Step 27: $\sqrt{2} - 1$ is the thermal kernel ratio. Double verdict; no marker moves. [verification/v527_seam_clock_silver_spectrum.py]

**Step 29 — the bit as a twist-class choice: the physical definition [E]/[C].** Step 26 harvested two bit-presupposing $\pi/2$ structures (the $\eta$ flip and the twist frustration). Both are now exact *equivalences* to the bit, both directions each. Facet #14: $\eta$-flip $\Leftrightarrow \delta = \pi/2$ ($\sigma \circ r = -\sigma$ on all four fixing axes; off $\pi/2$ on every axis $\sigma \circ r = +\sigma$; the cardinality lemma $\sigma \circ f = -\sigma \Rightarrow |A_+| = |A_-| \Rightarrow m = 4$; $g$-class census over all 224 axes: flip class exactly at $\{15, 31\}$ for $m = 4$). Facet #15: twist-frustration $\Leftrightarrow \delta = \pi/2$ ($\theta$ exchanges $D_{\text{short}} \leftrightarrow D_{\text{long}}$; Gram indefinite $(4,4,0)/(8,8,0)$ exactly at $\pi/2$; off reducible and PD; same cardinality lemma). The order parameter $O$ = flip-axis fraction equals $1/2$ at $m = 4$ and 0 everywhere else, with class alternation $(-,+,-,+)$ the nontrivial $\mathbb{Z}_2$ character of the $\mathbb{Z}_8$ clock-tower shadow; gauge-robust under every v525 freedom. Unification: $\eta$-flip = frustration = flag-transitivity are *one* $\mathbb{Z}_2$ invariant (same arc system); the v510 Kramers class is another (constant on the family) — Kramers = POSITION half (derived), twist class = MODULUS half (the input). The v512 web grows $13 \to 15$ facets. [C] measurement sketch (principle only): defect transport around the axis torsor accumulates the $\eta$ holonomy = a relative $\pi$ phase iff the bit is set — in principle interferometrically readable. Honest fence: the bit remains formal input; a definition replaces no derivation. No marker moves. [verification/v528_seam_bit_twist_class_definition.py]

**Step 30 — the interacting FK toy: the first firing Kill-Test-2 shadow [E]/[C]/[O].** The door left open by Steps 22/26 — a genuinely interacting $\mathcal{A}_{\text{hol}}$ — is probed on the minimal

Fidkowski–Kitaev quartic on the 16-Majorana NS seam circle (256-dim, exact). Triple verdict. F1 = KILL (the most important part): $\Theta$ exists exactly on the interacting model ($U_r^2 = 2^8$, $\Theta^2 = +1$, clock-tower inversion $\mathrm{conj}_V \cdot V = 4096$, $[\Theta, H_g] = 0$, $\Theta\Omega_g = \Omega_g$) — Kill-Test 1 does *not* fire; but Kill-Test 2 *fires at toy level*: reflection positivity breaks in the interacting ground state for every $g > 0$ — inertia ladder $(37,0,0) \to (33,4,0) \to (31,6,0) \to (29,8,0)$ (most negative eigenvalue $-4.5 \times 10^{-2}$); the Gibbs state at $\beta = 2$ carries the same ladder; the mechanism is interference (neither crossing nor half-local terms alone break); the STRADDLE LAW holds on all 24 entries — RP fails exactly on quartet-straddled cuts and stays PD on quartet-avoiding ones (the 24 entries cover *asymmetric* straddles only — refined in Step 31, since typed as the *odd-split law* in the refinement note below). Mandatory fence: one toy, one interaction class, [C] flat-band parent — this *narrows* the Woit route (every $\mathcal{A}_{\mathrm{hol}}$ must protect RP against exactly this mechanism; the straddle law is a concrete hurdle for $\beta_3/\gamma$), it does *not* close it; `WOIT.OS.TWISTOR.01` stays [O]. F2 = KILL: the *tenth* side-blind test on the v512 scoreboard ($9 \to 10$), the first with genuinely non-Wick-computable dynamics — the interaction sees $\delta$ massively (spectral spread 1.32), but the mark-axis mirror $\Theta_{15}$ maps $H(m) \to H(8-m)$ exactly; all side data are mirror-equal. F3 = ERFOLG: grading and the nonsplit $\mathbb{Z}_8$ tower survive ($\tilde{U}^2 = 256\Gamma = (-1)^F$, $V^2 = 256\tilde{U}$); the FK fixed-point anchor holds; the v524 OS quotient survives $g > 0$ constructively on the $\delta = \pi/2$ mark-anchored model. No marker moves. [`verification/v529_seam_interacting_toy_fk.py`]

**Step 31 — the straddle cone executed: reflection positivity *selects* the bit [E]/[C].** The review contract left by Step 30 (independent couplings $\lambda_q$ on the four mark quartets of $M(\delta)$; the RP cone on the equivariant family) is executed, double-verdicted. V1 = KILL: the literal *leading-order* cone is empty for every member and both signs (first-order perturbation of the OS Gram via the exact reduced resolvent, Hermitian to $10^{-15}$, central-difference regression $2 \times 10^{-8}$) — the linear pencil cannot carry the selection; the recoverable leading-order content is the straddled-cut drift sign pattern ($\pi/2$: $+1.4 \times 10^{-4}$ for $s{=}+$, $-3.7 \times 10^{-2}$ for $s{=}-$; the asymmetric members exactly reversed) — necessary, not sufficient. V2 = ERFOLG, *the selection law* (the central datum): among *all* well-posed members and both coupling signs, reflection positivity holds on *every* admissible cut over the whole grid $g \in \{1/32..8\}$ for *exactly one* combination — $\delta = \pi/2$ with *positive* coupling: all four Grams PD $(37,0,0)$ *including the previously untested straddled clock axes* 7/15, minimum eigenvalue *lifted* off the free RP boundary ($1.78 \times 10^{-6} \to 5.5 \times 10^{-5}$ near $g = 1/4$), Gibbs $\beta = 2$ identical; the asymmetric members die ($s{=}+$ already at $g = 1/32$, $s{=}-$ from 1/8: the protection at $\pi/2$ is *nonperturbative*). The Step-30 straddle law refines: asymmetric straddling kills RP, the *symmetric* straddling of the self-mirror member protects it. The geometry behind the dichotomy is exact: $\delta = \pi/2$ is the *unique* member whose four quartets partition the 16-circle and whose straddling quartets are reflection-closed across their cuts with sign $+1$ (stabilizer = the full clock+mirror group, four admissible cuts $\{3,7,11,15\}$); the equivariant Fix rays are uniform on $m = 2..6$ and NS-wrap-twisted on the tight members $m = 1/7$. Typed cross-link: the RP-selected member is exactly the carrier of the Step-29 twist-class order parameter $O = \frac{1}{2}$ — the dynamical selector and the kinematic definition point at the *same* member. Honest fence: one toy, one interaction class, [C] flat-band parent, deg $\leq 2$ OS bases — toy-level *evidence for* the bit's dynamical origin, not a derivation; `WOIT.OS.TWISTOR.01` stays [O]. No marker moves. [`verification/v534_seam_straddle_cone.py`]

**Steps 30/31 refined — the odd-split law: the mechanism made decidable, and the Kill-Test-2 counterweight narrowed [E]/[C].** The seam interaction front of 2026-08-27 upgrades the straddle law to a decidable criterion. *(i) The odd-split law:* the Jaffe–Pedrocchi sufficient condition for Majorana reflection positivity (Jaffe–Pedrocchi `arXiv:1406.1384`; Wei–Li–Xiang `arXiv:1601.01994`), made decidable on the toy, predicted RP survival on *center*-straddled cuts (2+2 mirror split; the $m = 4$ model on the clock axes 7/15) and direct measurement confirmed it ($(37,0,0)$ at $g = 0.5/1/2$): a straddled cut kills RP iff its quartet split is *odd* (1+3). The JP-certified vs measured RP confusion matrix over all 35 cells is perfect (23/12/0/0); the $\mu_4$-covariant quartic interaction space (dimension 464) contains *no* direction JP-certifiable on all cuts — the threat stays generic. *(ii) OS viability on the avoiding family:* the quartet-avoiding cut family of the

clock-invariant $m = 4$ member generates the *full* even seam algebra (rank 15, dimension 32768) and carries a *positive* OS/GNS inner product at $g = 1$ (minimum $+4.4338 \times 10^{-5}$, GNS 37/37, spectrum deforming continuously from $g = 0$) while the straddled control genuinely fails ($-6.936 \times 10^{-2}$); the $\mu_4$ clock maps the avoiding pair $\{3, 11\}$ onto itself exactly. Reading: the Kill-Test-2 counterweight is *narrowed* — the toy exhibits an OS-viable cut geometry selected by its own clock — *not* removed. All toy-level [E]; fence inherited (one toy, one interaction class, [C] flat-band parent); `WOIT.OS.TWISTOR.01` stays [O]. No marker moves. [`verification/v972_seam_interaction_front.py`]

**Step 32 — the $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality typed: the OS cut and the $(2, 2)$ signature are forced, and the kill branch is empty [E]/[C].** The WOIT-$\beta_3$ milestone, executed as a compatibility-class construction. A $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality is a nondegenerate Hermitian form $\Phi$ on $\mathbb{C}^4$; clock compatibility forces the two-block structure exactly ($16 \to 8$ real parameters; the deck adds nothing, the wrong clock lift breaks the split), and the pairing against Woit's Euclidean structure $\rho_{\text{tw}}$ splits the class into exactly two branches: the *Euclidean* branch ($\sigma_{\text{std}}$'s home; seam-line norm $a(|z|^2 + 1)$ — no null point, no OS cut, ever) and the *Minkowski* branch, whose null locus meets the seam line *exactly in the seam circle* (the OS cut is forced, not imposed) and whose total signature is $(2, 2)$ for every nondegenerate member (the second block is minus the conjugate of the first; charpoly mirrored): **Woit's Minkowski signature is a theorem of clock + $\rho_{\text{tw}}$ pairing**. The contract kill branch — "the duality forces a clock-centralising (family A) structure" — is *empty by algebra*: every compatible duality inverts clock and deck (family D; family A would need $\rho_4^2 =$ scalar, but $\rho_4^2 = \text{diag}(-1, 1, -1, 1)$), and family A independently has no quotient ($(4, 4, 0)$). Every seam-preserving Minkowski member induces *one* projective conjugation $z \to -\bar{z}$ ($\mu = -1$, fixed points $\pm i$), perpendicular to $\sigma_{\text{std}}$'s $z \to +\bar{z}$; under the declared RP-side placement (exact rational-$\pi$ arithmetic, offsets $\mp\pi/2$) that member *is* $\theta_\perp$ — the mirror whose descent Step 25 certified: $\Theta_{\text{phys}}$ is the shadow of the $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality (quotient regressions re-verified: Gram PD, anti-unitary with $\eta_\perp = +i$, $\Theta^2 = +1$ per sector, time reversal, torsor closure onto the deck). Role separation complete: $\sigma_{\text{std}} \sim$ the OS cut (the metric), the $\Phi$-twisted Minkowski duality $\sim$ the descending mirror. Honest fence: free/quotient level only, $\mathcal{A}_{\text{hol}}$ untouched, all seven kill tests live there; the declared conventions (seam line, seam preservation, placement, torsor relabeling) are cited from Steps 1/20/25, not invented; $\gamma$ is next. No marker moves. [`verification/v565_woit_beta3_pt_duality.py`]

**What remains, honestly (WP5e proper [O]).** The single open step is the *global* BCOV/Kodaira–Spencer quantisation on $\mathbb{PT}/\mathbb{Z}_4$: the boundary partition function $E_4/\eta^8$ *derived from the twistor side*, including the $q^{-1/3}$ prefactor whose CFT side Step 9 has pinned and whose equivariant skeleton Step 11 has verified — the exchange sub-branch is closed by Step 12, the full-tensor ledger is executed by Step 14, the level-from-flux dial is executed by Step 13, the bulk-axion slot is built by Step 15, all three back-reaction milestones it preregistered are executed by Steps 16–18 (the Step-15 fence M1–M3 is *fully worked off*), the $\delta_1$ chain is *decided* by Step 19 (kill under the derived measure), and the measure question is *decided at probe level* by Step 21 (the declared completion reading wins: single-valuedness derived from F-independence + the Quillen pairing under the typed premises TP-1..TP-4; the derived chiral measure was never a well-defined nonvanishing moduli integral), and the constructive BCOV derivation of the $w_m$ normalisation itself is *executed* by Step 24 ($1/\det_j$ computed from three independent sources; nothing in $w_m$ is declared any more at that level), so the named remaining target narrows to the *global BCOV integral on* $\mathbb{PT}/\mathbb{Z}_4$ *beyond the fibre zero-mode factor*; the cubic $d$-channel stays not needed and its physical-justification question ($c_d = 32 \times 60$; the $1920 = |W(D_5)|$ reading look-elsewhere-loaded and convention-contingent, v513) stays dissolved (Steps 19 and 21 do not revive it) — together with the continuum uplift of the Step-10 lattice witnesses (Xu's theorem for the abstract orbifold statement; the concrete seam quotient net and the interacting condensed $(E_8)_1$ net are Costello–Li territory). The route does *not* close `SEAM.EQUIV.01`; it *is* the keystone's second, quantitative route, named in its own right as `SEAM.EQUIV.TWISTOR.01` ([O]; the conditional MMST route is `SEAM.EQUIV.MMST.01`, closed modulo cited theorems; the parent closes if either route closes and stays [O] as an unconditional claim) — the same two-step shape as MMST (limit + maximal ideal) with the ideal quantitatively fixed, the deleting operator explicit, the limit state constructed, the index chain measured, the

KLM triple witnessed, both faces of the inflow anchored, the collapse confirmed full-tensorially with one cubic door open, one level a theorem with its value pinned by the sector counter, the bulk-axion slot a construction with $\tilde{\lambda} = 6$ triply pinned, the back-reacted $\Omega_N$ closed-form with $(2\pi i)^2$-integral lockstep periods and the source charge $4 = |\mu_4|$ forced, the twisted KS measure (on the declared completion reading) landing on the unique quartic-free weighting, the $a_0$ uplift coupled to the centre count on four scales, the $\delta_1$ chain decided — kill under the derived measure — the measure question decided at probe level for the declared reading (Step 21) and the $w_m$ normalisation itself derived constructively (Step 24) — while the global quantisation (now narrowed to the global BCOV integral beyond the fibre zero-mode factor) stays open. `SEAM.EQUIV.01` stays [O] throughout; no marker moves.

**The Woit bridge: chiral Wick rotation and the real structure [O].** One *external* programme is close enough in shape to deserve a named paragraph — named precisely so that proximity is not mistaken for confirmation. Woit's *Euclidean Twistor Unification* (`arXiv:2104.05099`) starts from the observation that the usual Wick rotation is problematic for *purely chiral* theories (the Euclidean continuation of a chiral fermion is not a chiral fermion; $SO(4) \cong SU(2)\times SU(2)$ splits what $SO^+(3,1)$ holds together), and proposes the counter-move: formulate the theory *Euclidean-holomorphically on projective twistor space* $\mathbb{PT}$, and reconstruct the Lorentzian theory from a conjugation structure / boundary values — the real form is *recovered*, not assumed. TFPT's celestial/twistor route has, independently, assembled the very ingredients such a reconstruction needs: reflection positivity of the seam collar (v379; the Osterwalder–Schrader axiom, machine-witnessed), the OS reconstruction step itself (the cited AMT/OS selector in `FORM.SEAM.MMST.01`), the order-4 clock $\rho = \mathrm{diag}(i, 1)$ acting on the twistor sphere (v492), and the seam reflection (Step 20 makes the referent precise: the seam-circle *reflection*, not the deck/covering involution — the deck, whose freeness is topology v510, carries the $(-1)^F$ Kramers class instead). What is *missing* is stated exactly: a *global anti-linear map* $\Theta : \mathcal{A}(\mathbb{PT}/\Gamma) \to \mathcal{A}(\mathbb{PT}/\Gamma)$ with $\Theta^2 = 1$ and $\Theta\rho\Theta = \rho^{-1}$ on the *interacting* open+closed twistorial algebra, together with reflection positivity of the interacting BCOV+SDYM functional — the two inputs of the new central contract `WOIT.OS.TWISTOR.01` in `tfpt_research_contracts`. The $\alpha$-stage status (Step 20, v519): *the missing global object now exists at the free level* — $\Theta$ with $\Theta^2 = +1$ and $\Theta\rho\Theta = \rho^{-1}$ on all four levels (sphere, $\mathbb{C}^4$, $\mathbb{Z}_8$ spin, Cl(16) Fock), with free RP on the seam system selecting the same family; Woit's two inequivalent real structures ($\rho_{\mathrm{tw}}$ and $\sigma_{\mathrm{std}}$) are exactly the two families of that classification, in *different* contract slots. The $\beta_1$-stage status (Step 23, v522) sharpens the dictionary once more: the $\mu_4$ clock is *Woit's euclidean rotation itself* (time-like: every seam mirror axis inverts it), the only gaugeable part of its tower is the GSO/fermion-parity $\mathbb{Z}_2$, and gauge-fixed RP holds under that typing — kill test (2)'s free shadow does not fire. The $\beta_2$-stage status (Step 25, v524) delivers the first *reconstructed* objects: the OS quotient of the free system is explicit — $(\mathcal{H}_{\mathrm{phys}}, \Omega)$ positive definite at both levels, the euclidean rotation a positive transfer step with spectral calculus and a reconstructed unitary rotation group (exact clock spectrum $\{1, \sqrt{2}-1\}$ at $N = 8$), the compact-circle reconstruction thermal (KMS) with exactly the pre-declared silver witnesses. The $\beta_3$-stage status (Step 32, v565) closes the dictionary at the free/quotient level: Woit's Minkowski conjugation ($\mathbb{PT} \to \mathbb{PT}^*$) is constructed as the compatibility class of Hermitian dualities, its Minkowski branch carries the forced $(2, 2)$ signature and the forced OS cut on the seam circle, the kill branch (a clock-centralising family-A structure) is empty by algebra, and the unique induced seam member *is* the $\beta_2$ mirror $\theta_\perp$ — $\Theta_{\mathrm{phys}}$ is the quotient shadow of the $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality, with $\sigma_{\mathrm{std}}$ playing the role of the OS cut itself. *The interacting statement stays [O]* — kill tests 1 and 2 are discharged only at the free level, and until $\Theta$ + RP exist on $\mathcal{A}_{\mathrm{hol}}$, the Woit connection remains a strong analogy, not a mathematical integration; nothing in this paragraph is evidence for either programme, and no marker moves.

## 18 Honest status: bootstrap, not creation from nothing

- **What is genuinely shown [E]:** $g_{\mathrm{car}} = 5$ is an *overdetermined* fixed point of the $E_8$ closure (two independent Lie-theory conditions); the integer "8" in $c_3$ is the carrier Coxeter = $E_8$

rank; the discrete data closes a loop. The earlier landscape scan confirms $g_{\text{car}} = 5$ is the *unique* value at which all hierarchies/family/gauge close — so the fixed point is unique, not one of many. **This uniqueness is now machine-formalised [E] in Lean 4** (Theorem A, `lean4-carrier-rigidity`): the division-free Pascal condition $2^g = g^2 + g + 2$ has the *unique* solution $g = 5$ (`carrier_rank_pascal_unique`; growth lemma for $g \geq 6$; $N_{\text{fam}} = (2^4{-}1)/5 = 3$, $g{+}N_{\text{fam}} = 8$), audited to use only the standard kernel axioms. The 2026-06-11 hardening round adds two formal modules: `AnchorLadder.lean` (the anchor power-sum law $p_n(a) = 2 + 2^n$ for *all* $n$, the ladder atoms $(3, 4, 6, 10)$, 240/8/248, and the binary ladder $p_{n+1}{-}p_n = 2^n$ as general theorems) and `PascalLadder.lean` (the odd-midpoint ladder identity $\sum_{k\leq K} \binom{2K+1}{k} = 2^{2K}$ for *all* $K$ via Mathlib, plus the kernel-decidable bounded uniqueness of the $K{=}2$ carrier case) — the solves-side of the Pascal Ladder Theorem ( [`verification/v108_pascal_ladder.py`]) is thereby [E].

- **What this is *not*:** a derivation from zero inputs. A self-consistency loop is consistent *at* a value; uniqueness is what makes it predictive (and that is supplied by the landscape scan). The continuous $\pi$ is not produced by the loop — it is the Möbius/boundary geometry, the one irreducible primitive.
- **Conceptual gain:** the theory's foundation tightens from "two free axioms" to "*the* $E_8$-closure fixed point + the geometric $\pi$." That is the Möbius self-consistency you intuited: the seed is the start of the cascade, and the cascade returns the seed's discrete content.
- **The current reduction state (2026-07-22; text consolidation, markers unchanged):** the compiler's dimensionless inputs reduce to *four marks* (derived from P1-side topology, v216), *one discrete symmetry-lift bit* (flag transitivity $\iff \tau = i$; equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle; v499/v506/v507/v510/v512), and $\pi$ — with the [C] identifications R1/R2 of v499 as the remaining conditional layer. `AX.P2.01` stays a declared axiom and `QGEO.REALIZE.01` stays [C]/[O]: this line describes the *reduction state*, not an abolition.

# Part IV
# Open items and the $E_8$-slice scan

## 19 The honest open-items list

This is the complete current ledger of what is *not* fully closed, graded by how hard the gap is.

| # | Open item | Grade | Hardening path / what closes it |
|---|---|---|---|
| *1. SM core — finite computations, no new physics* | | | |
| 1a | absolute quark $c$-scale (ratios closed) | [O] | quark *ratios* are integer Plücker readouts on the derived selector stratum (v49/v69/v71); only the *absolute* $U_f^\star$ normalisation remains — an anchor |
| 1b | $\theta_{12}$ solar angle | [E] | **texture closed (below):** explicit $\mu\tau$-symmetric texture verified; only $\varepsilon{=}\frac{3}{4}\varphi_0^{\rm ret}{\approx}c_3$ (0.23%) stays conditional |
| *2. Scheme layer — standard RG, no new input* | | | |
| 2a | $m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s$ | [C] | RG threshold matching $R_{\rm SM}$; $m_H$ on the closed UV quartic $1/16\pi^2$ |
| 2b | hadron pole spectrum (singlet) | [C] | singlet algebra $\mathcal{A}_{\rm sing}$ explicit; dynamics already closed (gap $6\log\frac{3}{2}$) |
| *3. Conditional — named analytic hypotheses* | | | |
| 3a | gravity: local eq. no free dimensionless Newton coupling under named QFT/entanglement premises [C]; ambient redundancy | [E]/[C] | full covariant $G_{ab}{+}\Lambda g_{ab}{=}c_3^{-1}T_{ab}$ (v358/v359); perturbative graviton unitarity established (Stelle ghost a truncation artefact, v304/v370/v380); ambient measure a [C] redundancy (v369) |
| 3b | strong CP / QFT closure | [C] | reflection positivity (RP) is the one remaining input |
| 3c | $\Lambda$ branch label (rec vs car) | [C] | one discrete choice; typed via $(8\pi)^2$, value not fixed |
| *4. Axioms — not eliminable, only hardenable* | | | |
| 4 | P1 ($c_3$), P2 ($g_{\rm car}{=}5$) | [O] | more Lean; P2 algebra already machine-verified; remain inputs |
| *5. Frontier — typed transfer targets, never compiler powers (see frontier doc)* | | | |
| 5 | $\eta_B$, $m_p/m_e$, Koide, dark matter, ambient QG measure | [C] | transfers now runnable solvers with kill tests (v371–v375); ambient measure a [C] redundancy (v369); not forced (scan results below) |
| *6. Cosmology late-time pipeline — not simplified by the compiler* | | | |
| 6 | $C_\ell$ transfer, $a_{\ell m}$ seam, baryogenesis | [C] | frontier-doc machinery; $a_{\ell m}$ "good CMB world, not yet this CMB world" (SMICA test) |

*Marker key:* ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`.

**Net.** Items 1 are finite arithmetic; 2 are standard RG; 3 hinge on named hypotheses (RP, the $\Lambda$ branch, the UV completion); 4 are the two irreducible axioms; 5–6 are honestly open. The inflation sector (the only cosmology piece the compiler *does* sharpen) is now a parameter-free prediction ($M{=}c_3^{7/2}\bar{M}_{\rm Pl}$, document 2).

**Item 1b closed: the explicit solar-angle Majorana texture [E]**

The last open SM angle is now realised by an *explicit* texture, not a fitted number. Take the $\mu\tau$-symmetric Majorana matrix (tri-bimaximal at $\varepsilon{=}0$)

$$M_\nu(\varepsilon) = \begin{pmatrix} -\eta(\varepsilon) & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad \eta(\varepsilon) = \frac{2\sqrt{2}}{\tan 2\theta_{12}} - 1, \quad \sin^2\theta_{12} = \tfrac{1}{3}(1-2\varepsilon). \tag{1}$$

$\mu\tau$-symmetry forces $\theta_{23} = 45°$ and $\theta_{13} = 0$ *exactly*; diagonalisation at the TFPT solar deviation $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}}$ gives

$$\boxed{\sin^2\theta_{12} = \tfrac{1}{3} - \tfrac{\varphi_0^{\text{ret}}}{2} = 0.306747, \quad \theta_{23} = 45°, \quad \theta_{13} = 0} \quad \textbf{[E]} \tag{2}$$

**Honest residual** ([`verification/v9_neutrino_texture.py`]). The texture and the angle are exact. The seam identification is $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = \frac{3}{4}\left(\frac{1}{6\pi} + \frac{3}{256\pi^4}\right) = c_3 + \frac{9}{1024\pi^4}$: the *leading geometric term is exactly* $c_3 = \frac{1}{8\pi}$, and the 0.23% residual is purely the $\varphi_0^{\text{ret}}$ seed tail (document 2). So $\varepsilon = c_3$ holds at leading order; the full $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}}$ is what enters the texture. The reactor angle $\theta_{13}$ ($\sin^2\theta_{13} = e^{-5/6}\varphi_0^{\text{ret}}$, document 2) enters through a separate $\mu\tau$-breaking term and is not part of this minimal solar texture. **Scope:** this realises the TFPT solar shift in the $\mu\tau$-symmetric *limit*; it is *not* the full PMNS matrix. The $\theta_{23}$=45° value is the symmetric-limit centre, *not* an octant prediction — current global fits (NuFIT 6.0) leave the octant unresolved — and the reactor angle plus the combined matrix require the separately stated $\mu\tau$-breaking channel.

## 20 The $E_8$-slice scan: results (honest hit / miss)

All seven maximal slices of 248 were scanned for the unassigned quantities. The rule was a meaningful (low-complexity) compiler/slice expression matching to $< 0.5\%$; matches that need a free fitted factor are reported as *misses*.

| Slice / target | Result | Verdict |
|---|---|---|
| $A_4{\times}A_4 = SU(5)^2$ | $\sin^2\theta_W\vert_{\text{GUT}} = 3/8$ sits in the $SU(5)$ hypercharge embedding; the 12 coset states are the proton-decay $X, Y$ bosons | **[E]** (known) |
| $F_4{\times}G_2$ / Koide | Koide $Q = 2/3 = \vert\mathbb{Z}_2\vert/N_{\text{fam}}$ (sheet over families); $J_3(\mathbb{O})$ has dim 27=26+1, 3×3 octonionic "generations" on the diagonal | **[C]** structural |
| $D_8 = SO(16)$ | $128 = (16,4) \oplus (\overline{16},\overline{4})$ — *same* spinor as $D_5{\times}A_3$; $\nu_R$ (seesaw) lives in the 16; **no new dark-matter state** (only the 4th $\mu_4$ direction) | **[E]** |
| $A_8 = SU(9)$, 9=3×3 | $\Lambda^3(3,3) = (10,1)\oplus(8,8)\oplus(1,10)$: the (8,8)=64 dominates — naive family×colour **fails** | miss |
| $A_8 \supset SU(5){\times}SU(4)$ | $\Lambda^3(5{+}4)$ contains the SM **10**: a genuine chiral-family-unification *lead*, but not TFPT-forced | lead |
| $E_6{\times}A_2$ / Jarlskog | cross term $2(\mathbf{1}^\top Ra)N_{\text{fam}} = 162 = 248{-}78{-}8$ exact; $J \sim \lambda_Y^6$ only to right *order*, no clean hit | weak |
| $E_8$ theta shells | root shells $240, 2160, 6720, \ldots$; $31{=}2^{g_{\text{car}}}{-}1$ is *not* a clean shell ratio | inconcl. |
| $m_p/m_e = 1836.15$ | no clean $E_8$/compiler expression (a QCD-confinement ratio); apparent hits need a free factor | **miss** |
| $\eta_B \approx 6.1{\times}10^{-10}$ | sits *between* $c_3^6{=}4{\times}10^{-9}$ and $c_3^7{=}1.6{\times}10^{-10}$; right order, no integer power | order only |

*Marker key:* ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`.

**Genuine new readings the scan produced**

- **Koide = $\vert\mathbb{Z}_2\vert/N_{\text{fam}}$.** The empirical Koide value 2/3 is exactly the sheet over the family count — a clean (if simple) compiler reading, promoting Koide from "near 2/3" to "democratic $\vert\mathbb{Z}_2\vert/N_{\text{fam}}$."
- **$E_8$ as the optimal 8-bit code.** $E_8$ is the optimal 8D lattice (kissing number 240). In byte framing $256{-}248 = 8 = \operatorname{rank} E_8$ and $256{-}240 = 16 = \dim S^+$ — a clean information-theoretic reading of the two load-bearing $E_8$ numbers.
- **Dark matter = no new state.** The $SO(16)$ scan shows the 128 is the *same* matter spinor; there is no spare $E_8$ singlet to be a WIMP. Any DM candidate must be the determinant-line axion (frontier doc), not a new $E_8$ rep — a useful *negative*.
- **$SU(9) \supset SU(5){\times}SU(4)$ chiral lead.** $\Lambda^3$ puts the SM **10** in the antisymmetric; the cleanest unexploited route to a family-unification reading, flagged for future work.

**Honest negatives**

$m_p/m_e$ has *no* clean $E_8$/compiler derivation (it is a confinement ratio — the apparent "hits" all need a free fitted factor); $\eta_B$ matches only at the order-of-magnitude level ($\sim c_3^{6.x}$); the $\Lambda$ rec/car branch is *not* selected by the theta series; the Jarlskog invariant is right only to order $\lambda_Y^6$. These are reported as open, not forced.

## 21 Following the $SU(9)$ lead: cross-slice consistency

The one scan lead worth following is $A_8 = SU(9) \supset SU(5) \times SU(4)$. Splitting $9 = (5,1) \oplus (1,4)$ the $E_8$ branching $248 = 80 \oplus 84 \oplus \overline{84}$ decomposes *exactly* as:

| piece | $SU(5) \times SU(4)$ content | reading |
|---|---|---|
| 80 (adj) | $(24,1)+(1,15)+(5,\bar{4})+(\bar{5},4)+(1,1)$ | $SU(5)_{\rm GUT}$ + $SU(4)_{\rm fam}$ gauge |
| $84 = \Lambda^3 9$ | $(10,1)+(10,4)+(5,6)+(1,4)$ | the SM **10** as family-singlet + family-**4** |
| $\overline{84}$ | $(\overline{10},1)+(\overline{10},\bar{4})+(\bar{5},6)+(1,\bar{4})$ | the conjugate matter |

(Dimension checks: 24+15+20+20+1 = 80; 10+40+30+4 = 84; total 248.)

**The genuine content: $A_8$ and $D_5 \times A_3$ agree on the family [E]**

The $SU(4)$ here is *the same* $A_3$ family group as in the construction slice $D_5 \times A_3$ — the four punctures $\mu_4$ of $\mathbb{P}^1 \backslash \mu_4$. The $SU(5)$ is the Georgi–Glashow GUT sitting inside the carrier $SO(10) = D_5$ ($SU(5) \subset SO(10)$). So the two maximal slices are *consistent*: both read the family as $SU(4)$=$A_3$, and $A_8$ additionally exposes the $SU(5)$ matter content — the SM **10** appears as $(10,1) \oplus (10,4)$, i.e. a family-singlet **10** plus a family-**4**, where the **4** of $A_3$ is exactly TFPT's "$4 = N_{\rm fam} + 1$" (three families + the $\mu_4$ direction).

**Honest limit: chirality is still the index, not $E_8$**

$84 \oplus \overline{84}$ is *vectorlike* ($10 \oplus \overline{10}$ etc.): $E_8$ being non-chiral, the slice cannot by itself produce three *chiral* generations. The chirality is the boundary index $\mathrm{Ind}(D^+_{\rm fam}) = 3$ — the *same* input TFPT already uses. So $SU(9)$ is a consistent re-reading (matter in $SU(5)$, family in $A_3$), not an independent derivation of $N_{\rm fam} = 3$. Each $SU(5)$ pair ($\mathbf{10} + \bar{\mathbf{5}}$) is anomaly-free, so the embedding is clean. [C]

## Appendix: computational verification

The audit/bridge identities of this document (marked [E], [E]) are re-derived from $\{c_3, g_{\rm car}\}$ and machine-checked by the suite in the repository folder `verification/`; the inline tags `[verification/...]` point to the exact script. Relevant here:

- `v1_e8_glue.py` — the $\mu_4$ glue $E_8$=$(D_5 \oplus A_3)$+$\mu_4$ (load-bearing).
- `v4_flavor_matrix.py` — the spectral selector $\det R$=$h(D_5)$=8, minors $\{2,3,5\}$.
- `v5_e8_cascade.py` — the cascade $D_n$=60−2$n$: endpoints, exponent rungs, IR tail, variance.
- `v6_bootstrap.py` — the self-consistency loop: $\mu^2$−5$\mu$+4=0, $g_{\rm car}$=5 three ways, 8= rank $E_8$.

The $E_8$-slice/atlas readings are explicitly [O] (audit-level), not part of the pass/fail suite. Run `python verification/run_all.py`; full map in `verification/README.md`.